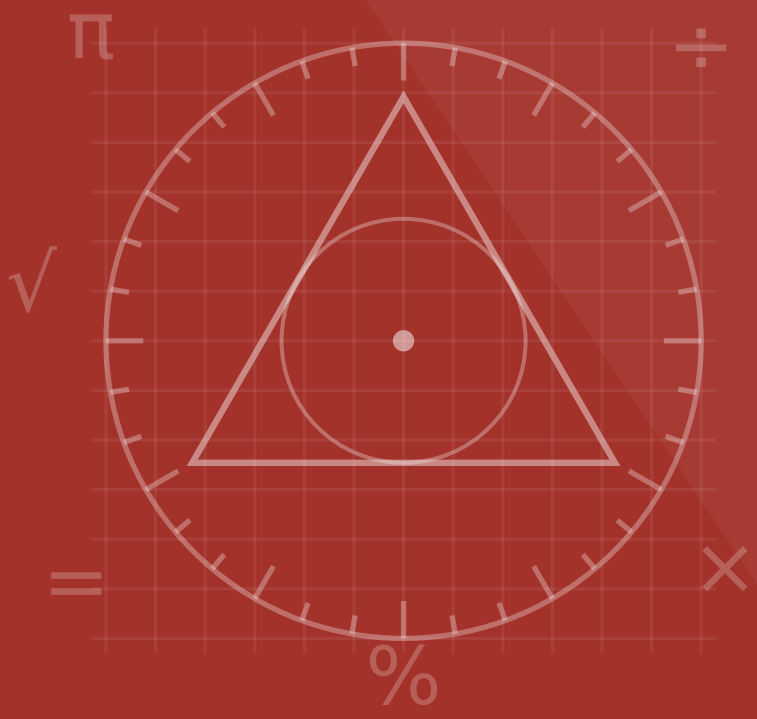
VENVLEARNING

STUDENT TEXTBOOK



Mathematics

GRADE 4

Lessons, examples, and practice for the whole year

2026–2027 School Year · 10 units

VENVLEARNING

Grade 4
Mathematics

STUDENT TEXTBOOK

2026–2027 School Year

Unit 1 Place Value, Rounding, and Whole Numbers
Unit 2 Multiplication
Unit 3 Division
Unit 4 Factors, Multiples, and Patterns
Unit 5 Fraction Equivalence and Comparison
Unit 6 Fraction Operations
Unit 7 Decimals
Unit 8 Geometry, Angles, and Line Plots
Unit 9 Measurement, Area, and Perimeter
Unit 10 Year in Review

This book belongs to _____

A VenvLearning family homeschool curriculum

How to use this book

Each lesson starts with an I can goal and a Let'S Explore question. Read the lesson with your teacher. Study the Worked Examples step by step, and try the Try It problems. Your practice pages are in your Student Workbook, labeled with the same lesson number.

- Key Idea boxes hold the most important rule or fact. Learn these.
- Watch Out boxes warn you about common mistakes.
- Every Friday there is a short Weekly Check in your workbook. The last Friday of a unit is the Unit Test.

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UNIT 1

Place Value, Rounding, and Whole Numbers

How our number system works, and how to read, compare, round, add, and subtract numbers up to one million.

Every number you will work with this year is built from place value. In this unit you will see how each place is worth ten times the place to its right, read and write numbers up to one million, compare and order them, round them to friendly numbers, and add and subtract them with the standard algorithm.

In this unit you will…

- Explain why each place is worth 10 times the place to its right.
- Read and write numbers to 1,000,000 in standard, word, and expanded form.
- Compare and order multi-digit numbers using >, =, and <.
- Round numbers to any place and use rounding to estimate.
- Add and subtract multi-digit numbers with the standard algorithm and check if answers are reasonable.

MATH IN REAL LIFE

Big numbers are everywhere: the number of people in Mississippi (about 2.9 million), the miles on a car, the price of a house, or the seats in a stadium. Place value helps you read them, and rounding helps you talk about them quickly.

UNIT VOCABULARY

place value	The value a digit has because of its place in a number.
period	A group of three places separated by a comma.
expanded form	A number written as the sum of the values of its digits.
round	To change a number to a nearby friendly number.
estimate	A close answer found with rounded numbers.
regroup	To trade 10 of one unit for 1 of the next bigger unit, or the reverse.

Lessons in this unit

Lesson	Title
1.1	How Place Value Works
1.2	Place Value to One Million
1.3	The Value of a Digit
1.4	Standard, Word, and Expanded Form
1.5	Renaming Numbers
1.6	Comparing Multi-Digit Numbers
1.7	Ordering Numbers
1.8	Large Numbers on a Number Line
1.9	Rounding with a Number Line
1.10	Rounding to Any Place
1.11	Estimating Sums and Differences
1.12	Is My Answer Reasonable?
1.13	Adding Multi-Digit Numbers
1.14	Subtracting with Regrouping
1.15	Subtracting Across Zeros
1.16	Solving Multi-Step Word Problems

UNIT 1 · LESSON 1.1 · 4.NBT.1

Lesson 1.1: How Place Value Works

I can explain that a digit in one place is worth ten times what it would be worth in the place to its right.

WORDS TO KNOW

digit	One of the ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 used to write numbers.
place value	The value a digit has because of where it is in a number.
base-ten	Our number system, where each place is worth 10 times the place to its right.

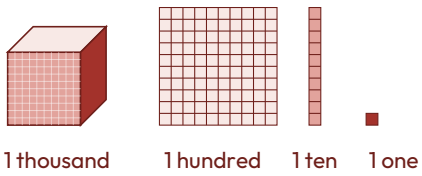
LET'S EXPLORE

Imagine you have a big pile of 1,000 paper clips. Counting them one at a time would take a long time! How could you bundle them to count faster? What size bundles would you make?

We write every number, no matter how big, with just ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. The trick that makes this work is place value. Where a digit sits tells us how much it is worth.

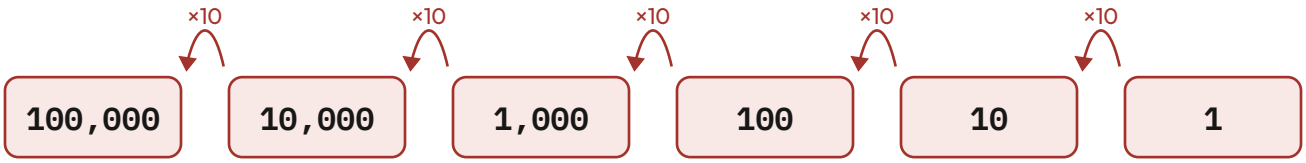
Bundling by tens

Our number system is called base-ten because we group things in tens. Ten ones make one ten. Ten tens make one hundred. Ten hundreds make one thousand. The pattern keeps going forever.



Base-ten blocks: a thousand cube, a hundred flat, a ten rod, and a one cube.

Look at the blocks. A ten rod is 10 one cubes stacked together. A hundred flat is 10 ten rods side by side. A thousand cube is 10 hundred flats stacked up.



Each place is worth 10 times the place to its right.

KEY IDEA

In a multi-digit number, a digit in one place is worth 10 times what it would be worth in the place to its right.

WORKED EXAMPLE · COMPARE TWO 7S

In the number 770, how many times greater is the first 7 than the second 7?

Hundreds	Tens	Ones
7	7	0

1 The first 7 is in the hundreds place. Its value is 700.

2 The second 7 is in the tens place. Its value is 70.

3 Ask: 70 times what number equals 700? 70 × 10 = 700.

Answer: The first 7 is worth 10 times as much as the second 7.

WORKED EXAMPLE · TEN TIMES AS MUCH

What number is 10 times as much as 400?

1 Think about place value. 400 is 4 hundreds.

2 Ten times 4 hundreds is 4 thousands, because each place is worth 10 times the place to its right.

3 Write 4 thousands as a number: 4,000.

Answer: 10 × 400 = 4,000

WATCH OUT

When you multiply a number by 10, the digits move one place to the left. A zero fills the ones place. It is not just “adding a zero”: 10 × 35 = 350 because 3 tens become 3 hundreds and 5 ones become 5 tens.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many tens are in 100?

2. What number is 10 times as much as 60?

3. In 5,550, how does the 5 in the thousands place compare with the 5 in the hundreds place?

More practice: Workbook, Lesson 1.1.

LESSON SUMMARY

Each place is worth 10 times the place to its right: 10 ones = 1 ten, 10 tens = 1 hundred, 10 hundreds = 1 thousand.

UNIT 1 · LESSON 1.2 · 4.NBT.2

Lesson 1.2: Place Value to One Million

I can read and write numbers up to one million using periods and commas.

WORDS TO KNOW

period	A group of three places in a number, such as the ones period or the thousands period.
standard form	The usual way to write a number with digits, such as 452,318.
one million	1,000,000. It is 10 times 100,000 and 1,000 times 1,000.

LET'S EXPLORE

The town of Hattiesburg has about 48,000 people. The state of Mississippi has close to 3 million. How do you read numbers this big out loud?

Big numbers are easier to read when we split the digits into groups of three called periods. A comma separates each period. Starting from the right, the periods are the ones period, the thousands period, and the millions period.

THOUSANDS PERIOD			ONES PERIOD		
Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
4	5	2	3	1	8

452,318 in a place-value chart.

Each period has the same three places: hundreds, tens, and ones. In the thousands period they are called hundred thousands, ten thousands, and thousands.

Reading a large number

- Read the number in the left period like a normal three-digit number.
- Say the period name when you reach the comma (“thousand”).
- Read the last period. You do not say “ones.”

WORKED EXAMPLE · READ A SIX-DIGIT NUMBER

Read 452,318 out loud.

1 The thousands period shows 452. Say “four hundred fifty-two thousand.”

2 The ones period shows 318. Say “three hundred eighteen.”

Answer: four hundred fifty-two thousand, three hundred eighteen

WORKED EXAMPLE · WRITE A NUMBER IN STANDARD FORM

Write “sixty thousand, four hundred five” in standard form.

1 The thousands period is sixty, so write 60 and then a comma.

2 The ones period is four hundred five: 405. It needs three digits.

3 Put them together.

Answer: 60,405

WATCH OUT

Every period after the first one needs exactly three digits. Sixty thousand, forty-five is 60,045, not 60,45. Use zeros to hold empty places.

KEY IDEA

One million is written 1,000,000. It is 10 times 100,000. That is the largest number we work with this year.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Read 307,250 out loud. Write the words.

2. Write “ninety-one thousand, six” in standard form.

3. How many digits does one million have?

More practice: Workbook, Lesson 1.2.

LESSON SUMMARY

Periods are groups of three digits separated by commas. Read each period, then say its name.

UNIT 1 · LESSON 1.3 · 4.NBT.1, 4.NBT.2

Lesson 1.3: The Value of a Digit

I can name the place and the value of any digit in a number up to one million.

WORDS TO KNOW

value	How much a digit is worth in a number. In 7,482 the 4 has a value of 400.

LET'S EXPLORE

Two numbers both have a 9 in them: 59 and 95,000. Is the 9 worth the same amount in both numbers?

Every digit has a place and a value. The place is the name of its column. The value is the digit times the place.

THOUSANDS PERIOD			ONES PERIOD		
Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
3	8	6	2	5	4

In 386,254 the 8 is in the ten thousands place.

In 386,254 the digit 8 is in the ten thousands place. Its value is 8 × 10,000 = 80,000.

WORKED EXAMPLE · FIND PLACE AND VALUE

What is the place and the value of the 5 in 652,019?

THOUSANDS PERIOD			ONES PERIOD		
Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
6	5	2	0	1	9

1. Count places from the right: ones (9), tens (1), hundreds (0), thousands (2), ten thousands (5).
2. The 5 is in the ten thousands place.
3. Value = 5 × 10,000.

Answer: ten thousands place; value 50,000

WORKED EXAMPLE · COMPARE VALUES

In 444,000, how does the value of the first 4 compare with the second 4?

1 First 4: hundred thousands place, value 400,000.

2 Second 4: ten thousands place, value 40,000.

3 40,000 × 10 = 400,000.

Answer: The first 4 is worth 10 times the second 4.

KEY IDEA

Value = digit × place. A zero has a value of 0, but it still holds its place.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. What is the value of the 7 in 27,315?

2. In which place is the 3 in 830,642?

3. Which digit is in the hundred thousands place in 914,208?

More practice: Workbook, Lesson 1.3.

LESSON SUMMARY

To find a digit's value, multiply the digit by the value of its place. Zeros hold empty places.

UNIT 1 · LESSON 1.4 · 4.NBT.2

Lesson 1.4: Standard, Word, and Expanded Form

I can write a number in standard form, word form, and expanded form.

WORDS TO KNOW

word form	A number written with words, such as thirty-two thousand, five.
expanded form	A number written as the sum of the values of its digits, such as 30,000 + 2,000 + 5.

LET'S EXPLORE

Your grandmother writes a check for $1,250 and must write the amount in words. How would she write it?

There are three common ways to show the same number.

Form	Example: 32,605
Standard form	32,605
Word form	thirty-two thousand six hundred five
Expanded form	30,000 + 2,000 + 600 + 5

Expanded form shows the value of each digit. Skip any zeros, because their value is 0.

WORKED EXAMPLE · WRITE IN EXPANDED FORM

Write 470,903 in expanded form.

1 List the value of each digit that is not zero: 4 → 400,000; 7 → 70,000; 9 → 900; 3 → 3.

2 Join the values with plus signs.

Answer: 400,000 + 70,000 + 900 + 3

WORKED EXAMPLE · FROM EXPANDED FORM TO STANDARD FORM

Write 80,000 + 6,000 + 40 + 2 in standard form.

THOUSANDS PERIOD		ONES PERIOD		
Ten Thousands	Thousands	Hundreds	Tens	Ones
8	6	0	4	2

1 Find each place: 8 ten thousands, 6 thousands, 0 hundreds, 4 tens, 2 ones.

2 Write a zero for the missing hundreds.

Answer: 86,042

WATCH OUT

In expanded form, the hundreds place in 86,042 is empty. When you go back to standard form, put a 0 there or you will get 8,642, which is a much smaller number.

KEY IDEA

Standard form, word form, and expanded form are three names for the same number.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Write 25,019 in expanded form.
2. Write 300,000 + 50,000 + 700 + 1 in standard form.
3. Write 41,300 in word form.

More practice: Workbook, Lesson 1.4.

LESSON SUMMARY

Standard form uses digits. Word form uses words. Expanded form adds the value of each digit.

UNIT 1 · LESSON 1.5 · 4.NBT.1

Lesson 1.5: Renaming Numbers

I can rename a number using different place-value units, such as 3,400 = 34 hundreds.

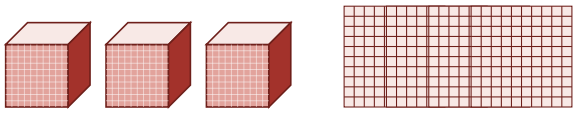
WORDS TO KNOW

rename	To write the same amount using different units, such as 1 thousand = 10 hundreds.
regroup	To trade 10 of one unit for 1 of the next bigger unit, or 1 for 10 of the next smaller unit.

LET'S EXPLORE

A bank gives you $3,400 in $100 bills. How many $100 bills do you get?

Because 10 of one unit equals 1 of the next bigger unit, we can rename numbers in different ways. This will help a lot when we add, subtract, multiply, and divide.



3 thousands 4 hundreds

3,400 is 3 thousands and 4 hundreds.

Each thousand is 10 hundreds, so 3 thousands is 30 hundreds. Add the 4 hundreds: 3,400 = 34 hundreds. That means you get 34 bills of $100.

WORKED EXAMPLE · RENAME WITH TENS

How many tens are in 2,560?

1 Cover the ones digit: 2,560 → 256.

2 There are 256 tens in 2,560. Check: 256 × 10 = 2,560.

Answer: 256 tens

WORKED EXAMPLE · RENAME WITH THOUSANDS

Write 45 thousands in standard form.

1 1 thousand = 1,000, so 45 thousands = 45 × 1,000.

2 45 thousands is 4 ten thousands and 5 thousands.

Answer: 45,000

WORKED EXAMPLE · MANY WAYS TO SHOW ONE NUMBER

Show 1,200 in three ways.

1 1 thousand + 2 hundreds

2 12 hundreds

3 120 tens

Answer: All three name 1,200.

KEY IDEA

To find how many of a unit are in a number, look at all the digits from the left down to that unit's place. There are 56 hundreds in 5,600 and 560 tens in 5,600.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many hundreds are in 7,000?

2. Write 120 tens in standard form.

3. Fill in: 6,300 = ___ hundreds

More practice: Workbook, Lesson 1.5.

LESSON SUMMARY

The same number can be named with different units: 3,400 = 3 thousands 4 hundreds = 34 hundreds = 340 tens.

UNIT 1 · LESSON 1.6 · 4.NBT.2

Lesson 1.6: Comparing Multi-Digit Numbers

I can compare two multi-digit numbers using >, =, and <.

WORDS TO KNOW

greater than (>)	Shows that the first number is larger: 9 > 4.
less than (<)	Shows that the first number is smaller: 4 < 9.
equal to (=)	Shows two amounts are the same.

LET'S EXPLORE

Two video games sold 245,600 copies and 254,600 copies. They have the same digits! Which sold more?

To compare numbers, compare the digits in the same place, starting with the greatest place.

Steps to compare

- Line the numbers up by place value (ones under ones).
- If one number has more digits, it is greater.
- If they have the same number of digits, start at the left. Find the first place where the digits are different.
- The number with the greater digit in that place is greater.

WORKED EXAMPLE · COMPARE

Compare 245,600 and 254,600.

	HTh	TTh	Th	H	T	O
245,600	2	4	5	6	0	0
254,600	2	5	4	6	0	0

1 Both have six digits.

2 Hundred thousands: 2 and 2, the same.

3 Ten thousands: 4 and 5. They are different, and 4 < 5.

Answer: 245,600 < 254,600

WORKED EXAMPLE · DIFFERENT NUMBERS OF DIGITS

Compare 98,999 and 101,000.

1 98,999 has five digits. 101,000 has six digits.

2 A six-digit number is always greater than a five-digit number.

Answer: 98,999 < 101,000

KEY IDEA

The symbol opens toward the greater number, like a mouth that wants the bigger meal: 7 > 3 and 3 < 7.

WATCH OUT

Do not compare the first digits you see if the numbers have different lengths. 98,999 starts with a 9, but it is less than 101,000.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Compare: 63,417 ___ 63,471

2. Compare: 500,010 ___ 499,999

3. Compare: 80,000 + 700 ___ 80,700

More practice: Workbook, Lesson 1.6.

LESSON SUMMARY

Compare digits in the same place from left to right. The first place with different digits decides.

UNIT 1 · LESSON 1.7 · 4.NBT.2

Lesson 1.7: Ordering Numbers

I can order three or more multi-digit numbers from least to greatest or greatest to least.

WORDS TO KNOW

order	To arrange numbers from least to greatest or from greatest to least.
least	The smallest number in a group.
greatest	The largest number in a group.

LET'S EXPLORE

Three Mississippi rivers have these lengths in miles: Pascagoula 81, Big Black 330, Yazoo 188. Which is longest? How would you list them from shortest to longest?

Ordering is comparing again and again. Line up the numbers by place value and compare one place at a time.

WORKED EXAMPLE · ORDER FROM LEAST TO GREATEST

Order 36,812; 36,281; 38,126.

Number	TTh	Th	H	T	O
36,812	3	6	8	1	2
36,281	3	6	2	8	1
38,126	3	8	1	2	6

1 Ten thousands: all are 3.

2 Thousands: 6, 6, 8. So 38,126 is greatest.

3 Compare the other two at the hundreds place: 36,812 and 36,281. 2 < 8.

Answer: 36,281; 36,812; 38,126

WORKED EXAMPLE · ORDER FROM GREATEST TO LEAST

Order 902,415; 92,415; 920,415.

1 92,415 has only five digits, so it is least.

2 Compare 902,415 and 920,415: ten thousands place 0 < 2.

3 Greatest first.

Answer: 920,415; 902,415; 92,415

KEY IDEA

Read the directions carefully: least to greatest starts with the smallest number. Greatest to least starts with the largest.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Order from least to greatest: 4,509; 4,950; 4,095
2. Order from greatest to least: 71,300; 17,300; 73,100
3. Which number is greatest: 650,001; 605,100; 650,010?

More practice: Workbook, Lesson 1.7.

LESSON SUMMARY

To order numbers, line them up by place value and compare from the greatest place. Check the direction: least to greatest or greatest to least.

UNIT 1 · LESSON 1.8 · 4.NBT.2

Lesson 1.8: Large Numbers on a Number Line

I can place and find large numbers on a number line.

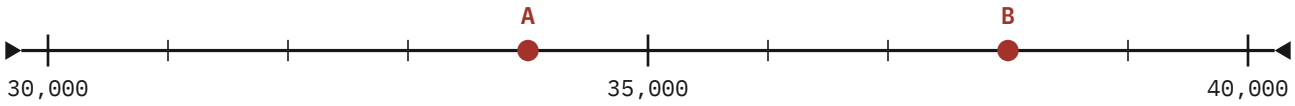
WORDS TO KNOW

number line	A line with evenly spaced marks that shows numbers in order.
interval	The amount between two marks next to each other on a number line.

LET'S EXPLORE

A number line goes from 0 to 10,000. Where would you put 5,000? Where would 9,900 go?

A number line shows numbers in order. Numbers get greater as you move right. To use a number line with big numbers, first find the interval: how much each jump between marks is worth.



Each mark is 1,000 more than the mark before it.

WORKED EXAMPLE · FIND THE INTERVAL

On the number line above, what numbers are at A and B?

1 From 30,000 to 40,000 there are 10 jumps.

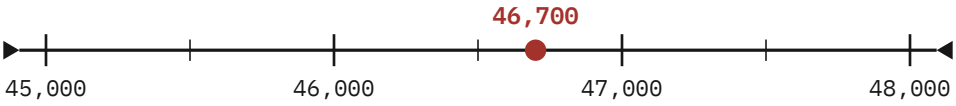
2 10,000 ÷ 10 = 1,000, so each jump is 1,000.

3 A is 4 jumps past 30,000: 34,000. B is 8 jumps past 30,000: 38,000.

Answer: A = 34,000 and B = 38,000

WORKED EXAMPLE · PLACE A NUMBER

Between which two marks does 46,700 go on a number line marked by thousands?



1 46,700 is more than 46,000 and less than 47,000.

2 700 is a little more than halfway (500) between them.

Answer: Between 46,000 and 47,000, a bit past the middle

KEY IDEA

Number lines help us see which numbers are close to each other. Tomorrow's skill, rounding, uses this idea.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A number line goes from 0 to 1,000 with 10 equal jumps. How much is each jump?

2. Which thousand is 52,900 closer to: 52,000 or 53,000?

3. What number is halfway between 200,000 and 300,000?

More practice: Workbook, Lesson 1.8.

LESSON SUMMARY

On a number line, find the interval first. Numbers increase to the right, and the line shows which benchmark a number is closer to.

UNIT 1 · LESSON 1.9 · 4.NBT.3

Lesson 1.9: Rounding with a Number Line

I can use a number line to round a number to the nearest ten, hundred, or thousand.

WORDS TO KNOW

round	To find a nearby, friendly number that is easier to use. 47 rounded to the nearest ten is 50.
benchmark	A friendly number, like a multiple of 10, 100, or 1,000, used to round or estimate.
halfway	The point exactly in the middle of two benchmarks, like 450 between 400 and 500.

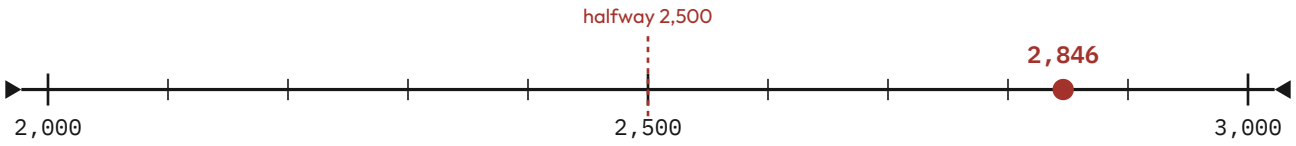
LET'S EXPLORE

A newspaper headline says "About 3,000 people came to the county fair." The real count was 2,846. Why might the newspaper use 3,000 instead of the exact number?

We round numbers when we do not need an exact answer. A rounded number is close to the real number but easier to say, remember, and work with.

Rounding is finding the closer benchmark

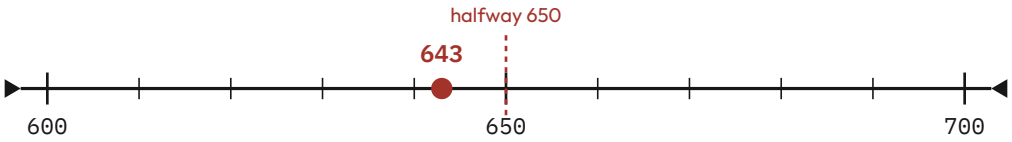
To round 2,846 to the nearest thousand, ask: is it closer to 2,000 or to 3,000? A number line shows the answer.



2,846 is past the halfway point (2,500), so it is closer to 3,000.

WORKED EXAMPLE · ROUND TO THE NEAREST HUNDRED

Round 643 to the nearest hundred.



1 Find the two hundreds that 643 is between: 600 and 700.

2 Find halfway: 650.

3 643 is less than 650, so it is closer to 600.

Answer: 643 rounds to 600

WORKED EXAMPLE · A NUMBER EXACTLY HALFWAY

Round 75 to the nearest ten.

1 75 is between 70 and 80.

2 75 is exactly halfway.

3 The rule: when a number is exactly halfway, we round up.

Answer: 75 rounds to 80

KEY IDEA

To round: (1) find the two benchmarks the number is between, (2) find the halfway point, (3) choose the closer benchmark. Halfway numbers round up.

WATCH OUT

Rounding to the nearest hundred does not mean the answer has to end in one zero. It ends in two zeros, like 600 or 1,200.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Round 38 to the nearest ten.

2. Round 1,250 to the nearest hundred.

3. Round 7,480 to the nearest thousand.

More practice: Workbook, Lesson 1.9.

LESSON SUMMARY

To round, find the two benchmarks, find the halfway point, and choose the closer benchmark. Halfway rounds up.

UNIT 1 · LESSON 1.10 · 4.NBT.3

Lesson 1.10: Rounding to Any Place

I can use place value to round a number to any place, up to the hundred thousands.

WORDS TO KNOW

rounding place	The place you are rounding to. If you round to the nearest thousand, the rounding place is the thousands place.

LET'S EXPLORE

Drawing a number line every time takes a while. Is there a faster way to round 46,281 to the nearest ten thousand?

Number lines show why rounding works. Place value gives us a quick way to do it.

The place-value rounding rule

- Underline the digit in the rounding place.
- Look at the digit just to its right.
- If that digit is 5 or more, add 1 to the underlined digit. If it is 4 or less, keep the underlined digit the same.
- Change every digit to the right of the rounding place to 0.

WORKED EXAMPLE · ROUND TO THE NEAREST TEN THOUSAND

Round 46,281 to the nearest ten thousand.

Ten Thousands	Thousands	Hundreds	Tens	Ones
4	6	2	8	1

1 Underline the ten thousands digit: 4̲6,281.

2 Look right: the thousands digit is 6.

3 6 is 5 or more, so 4 becomes 5.

4 Change the digits to the right to zeros.

Answer: 50,000

WORKED EXAMPLE · ROUND TO THE NEAREST HUNDRED THOUSAND

Round 734,900 to the nearest hundred thousand.

1 Underline the hundred thousands digit: 7̲34,900.

2 Look right: the ten thousands digit is 3.

3 3 is 4 or less, so 7 stays 7.

4 Change the digits to the right to zeros.

Answer: 700,000

WORKED EXAMPLE · WHEN ROUNDING UP MAKES A 9 BECOME 10

Round 3,970 to the nearest hundred.

1 Underline the hundreds digit: 3,9̲70.

2 Look right: 7 is 5 or more, so round up.

3 9 hundreds + 1 hundred = 10 hundreds = 1 thousand. Regroup: 3 thousands become 4 thousands.

Answer: 4,000

KEY IDEA

Look one place to the right. 5 or more: round up. 4 or less: stay the same. Then fill with zeros.

WATCH OUT

Only the digit in the rounding place can change (unless it regroups). Never change digits to the left for no reason. Round 46,281 to the nearest thousand: 46,000, not 40,000.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Round 58,412 to the nearest thousand.

2. Round 58,412 to the nearest ten thousand.

3. Round 249,999 to the nearest hundred thousand.

More practice: Workbook, Lesson 1.10.

LESSON SUMMARY

Underline the rounding place, look one place to the right, round up for 5 or more, and replace the digits after it with zeros.

UNIT 1 · LESSON 1.11 · 4.NBT.3

Lesson 1.11: Estimating Sums and Differences

I can round numbers to estimate sums and differences.

WORDS TO KNOW

estimate	An answer that is close to the exact answer, often found with rounded numbers.
sum	The answer to an addition problem.
difference	The answer to a subtraction problem.

LET'S EXPLORE

Your family drives 386 miles on Saturday and 412 miles on Sunday. Did you drive more or less than 1,000 miles? Can you tell without finding the exact answer?

An estimate is a smart guess that is close to the exact answer. Estimating is useful when you only need to know "about how many," and it helps you check your work later.

Round, then compute

WORKED EXAMPLE · ESTIMATE A SUM

Estimate 386 + 412.

1. Round each number to the nearest hundred: 386 → 400 and 412 → 400.
2. Add the rounded numbers: 400 + 400 = 800.

Answer: about 800 miles, so less than 1,000 miles

WORKED EXAMPLE · ESTIMATE A DIFFERENCE

Estimate 7,815 − 2,290.

1. Round to the nearest thousand: 7,815 → 8,000 and 2,290 → 2,000.
2. Subtract: 8,000 − 2,000 = 6,000.

Answer: about 6,000

Choosing a place to round to

Rounding to a greater place makes the math easier. Rounding to a smaller place makes the estimate closer to the exact answer. Usually, round to the greatest place of the smaller number.

Estimate 4,628 + 1,357	Rounded numbers	Estimate
Nearest thousand	5,000 + 1,000	6,000
Nearest hundred	4,600 + 1,400	6,000
Exact answer	4,628 + 1,357	5,985

KEY IDEA

To estimate, round each number first, then add or subtract the rounded numbers. Use the word about in your answer.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Estimate 52 + 37 by rounding to the nearest ten.
2. Estimate 6,120 − 3,890 by rounding to the nearest thousand.
3. Estimate 24,610 + 13,250 by rounding to the nearest thousand.

More practice: Workbook, Lesson 1.11.

LESSON SUMMARY

Round each number, then add or subtract the rounded numbers. An estimate tells you about how much.

UNIT 1 · LESSON 1.12 · 4.NBT.3, 4.OA.3

Lesson 1.12: Is My Answer Reasonable?

I can use estimation to decide whether an answer makes sense.

WORDS TO KNOW

reasonable	Makes sense; close to what we expect from an estimate.

LET'S EXPLORE

A classmate says 4,895 + 3,210 = 18,105. Without adding, how can you tell something is wrong?

Even good mathematicians make mistakes. A quick estimate is your safety net. If your exact answer is far from your estimate, check your work.

WORKED EXAMPLE · CHECK AN ANSWER

Is 4,895 + 3,210 = 18,105 reasonable?

1. Estimate: 5,000 + 3,000 = 8,000.
2. 18,105 is much more than 8,000.
3. The answer is not reasonable. The exact sum is 8,105.

Answer: No. It should be close to 8,000.

WORKED EXAMPLE · USE AN ESTIMATE IN A STORY

Mom has $500. She wants a $289 tablet and a $178 bike helmet. Does she have enough money?

1. Round to the nearest hundred: 300 + 200 = 500.
2. Both numbers were rounded up, so the exact total is less than $500.
3. Check: 289 + 178 = 467.

Answer: Yes. $467 is less than $500.

WATCH OUT

When both numbers are rounded up, the estimate is greater than the exact sum. When both are rounded down, the estimate is less. Thinking about this tells you whether you have enough.

KEY IDEA

Estimate before or after you solve. If the exact answer and the estimate are far apart, look for a mistake.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Is 612 + 297 = 909 reasonable? Use an estimate.
2. Is 8,050 − 2,980 = 6,070 reasonable?
3. You have $60. Books cost $18, $24, and $15. Do you have enough? Use estimation.

More practice: Workbook, Lesson 1.12.

LESSON SUMMARY

Use an estimate to check if an answer makes sense. Far from the estimate? Look for a mistake.

UNIT 1 · LESSON 1.13 · 4.NBT.4

Lesson 1.13: Adding Multi-Digit Numbers

I can add multi-digit numbers using the standard algorithm.

WORDS TO KNOW

standard algorithm	A step-by-step way to add or subtract by lining up places and working from right to left.
regroup	When a place has 10 or more, trade 10 of them for 1 of the next bigger place.

LET'S EXPLORE

On Saturday 2,768 people visited the aquarium. On Sunday 3,456 people visited. How many visitors came in all?

You already know how to add 2- and 3-digit numbers. The same steps work for numbers of any size.

The standard algorithm for addition

- Line up the numbers by place value: ones under ones, tens under tens.
- Add the ones first. If the sum is 10 or more, regroup: write the ones digit and carry 1 ten to the tens column.
- Add each place, moving left, and regroup whenever a sum is 10 or more.
- Estimate to check that your answer is reasonable.

WORKED EXAMPLE · ADD WITH REGROUPING

Find 2,768 + 3,456.

$$\begin{array}{r} {\scriptstyle 1\ 1\ 1\ \ } \\ 2,768 \\ +3,456 \\ \hline 6,224 \end{array}$$

1 Ones: 8 + 6 = 14. Write 4, regroup 1 ten.

2 Tens: 1 + 6 + 5 = 12. Write 2, regroup 1 hundred.

3 Hundreds: 1 + 7 + 4 = 12. Write 2, regroup 1 thousand.

4 Thousands: 1 + 2 + 3 = 6.

5 Check: 3,000 + 3,000 = 6,000. 6,224 is close.

Answer: 6,224 visitors

WORKED EXAMPLE · ADD THREE NUMBERS

Find 14,205 + 8,390 + 1,076.

1 Line up all three numbers by place value.

2 Ones: 5 + 0 + 6 = 11 → write 1, regroup 1.

3 Tens: 1 + 0 + 9 + 7 = 17 → write 7, regroup 1.

4 Hundreds: 1 + 2 + 3 + 0 = 6. Thousands: 4 + 8 + 1 = 13 → write 3, regroup 1. Ten thousands: 1 + 1 = 2.

Answer: 23,671

WATCH OUT

Line the numbers up on the right. When numbers have different lengths, the ones must still be in the same column.

KEY IDEA

Add from right to left. When a place adds to 10 or more, regroup 10 of that unit as 1 of the next unit.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4,526 + 3,371

2. 6,859 + 2,764

3. 45,308 + 27,996

More practice: Workbook, Lesson 1.13.

LESSON SUMMARY

Line up places, add from right to left, and regroup when a place has 10 or more. Estimate to check.

UNIT 1 · LESSON 1.14 · 4.NBT.4

Lesson 1.14: Subtracting with Regrouping

I can subtract multi-digit numbers using the standard algorithm, regrouping when needed.

WORDS TO KNOW

regroup (subtraction)	Trade 1 of a bigger unit for 10 of the next smaller unit so you have enough to subtract.

LET'S EXPLORE

A school raised $6,215 for a new playground. It has spent $3,748. How much money is left?

When you subtract, start in the ones place. If the top digit is smaller than the bottom digit, you need to regroup (some people say “borrow”) from the next place to the left.

WORKED EXAMPLE · SUBTRACT WITH REGROUPING

Find 6,215 − 3,748.

$$\begin{array}{r} {\scriptstyle 5\ 1\ 11\ 015} \\ 6,215 \\ -3,748 \\ \hline 2,467 \end{array}$$

1. Ones: 5 − 8 won't work. Regroup 1 ten: the tens become 0 and the ones become 15. 15 − 8 = 7.
2. Tens: 0 − 4 won't work. Regroup 1 hundred: hundreds become 1, tens become 10. 10 − 4 = 6.
3. Hundreds: 1 − 7 won't work. Regroup 1 thousand: thousands become 5, hundreds become 11. 11 − 7 = 4.
4. Thousands: 5 − 3 = 2.
5. Check by adding: 2,467 + 3,748 = 6,215. ✓

Answer: $2,467 is left

Check subtraction with addition

Subtraction and addition are opposites. If 6,215 − 3,748 = 2,467, then 2,467 + 3,748 must equal 6,215. Always check your subtraction this way.

KEY IDEA

Subtract from right to left. If the top digit is too small, regroup 1 from the next place to make 10 more. Check with addition.

WATCH OUT

Do not subtract the smaller digit from the bigger one when the bigger one is on the bottom. 5 − 8 is not 3! Regroup instead.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 8,652 − 3,421
2. 7,314 − 2,856
3. 52,610 − 18,374

More practice: Workbook, Lesson 1.14.

LESSON SUMMARY

Subtract from right to left. Regroup when the top digit is too small. Check by adding the answer to the number you subtracted.

UNIT 1 · LESSON 1.15 · 4.NBT.4

Lesson 1.15: Subtracting Across Zeros

I can subtract from numbers that have zeros, such as 5,000 − 2,368.

WORDS TO KNOW

subtract across zeros Regrouping through one or more zeros to get enough to subtract.

LET'S EXPLORE

A store had 5,000 school folders. It sold 2,368. How many folders are left?

Zeros can make subtraction look tricky. When the place you need to regroup from is 0, keep moving left until you find a digit you can regroup.

WORKED EXAMPLE · REGROUP ACROSS ZEROS

Find 5,000 − 2,368.

```
  4 9 9 10
  5,000
- 2,368
-------
  2,632
```

1. Ones: 0 − 8. The tens and hundreds are 0, so go to the thousands.
2. Think of 5,000 as 500 tens. Regroup 1 ten: 500 tens becomes 499 tens and 10 ones.
3. Now the top number reads 4 thousands, 9 hundreds, 9 tens, 10 ones.
4. Subtract each place: 10 − 8 = 2, 9 − 6 = 3, 9 − 3 = 6, 4 − 2 = 2.

Answer: 2,632 folders

A shortcut: subtract 1 from both numbers

5,000 − 2,368 has the same answer as 4,999 − 2,367, because both numbers moved down by 1. With 4,999 on top, you never need to regroup!

WORKED EXAMPLE · USE THE SHORTCUT

Find 3,000 − 1,452.

1. Subtract 1 from both: 2,999 − 1,451.

2 No regrouping: 9 − 1 = 8, 9 − 5 = 4, 9 − 4 = 5, 2 − 1 = 1.

Answer: 1,548

KEY IDEA

When you meet zeros, regroup from the first nonzero digit to the left. Every zero you pass becomes a 9.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 600 − 274
2. 4,000 − 1,635
3. 20,050 − 8,317

More practice: Workbook, Lesson 1.15.

LESSON SUMMARY

Regroup from the first nonzero digit on the left; zeros you pass become 9s. Or subtract 1 from both numbers first.

UNIT 1 · LESSON 1.16 · 4.NBT.4, 4.OA.3

Lesson 1.16: Solving Multi-Step Word Problems

I can solve two-step word problems with addition and subtraction and check that my answer is reasonable.

WORDS TO KNOW

bar model	A drawing with bars that shows how the numbers in a problem are related.
multi-step problem	A problem that takes more than one calculation to solve.

LET'S EXPLORE

A food bank had 4,250 cans. On Monday it received 1,875 more cans. On Tuesday it gave away 2,690 cans. How many cans are left?

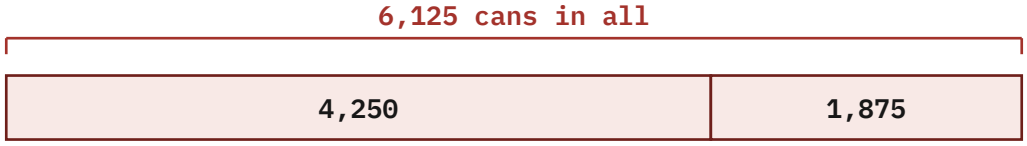
Some problems need more than one step. A plan helps.

Four-step plan

- Understand: What do you know? What do you need to find?
- Plan: Draw a bar model. Decide which operations to use and in what order.
- Solve: Do each step carefully.
- Check: Estimate. Is the answer reasonable?

WORKED EXAMPLE · TWO STEPS

Solve the food bank problem.

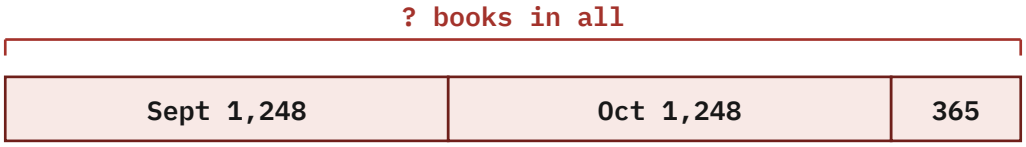


1. Step 1, add the new cans: 4,250 + 1,875 = 6,125.
2. Step 2, subtract the cans given away: 6,125 − 2,690 = 3,435.
3. Check: 4,000 + 2,000 − 3,000 = 3,000. 3,435 is close.

Answer: 3,435 cans

WORKED EXAMPLE · COMPARE, THEN COMBINE

A bookstore sold 1,248 books in September. It sold 365 more books in October than in September. How many books did it sell in the two months?



1 Find October: 1,248 + 365 = 1,613.

2 Combine both months: 1,248 + 1,613 = 2,861.

3 Check: 1,200 + 1,600 = 2,800. Reasonable.

Answer: 2,861 books

KEY IDEA

Read the whole problem first. Find the hidden question (the first step), solve it, then use that answer for the next step.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A library has 3,500 books. It buys 1,280 more and then donates 945. How many books does it have now?

2. Ella's family drove 1,320 miles. Jon's family drove 486 fewer miles. How many miles did both families drive in all?

3. A theater has 2,400 seats. 1,165 tickets were sold online and 738 at the door. How many seats are empty?

More practice: Workbook, Lesson 1.16.

LESSON SUMMARY

Use the four-step plan: Understand, Plan, Solve, Check. Solve the hidden first question, then the main question.

UNIT 1 REVIEW

Unit 1 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember...
1.1	Each place is worth 10 times the place to its right: 10 ones = 1 ten, 10 tens = 1 hundred, 10 hundreds = 1 thousand.
1.2	Periods are groups of three digits separated by commas. Read each period, then say its name.
1.3	To find a digit's value, multiply the digit by the value of its place. Zeros hold empty places.
1.4	Standard form uses digits. Word form uses words. Expanded form adds the value of each digit.
1.5	The same number can be named with different units: 3,400 = 3 thousands 4 hundreds = 34 hundreds = 340 tens.
1.6	Compare digits in the same place from left to right. The first place with different digits decides.
1.7	To order numbers, line them up by place value and compare from the greatest place. Check the direction: least to greatest or greatest to least.
1.8	On a number line, find the interval first. Numbers increase to the right, and the line shows which benchmark a number is closer to.
1.9	To round, find the two benchmarks, find the halfway point, and choose the closer benchmark. Halfway rounds up.
1.10	Underline the rounding place, look one place to the right, round up for 5 or more, and replace the digits after it with zeros.
1.11	Round each number, then add or subtract the rounded numbers. An estimate tells you about how much.
1.12	Use an estimate to check if an answer makes sense. Far from the estimate? Look for a mistake.
1.13	Line up places, add from right to left, and regroup when a place has 10 or more. Estimate to check.
1.14	Subtract from right to left. Regroup when the top digit is too small. Check by adding the answer to the number you subtracted.
1.15	Regroup from the first nonzero digit on the left; zeros you pass become 9s. Or subtract 1 from both numbers first.

Lesson	Remember...
1.16	Use the four-step plan: Understand, Plan, Solve, Check. Solve the hidden first question, then the main question.

UNIT 2

Multiplication

Comparing with multiplication, place-value patterns, area models, partial products, and the standard algorithm.

Multiplication is a fast way to combine equal groups, and it can also compare amounts. In this unit you will multiply numbers up to four digits by one-digit numbers and two-digit numbers by two-digit numbers, using models that show why the methods work.

In this unit you will…

- Use multiplication to compare amounts (“times as many”).
- Multiply by 10, 100, 1,000 and multiples of ten using place value.
- Multiply with area models, partial products, and the standard algorithm.
- Multiply two two-digit numbers.
- Solve multi-step word problems and check with estimates.

MATH IN REAL LIFE

Multiplication tells a farmer how many plants fit in a field, a coach how much jerseys will cost, and a family how many plates to buy for a party.

UNIT VOCABULARY

product	The answer to a multiplication problem.
factor	A number being multiplied.
area model	A rectangle split into parts to show multiplication.
partial products	Smaller products added to find the whole product.
estimate	A close answer found with rounded numbers.

Lessons in this unit

Lesson	Title
2.1	Multiplication as Comparison

Lesson	Title
2.2	Multiplying by 10, 100, and 1,000
2.3	Multiplying with Area Models
2.4	Partial Products
2.5	Multiplying Three-Digit Numbers
2.6	The Standard Algorithm for Multiplication
2.7	Multiplying Four-Digit Numbers
2.8	Estimating Products
2.9	Multiplying Multiples of Ten
2.10	Two-Digit by Two-Digit: Area Models
2.11	Two-Digit by Two-Digit: Partial Products
2.12	Choosing and Checking a Strategy
2.13	Multiplication Word Problems
2.14	Multi-Step Problems with Multiplication
2.15	Comparison Word Problems
2.16	Multiplication Problem Solving

UNIT 2 · LESSON 2.1 · 4.OA.1, 4.OA.2

Lesson 2.1: Multiplication as Comparison

I can read and write multiplication equations that compare two amounts, like "24 is 3 times as many as 8."

WORDS TO KNOW

product	The answer to a multiplication problem.
factor	A number being multiplied.
times as many	A comparison that uses multiplication: 3 times as many as 8 is 24.

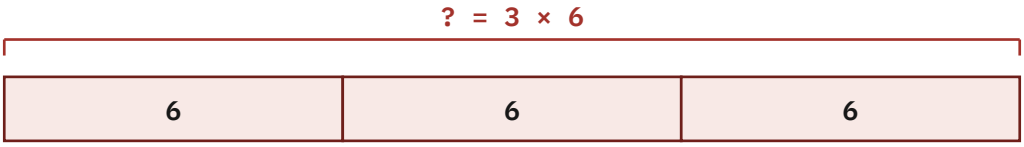
LET'S EXPLORE

Ellie has 6 stickers. Her big sister has 3 times as many stickers. How many stickers does her sister have?

You know multiplication as equal groups: 3 groups of 6. Multiplication can also compare two amounts.



Ellie: 6 stickers



Sister: 3 times as many

The sister's bar is 3 times as long as Ellie's bar. 3 × 6 = 18, so her sister has 18 stickers. We say: 18 is 3 times as many as 6.

WORKED EXAMPLE · WRITE A COMPARISON EQUATION

Write an equation for: 35 is 5 times as many as 7.

1 "Is" means equals.

2 "5 times as many as 7" means 5 × 7.

Answer: 35 = 5 × 7

WORKED EXAMPLE · FIND THE UNKNOWN AMOUNT

A tree is 4 times as tall as a bush. The bush is 2 feet tall. How tall is the tree?

1 The tree is the bigger amount. It is 4 times the bush.

2 $4 \times 2 = 8$.

Answer: 8 feet

WATCH OUT

"4 times as many" is not the same as "4 more." 4 more than 2 is 6. 4 times as many as 2 is 8.

KEY IDEA

In a comparison, one amount is a number of times as many as another. Use a bar model: the larger bar is made of copies of the smaller bar.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Write an equation: 42 is 6 times as many as 7.
2. Sam ran 3 miles. Ana ran 5 times as far. How far did Ana run?
3. Which is greater: 2 more than 9, or 2 times as many as 9?

More practice: Workbook, Lesson 2.1.

LESSON SUMMARY

Multiplication can compare: 24 is 3 times as many as 8 means $24 = 3 \times 8$. The bigger bar is copies of the smaller bar.

UNIT 2 · LESSON 2.2 · 4.NBT.1, 4.NBT.5

Lesson 2.2: Multiplying by 10, 100, and 1,000

I can use place value patterns to multiply by 10, 100, and 1,000 and by multiples of 10.

WORDS TO KNOW

multiple of 10 — A number you get by multiplying a whole number by 10, like 30, 70, or 120.

LET'S EXPLORE

A box holds 100 crayons. How many crayons are in 7 boxes? In 70 boxes?

In Unit 1 you learned that each place is worth 10 times the place to its right. That makes multiplying by 10, 100, and 1,000 fast.

Fact	× 10	× 100	× 1,000
7 × 1 = 7	7 × 10 = 70	7 × 100 = 700	7 × 1,000 = 7,000
4 × 6 = 24	4 × 60 = 240	4 × 600 = 2,400	4 × 6,000 = 24,000

When you multiply by 10, every digit moves one place to the left. Multiplying by 100 moves digits two places. Multiplying by 1,000 moves them three places.

WORKED EXAMPLE · USE A BASIC FACT

Find 5 × 800.

1. 800 is 8 hundreds.
2. 5 × 8 hundreds = 40 hundreds.
3. 40 hundreds = 4,000.

Answer: 4,000

WATCH OUT

Count the zeros carefully when the basic fact already ends in zero: 5 × 800 = 4,000 has three zeros, because 5 × 8 = 40.

KEY IDEA

To multiply by a multiple of 10, 100, or 1,000: multiply the basic fact, then think about place value (how many tens, hundreds, or thousands).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 9 × 100
2. 6 × 70
3. 8 × 5,000

More practice: Workbook, Lesson 2.2.

LESSON SUMMARY

Multiply the basic fact, then use place value: 6 × 300 = 6 × 3 hundreds = 18 hundreds = 1,800.

UNIT 2 · LESSON 2.3 · 4.NBT.5

Lesson 2.3: Multiplying with Area Models

I can use an area model to multiply a two-digit number by a one-digit number.

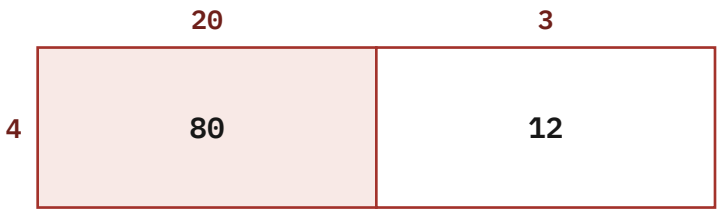
WORDS TO KNOW

area model	A rectangle split into parts to show multiplication.
distributive property	Breaking a factor into parts, multiplying each part, and adding: 4 × 23 = 4 × 20 + 4 × 3.

LET'S EXPLORE

A garden has 4 rows with 23 tomato plants in each row. How many plants are there?

An array of 4 rows of 23 would take a long time to draw. Instead, split 23 into 20 + 3 and draw a rectangle.

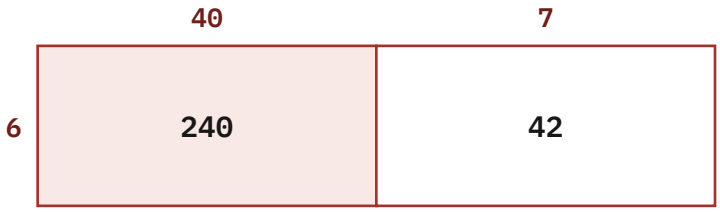


4 × 23 split into 4 × 20 and 4 × 3.

The big part is 4 × 20 = 80. The small part is 4 × 3 = 12. Add the parts: 80 + 12 = 92 plants. Breaking a number apart like this uses the distributive property.

WORKED EXAMPLE · USE AN AREA MODEL

Find 6 × 47.



1 Split 47 into 40 + 7.

2 6 × 40 = 240.

3 6 × 7 = 42.

4 Add: 240 + 42 = 282.

Answer: 282

KEY IDEA

Break the larger factor into tens and ones. Multiply each part. Add the parts.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3 × 25

2. 5 × 64

3. 8 × 39

More practice: Workbook, Lesson 2.3.

LESSON SUMMARY

An area model splits a factor into tens and ones. Multiply each part, then add: 6 × 47 = 240 + 42 = 282.

UNIT 2 · LESSON 2.4 · 4.NBT.5

Lesson 2.4: Partial Products

I can multiply using partial products and record my work in a vertical list.

WORDS TO KNOW

partial products	The smaller products you add together to get the whole product.

LET'S EXPLORE

You found 6 × 47 with an area model. Is there a way to write the same steps without drawing a rectangle?

Partial products is the area model written as a list. Each box of the area model becomes one line.

WORKED EXAMPLE · TWO-DIGIT BY ONE-DIGIT

Find 6 × 47 with partial products.

$$\begin{array}{rr} & 47 \\ \times & 6 \\ \hline 6 \times 7 & 42 \\ 6 \times 40 & 240 \\ \hline & 282 \end{array}$$

1. Multiply the ones: 6 × 7 = 42.
2. Multiply the tens: 6 × 40 = 240.
3. Add the partial products: 42 + 240 = 282.

Answer: 282

WORKED EXAMPLE · THREE-DIGIT BY ONE-DIGIT

Find 3 × 254 with partial products.

	254
	× 3
3 × 4	12
3 × 50	150
3 × 200	600
	762

1 3 × 4 = 12

2 3 × 50 = 150

3 3 × 200 = 600

4 Add: 12 + 150 + 600 = 762

Answer: 762

KEY IDEA

Each partial product matches one part of the area model. Write the factors you multiplied next to each line so you can check your work.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4 × 73

2. 7 × 126

3. 5 × 308

More practice: Workbook, Lesson 2.4.

LESSON SUMMARY

Partial products list each part: 6 × 47 = (6 × 7) + (6 × 40) = 42 + 240 = 282.

UNIT 2 · LESSON 2.5 · 4.NBT.5

Lesson 2.5: Multiplying Three-Digit Numbers

I can multiply a three-digit number by a one-digit number with an area model or partial products.

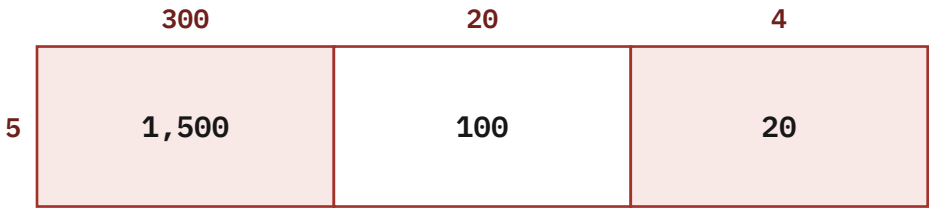
WORDS TO KNOW

hundreds part	In 324, the hundreds part is 300.

LET'S EXPLORE

A school bus can carry 48 students. A Mississippi school district has 215 buses. How could we find how many students they can carry?

Three-digit numbers split into three parts: hundreds, tens, and ones. The area model has three boxes.



$5 \times 324 = 1{,}500 + 100 + 20 = 1{,}620$

WORKED EXAMPLE · AREA MODEL AND PARTIAL PRODUCTS

Find 5×324.

1. $5 \times 300 = 1{,}500$
2. $5 \times 20 = 100$
3. $5 \times 4 = 20$
4. Add: $1{,}500 + 100 + 20 = 1{,}620$

Answer: 1,620

WORKED EXAMPLE · A NUMBER WITH A ZERO

Find 7×609.

1. $609 = 600 + 9$. There are no tens, so there are only two parts.
2. $7 \times 600 = 4{,}200$

3 $7 \times 9 = 63$

4 $4{,}200 + 63 = 4{,}263$

Answer: 4,263

KEY IDEA

A zero in a factor means that part of the product is zero. You can skip it, but keep the other places lined up.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3×412
2. 6×275
3. 4×708

More practice: Workbook, Lesson 2.5.

LESSON SUMMARY

Split a three-digit number into hundreds, tens, and ones; multiply each part and add.

UNIT 2 · LESSON 2.6 · 4.NBT.5

Lesson 2.6: The Standard Algorithm for Multiplication

I can multiply a multi-digit number by a one-digit number using the standard algorithm.

WORDS TO KNOW

standard algorithm (multiplication)	Multiplying each place from right to left and regrouping, writing the product on one line.

LET'S EXPLORE

Partial products work every time, but they take a lot of lines. Can we write 6 × 47 on just one line?

The standard algorithm does the same multiplying as partial products, but it regroups as it goes so the answer fits on one line.

WORKED EXAMPLE · STANDARD ALGORITHM

Find 6 × 47.

$$\begin{array}{r} 47 \\ \times\ \ 6 \\ \hline 282 \end{array}$$

1. Ones: 6 × 7 = 42. Write 2 in the ones place. Regroup 4 tens: write a small 4 above the tens.
2. Tens: 6 × 4 tens = 24 tens. Add the regrouped 4 tens: 28 tens.
3. Write 28 in front of the 2.

Answer: 282

WORKED EXAMPLE · THREE DIGITS

Find 4 × 386.

$$\begin{array}{r} 386 \\ \times \quad 4 \\ \hline 1,544 \end{array}$$

1 Ones: 4 × 6 = 24. Write 4, regroup 2 tens.

2 Tens: 4 × 8 = 32, plus 2 = 34 tens. Write 4, regroup 3 hundreds.

3 Hundreds: 4 × 3 = 12, plus 3 = 15 hundreds. Write 15.

Answer: 1,544

WATCH OUT

Multiply first, then add the regrouped number. Do not add the regrouped digit before multiplying.

KEY IDEA

Standard algorithm: multiply the ones, regroup; multiply the tens, add the regrouped tens, regroup; keep going left.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 5 × 68
2. 3 × 457
3. 8 × 209

More practice: Workbook, Lesson 2.6.

LESSON SUMMARY

Multiply each place from right to left. Add any regrouped number after multiplying.

UNIT 2 · LESSON 2.7 · 4.NBT.5

Lesson 2.7: Multiplying Four-Digit Numbers

I can multiply a four-digit number by a one-digit number.

LET'S EXPLORE

A farm ships 1,250 crates of sweet potatoes each week. How many crates does it ship in 6 weeks?

The same steps work for four-digit numbers. There is one more place, so one more step.

WORKED EXAMPLE · FOUR DIGITS

Find 6 × 1,250.

$$\begin{array}{r} 1{,}250 \\ \times \quad 6 \\ \hline 7{,}500 \end{array}$$

1. Ones: 6 × 0 = 0.
2. Tens: 6 × 5 = 30 tens. Write 0, regroup 3 hundreds.
3. Hundreds: 6 × 2 = 12, plus 3 = 15 hundreds. Write 5, regroup 1 thousand.
4. Thousands: 6 × 1 = 6, plus 1 = 7 thousands.

Answer: 7,500 crates

WORKED EXAMPLE · CHECK WITH AN ESTIMATE

Find 4 × 2,397 and check.

1. Standard algorithm gives 9,588.
2. Estimate: 4 × 2,000 = 8,000 and 4 × 2,400 = 9,600. 9,588 is close.

Answer: 9,588

KEY IDEA

Estimate before you multiply. If your answer is far from the estimate, check each step.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3 × 2,143
2. 5 × 1,608
3. 7 × 3,425

More practice: Workbook, Lesson 2.7.

LESSON SUMMARY

Four-digit numbers use the same steps; estimate first to check your answer.

UNIT 2 · LESSON 2.8 · 4.NBT.3, 4.OA.3

Lesson 2.8: Estimating Products

I can round to estimate products and decide if an answer is reasonable.

WORDS TO KNOW

estimate (product) Round a factor, then multiply to get a close answer.

LET'S EXPLORE

A ticket costs $38. About how much will 7 tickets cost? Is $200 enough?

To estimate a product, round the larger factor to its greatest place, then multiply in your head.

WORKED EXAMPLE · ESTIMATE

Estimate 7 × 38.

1 Round 38 to 40.

2 7 × 40 = 280.

Answer: about $280, so $200 is not enough

WORKED EXAMPLE · OVER OR UNDER?

Estimate 4 × 2,684. Is the exact answer more or less than the estimate?

1 Round 2,684 to 3,000.

2 4 × 3,000 = 12,000.

3 We rounded up, so the exact answer is less than 12,000. (Exact: 10,736.)

Answer: about 12,000

KEY IDEA

Rounding up gives an overestimate. Rounding down gives an underestimate.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Estimate 6 × 52
2. Estimate 3 × 789
3. Is 5 × 412 = 20,060 reasonable?

More practice: Workbook, Lesson 2.8.

LESSON SUMMARY

Round the larger factor and use a basic fact. Rounding up overestimates; rounding down underestimates.

UNIT 2 · LESSON 2.9 · 4.NBT.5

Lesson 2.9: Multiplying Multiples of Ten

I can multiply two multiples of ten, like 30 × 40, using basic facts and place value.

LET'S EXPLORE

A classroom has 30 rows of seats in an auditorium with 40 seats in each row. How many seats is that?

30 × 40 means 3 tens times 4 tens. 3 × 4 = 12, and tens times tens makes hundreds. So 30 × 40 = 12 hundreds = 1,200.

Problem	Basic fact	Think	Product
30 × 40	3 × 4 = 12	tens × tens = hundreds	1,200
20 × 60	2 × 6 = 12	12 hundreds	1,200
50 × 80	5 × 8 = 40	40 hundreds	4,000

WORKED EXAMPLE · TENS TIMES TENS

Find 70 × 90.

1 7 × 9 = 63.

2 tens × tens = hundreds: 63 hundreds.

Answer: 6,300

WORKED EXAMPLE · TENS TIMES ONES

Find 40 × 6.

1 4 × 6 = 24.

2 tens × ones = tens: 24 tens.

Answer: 240

KEY IDEA

Multiply the basic fact. Then count: tens × tens = hundreds. A quick check: the product has as many zeros as the two factors together, plus any zero from the fact.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 20 × 30

2. 60 × 50

3. 80 × 70

More practice: Workbook, Lesson 2.9.

LESSON SUMMARY

Tens × tens = hundreds: 30 × 40 = 12 hundreds = 1,200.

UNIT 2 · LESSON 2.10 · 4.NBT.5

Lesson 2.10: Two-Digit by Two-Digit: Area Models

I can use an area model to multiply two two-digit numbers.

LET'S EXPLORE

An orchard has 23 rows of pecan trees with 14 trees in each row. How many trees are there?

When both factors have two digits, split both into tens and ones. The area model has four boxes.

	20	3
10	200	30
4	80	12

23 × 14: split 23 into 20 + 3 and 14 into 10 + 4.

WORKED EXAMPLE · FOUR BOXES

Find 23 × 14.

1. 10 × 20 = 200
2. 10 × 3 = 30
3. 4 × 20 = 80
4. 4 × 3 = 12
5. Add all four: 200 + 30 + 80 + 12 = 322

Answer: 322 trees

WORKED EXAMPLE · ANOTHER ONE

Find 36 × 25.

	30	6
20	600	120
5	150	30

1 20 × 30 = 600 and 20 × 6 = 120

2 5 × 30 = 150 and 5 × 6 = 30

3 600 + 120 + 150 + 30 = 900

Answer: 900

KEY IDEA

Two-digit × two-digit makes four partial products. Add all four.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 12 × 13

2. 21 × 34

3. 45 × 16

More practice: Workbook, Lesson 2.10.

LESSON SUMMARY

Split both factors into tens and ones. Four boxes, four products, add them.

UNIT 2 · LESSON 2.11 · 4.NBT.5

Lesson 2.11: Two-Digit by Two-Digit: Partial Products

I can multiply two two-digit numbers using partial products.

LET'S EXPLORE

Can the four boxes of the area model be written as a list, like we did with one-digit numbers?

Yes! Each box becomes one partial product. Label each line so you know which digits you multiplied.

WORKED EXAMPLE · PARTIAL PRODUCTS

Find 34 × 26.

	34
	× 26
6 × 4	24
6 × 30	180
20 × 4	80
20 × 30	600
	884

1. 6 × 4 = 24
2. 6 × 30 = 180
3. 20 × 4 = 80
4. 20 × 30 = 600
5. Add: 24 + 180 + 80 + 600 = 884

Answer: 884

KEY IDEA

Multiply each part of one factor by each part of the other: ones × ones, ones × tens, tens × ones, tens × tens. Then add.

WATCH OUT

Line up the partial products by place value before adding.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 24 × 13
2. 52 × 41
3. 67 × 28

More practice: Workbook, Lesson 2.11.

LESSON SUMMARY

Two-digit × two-digit gives four partial products: ones×ones, ones×tens, tens×ones, tens×tens.

UNIT 2 · LESSON 2.12 · 4.NBT.5, 4.OA.3

Lesson 2.12: Choosing and Checking a Strategy

I can choose a multiplication strategy and check my answer with an estimate.

WORDS TO KNOW

strategy	A plan or method for solving a problem.

LET'S EXPLORE

Three students solved 25 × 16 in different ways and all got 400. How could that be?

There is more than one good way to multiply. Choose the one that fits the numbers.

- Area model: great for seeing the parts.
- Partial products: organized and easy to check.
- Mental math with friendly numbers: 25 × 16 = 25 × 4 × 4 = 100 × 4 = 400.
- Standard algorithm: quick for one-digit factors.

WORKED EXAMPLE · FRIENDLY NUMBERS

Find 25 × 12.

1 25 × 4 = 100, and 12 = 4 × 3.

2 So 25 × 12 = 100 × 3.

Answer: 300

WORKED EXAMPLE · ESTIMATE TO CHECK

Is 38 × 21 = 798 reasonable?

1 Estimate: 40 × 20 = 800.

2 798 is close to 800.

Answer: Yes

KEY IDEA

Pick a strategy that fits the numbers, then estimate to check.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 50 × 18

2. 19 × 30 (hint: 20 × 30 − 30)

3. Is 42 × 31 = 1,302 reasonable?

More practice: Workbook, Lesson 2.12.

LESSON SUMMARY

Choose a strategy that fits the numbers. Always estimate to check.

UNIT 2 · LESSON 2.13 · 4.OA.3, 4.NBT.5

Lesson 2.13: Multiplication Word Problems

I can solve word problems that use multiplication, including equal groups, arrays, and area.

WORDS TO KNOW

equal groups	Groups that all have the same number of items.

LET'S EXPLORE

A pack of trading cards has 12 cards. How many cards are in 15 packs?

Look for equal groups, rows and columns, or length and width. These are clues to multiply.

WORKED EXAMPLE · EQUAL GROUPS

15 packs with 12 cards each. How many cards?

1 Equal groups: 15 groups of 12.

2 15 × 12 = 180.

Answer: 180 cards

WORKED EXAMPLE · AREA

A rug is 9 feet long and 12 feet wide. What is its area?

1 Area of a rectangle = length × width.

2 9 × 12 = 108.

Answer: 108 square feet

KEY IDEA

Ask: Are there equal groups? Rows and columns? Length and width? If yes, multiply.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A team bought 14 jerseys at $23 each. What was the cost?

2. A parking lot has 18 rows of 25 spaces. How many spaces?

3. A garden is 16 feet by 7 feet. What is its area?

More practice: Workbook, Lesson 2.13.

LESSON SUMMARY

Equal groups, rows and columns, and length × width are all multiplication situations.

UNIT 2 · LESSON 2.14 · 4.OA.3

Lesson 2.14: Multi-Step Problems with Multiplication

I can solve multi-step word problems that use multiplication with addition or subtraction.

LET'S EXPLORE

Mom buys 3 shirts for $14 each and a pair of shoes for $39. How much does she spend?

Many real problems mix operations. Find the hidden first step, solve it, then finish.

WORKED EXAMPLE · MULTIPLY, THEN ADD

3 shirts at $14 and shoes for $39. Total?

1 Shirts: 3 × 14 = 42.

2 Add the shoes: 42 + 39 = 81.

Answer: $81

WORKED EXAMPLE · MULTIPLY, THEN SUBTRACT

A store had 500 bottles of water. It sold 18 cases of 24 bottles. How many bottles are left?

1 Sold: 18 × 24 = 432.

2 Left: 500 − 432 = 68.

Answer: 68 bottles

KEY IDEA

Write an equation for each step. Use a letter for the unknown: 3 × 14 + 39 = t.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4 tickets at $12 and a $9 parking fee. Total?

2. A farmer had 1,000 plants. He planted 25 rows of 36. How many plants are left?

3. Ana earns $15 a week for 12 weeks and spends $95. How much is left?

More practice: Workbook, Lesson 2.14.

LESSON SUMMARY

Find the hidden first step. Write an equation for each step and check with an estimate.

UNIT 2 · LESSON 2.15 · 4.OA.2

Lesson 2.15: Comparison Word Problems

I can tell the difference between "times as many" and "more than" problems and solve both.

WORDS TO KNOW

multiplicative comparison	A comparison using "times as many."
additive comparison	A comparison using "more than" or "fewer than."

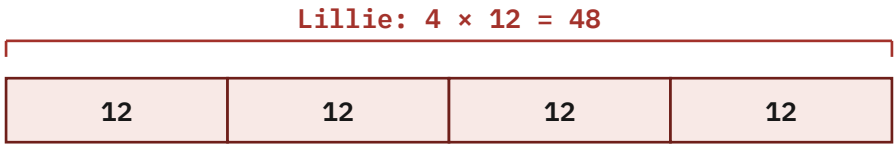
LET'S EXPLORE

Lillie has 4 times as many shells as Ellie. Ellie has 12 shells. How many shells do they have together?

Some comparison problems use multiplication (times as many) and some use addition or subtraction (more than, fewer than). Read carefully!

WORKED EXAMPLE · TIMES AS MANY, THEN ADD

Ellie has 12 shells. Lillie has 4 times as many. How many in all?



1 Lillie: 4 × 12 = 48.

2 Together: 12 + 48 = 60.

Answer: 60 shells

WORKED EXAMPLE · MORE THAN

A blue whale is 98 feet long. A school bus is 58 feet shorter. How long is the bus?

1 "Shorter" means subtract: 98 − 58 = 40.

Answer: 40 feet

WATCH OUT

The word “times” means multiply. The words “more” and “fewer” by themselves mean add or subtract.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A pencil costs 30¢. A pen costs 3 times as much. How much does the pen cost?
2. Tom has 25 marbles. Kim has 8 more than Tom. How many does Kim have?
3. A giraffe is 18 feet tall. That is 3 times as tall as a man. How tall is the man?

More practice: Workbook, Lesson 2.15.

LESSON SUMMARY

“Times as many” means multiply. “More than” and “fewer than” mean add or subtract.

UNIT 2 · LESSON 2.16 · 4.NBT.5, 4.OA.3

Lesson 2.16: Multiplication Problem Solving

I can solve a variety of multiplication problems and explain my thinking.

LET'S EXPLORE

You are planning a family reunion for 64 people. Each table seats 8. Each person needs 2 cups. Cups come in packs of 20. What do you need to buy?

Today you will use everything from this unit: comparisons, place value patterns, area models, partial products, the standard algorithm, and estimation.

WORKED EXAMPLE · PLAN A PARTY

How many cups are needed for 64 people if each person gets 2 cups?

1 64 × 2 = 128 cups.

Answer: 128 cups

WORKED EXAMPLE · PACKS

Cups come in packs of 20. Is 6 packs enough?

1 6 × 20 = 120.

2 120 is less than 128.

Answer: No. 7 packs (140 cups) are needed.

KEY IDEA

Read carefully, choose a strategy, show your work, and check with an estimate.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many tables for 64 people at 8 per table?
2. Plates come in packs of 25. How many plates in 3 packs?
3. Each person eats 3 rolls. How many rolls for 64 people?

More practice: Workbook, Lesson 2.16.

LESSON SUMMARY

Use the strategy that fits: patterns for multiples of 10, area models or partial products for big numbers, and estimation to check.

UNIT 2 REVIEW

Unit 2 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
2.1	Multiplication can compare: 24 is 3 times as many as 8 means 24 = 3 × 8. The bigger bar is copies of the smaller bar.
2.2	Multiply the basic fact, then use place value: 6 × 300 = 6 × 3 hundreds = 18 hundreds = 1,800.
2.3	An area model splits a factor into tens and ones. Multiply each part, then add: 6 × 47 = 240 + 42 = 282.
2.4	Partial products list each part: 6 × 47 = (6 × 7) + (6 × 40) = 42 + 240 = 282.
2.5	Split a three-digit number into hundreds, tens, and ones; multiply each part and add.
2.6	Multiply each place from right to left. Add any regrouped number after multiplying.
2.7	Four-digit numbers use the same steps; estimate first to check your answer.
2.8	Round the larger factor and use a basic fact. Rounding up overestimates; rounding down underestimates.
2.9	Tens × tens = hundreds: 30 × 40 = 12 hundreds = 1,200.
2.10	Split both factors into tens and ones. Four boxes, four products, add them.
2.11	Two-digit × two-digit gives four partial products: ones×ones, ones×tens, tens×ones, tens×tens.
2.12	Choose a strategy that fits the numbers. Always estimate to check.
2.13	Equal groups, rows and columns, and length × width are all multiplication situations.
2.14	Find the hidden first step. Write an equation for each step and check with an estimate.
2.15	“Times as many” means multiply. “More than” and “fewer than” mean add or subtract.
2.16	Use the strategy that fits: patterns for multiples of 10, area models or partial products for big numbers, and estimation to check.

UNIT 3

Division

Sharing and grouping, dividing with place value, partial quotients, long division, and remainders.

Division splits a total into equal groups. In this unit you will use multiplication facts and place value to divide numbers up to four digits by one-digit numbers, and decide what to do with leftovers.

In this unit you will…

- Explain division as sharing and as grouping.
- Divide multiples of 10, 100, and 1,000.
- Divide with area models, partial quotients, and long division.
- Find and check remainders.
- Decide what a remainder means in a word problem.

MATH IN REAL LIFE

Division helps split a pizza fairly, plan how many cars a trip needs, and find how far to drive each day of a road trip.

UNIT VOCABULARY

dividend	The number being divided.
divisor	The number you divide by.
quotient	The answer to a division problem.
remainder	The amount left over.

Lessons in this unit

Lesson	Title
3.1	What Division Means
3.2	Dividing Multiples of 10, 100, and 1,000

Lesson	Title
3.3	Dividing with Area Models
3.4	Partial Quotients
3.5	Understanding Remainders
3.6	Dividing Larger Numbers
3.7	What Do I Do With the Remainder?
3.8	Division Problem Solving

UNIT 3 · LESSON 3.1 · 4.OA.2, 4.NBT.6

Lesson 3.1: What Division Means

I can explain division as sharing equally or making equal groups, and use multiplication facts to divide.

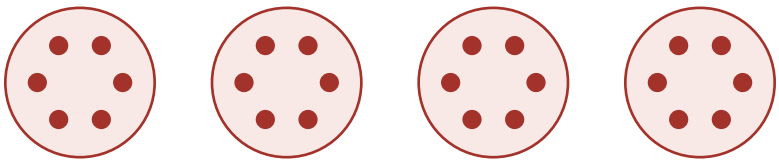
WORDS TO KNOW

dividend	The number being divided.
divisor	The number you divide by.
quotient	The answer to a division problem.
fact family	Related multiplication and division facts, like 6 × 4 = 24, 4 × 6 = 24, 24 ÷ 4 = 6, 24 ÷ 6 = 4.

LET'S EXPLORE

Grandma baked 24 cookies for 4 grandchildren. How many cookies does each child get if she shares them fairly?

Division answers two kinds of questions. Sharing: 24 cookies shared by 4 children, how many each? Grouping: 24 cookies put in bags of 4, how many bags? Both are 24 ÷ 4 = 6.



24 shared into 4 equal groups: 6 in each group.

Division is the opposite of multiplication. If you know 4 × 6 = 24, you know 24 ÷ 4 = 6 and 24 ÷ 6 = 4.

Dividend	÷	Divisor	=	Quotient
24	÷	4	=	6

WORKED EXAMPLE · USE A FACT FAMILY

Find 56 ÷ 7.

1 Think: 7 × ? = 56.

2 7 × 8 = 56.

Answer: 8

KEY IDEA

To divide, think of the related multiplication fact: 56 ÷ 7 = ? is the same as 7 × ? = 56.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 36 ÷ 9
2. Write the fact family for 5, 8, and 40.
3. 42 apples go in bags of 6. How many bags?

More practice: Workbook, Lesson 3.1.

LESSON SUMMARY

Division shares or groups equally. Use the related multiplication fact to divide.

UNIT 3 · LESSON 3.2 · 4.NBT.6

Lesson 3.2: Dividing Multiples of 10, 100, and 1,000

I can use basic facts and place value to divide numbers like 2,400 ÷ 6.

LET'S EXPLORE

A school has 2,400 pencils to share equally among 6 classrooms. How many pencils does each room get?

Use a basic fact and think about the unit. 2,400 is 24 hundreds. 24 hundreds ÷ 6 = 4 hundreds = 400.

Basic fact	Tens	Hundreds	Thousands
24 ÷ 6 = 4	240 ÷ 6 = 40	2,400 ÷ 6 = 400	24,000 ÷ 6 = 4,000

WORKED EXAMPLE · THINK IN UNITS

Find 3,500 ÷ 7.

1 3,500 = 35 hundreds.

2 35 ÷ 7 = 5, so 35 hundreds ÷ 7 = 5 hundreds.

Answer: 500

WATCH OUT

Check by multiplying: 7 × 500 = 3,500. If you get a different number, recount the zeros.

KEY IDEA

Name the dividend in tens, hundreds, or thousands, use the basic fact, then rename the quotient.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 320 ÷ 4
2. 4,500 ÷ 9
3. 6,000 ÷ 3

More practice: Workbook, Lesson 3.2.

LESSON SUMMARY

Use the basic fact and the unit: 2,400 ÷ 6 = 24 hundreds ÷ 6 = 4 hundreds = 400.

UNIT 3 · LESSON 3.3 · 4.NBT.6

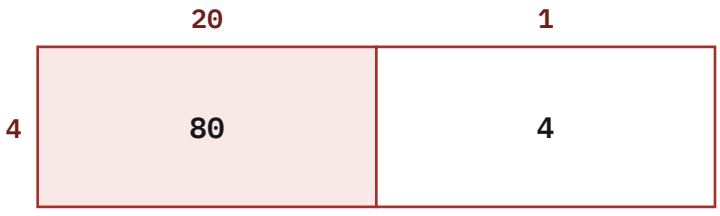
Lesson 3.3: Dividing with Area Models

I can use an area model to divide a two- or three-digit number by a one-digit number.

LET'S EXPLORE

A rectangle has an area of 84 square feet. One side is 4 feet long. How long is the other side?

Division can be pictured as a rectangle where you know the area (the dividend) and one side (the divisor). The missing side is the quotient.



84 ÷ 4: 80 ÷ 4 = 20 and 4 ÷ 4 = 1, so the side is 20 + 1 = 21.

WORKED EXAMPLE · SPLIT THE DIVIDEND INTO FRIENDLY PARTS

Find 84 ÷ 4.

1 Split 84 into 80 + 4, numbers that are easy to divide by 4.

2 80 ÷ 4 = 20.

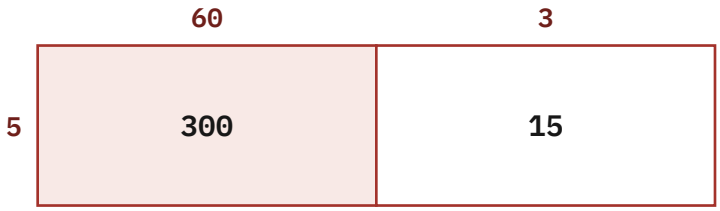
3 4 ÷ 4 = 1.

4 Add: 20 + 1 = 21.

Answer: 21

WORKED EXAMPLE · THREE DIGITS

Find 315 ÷ 5.



1 Split 315 into 300 + 15.

2 300 ÷ 5 = 60.

3 15 ÷ 5 = 3.

4 60 + 3 = 63.

Answer: 63

KEY IDEA

Break the dividend into parts that are easy to divide (multiples of the divisor). Divide each part, then add the answers.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 96 ÷ 3

2. 78 ÷ 6

3. 426 ÷ 6

More practice: Workbook, Lesson 3.3.

LESSON SUMMARY

Split the dividend into friendly parts, divide each part, and add the quotients.

UNIT 3 · LESSON 3.4 · 4.NBT.6

Lesson 3.4: Partial Quotients

I can divide using partial quotients and check my answer with multiplication.

WORDS TO KNOW

partial quotients	A way to divide by subtracting easy multiples of the divisor and adding up how many groups you took away.

LET'S EXPLORE

There are 156 students going on a field trip. Each van holds 6 students. How many vans are needed?

With partial quotients, you take away big, easy groups first, then smaller groups, until nothing (or a little) is left.

WORKED EXAMPLE · PARTIAL QUOTIENTS

Find 156 ÷ 6.

1. Take away 10 groups of 6: 156 − 60 = 96. (10)
2. Take away 10 more groups: 96 − 60 = 36. (10)
3. Take away 6 groups: 36 − 36 = 0. (6)
4. Add the groups: 10 + 10 + 6 = 26.
5. Check: 26 × 6 = 156. ✓

Answer: 26 vans

Left to divide	Subtract	Groups of 6
156	− 60 (10 × 6)	10
96	− 60 (10 × 6)	10
36	− 36 (6 × 6)	6
0		Total: 26

KEY IDEA

Any easy multiple works: 10 groups, 20 groups, 5 groups. Different students may take different steps and still get the same quotient.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 144 ÷ 4
2. 215 ÷ 5
3. 738 ÷ 9

More practice: Workbook, Lesson 3.4.

LESSON SUMMARY

Subtract easy multiples of the divisor, keep track of the groups, and add them up. Check by multiplying.

UNIT 3 · LESSON 3.5 · 4.NBT.6

Lesson 3.5: Understanding Remainders

I can find a quotient with a remainder and explain what the remainder means.

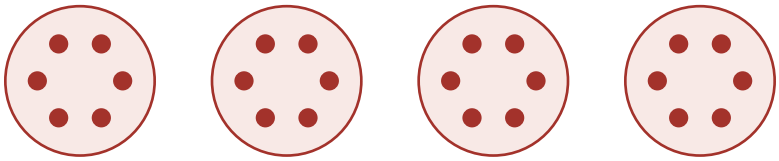
WORDS TO KNOW

remainder	The amount left over when a number cannot be divided evenly.

LET'S EXPLORE

Ellie has 26 stickers to share equally among 4 friends. How many does each friend get? Are any left?

Sometimes the groups come out even and sometimes there are some left over. The leftover amount is the remainder.



26 ÷ 4: each friend gets 6 (24 stickers), and 2 are left.

WORKED EXAMPLE · WRITE A REMAINDER

Find 26 ÷ 4.

1. 4 × 6 = 24, the closest product that is not too big.
2. 26 − 24 = 2 left over.
3. Write the answer as 6 R2.

Answer: 6 R2

WATCH OUT

The remainder must always be less than the divisor. If you have 5 left when dividing by 4, you can make another group!

WORKED EXAMPLE · CHECK A REMAINDER

Check 47 ÷ 5 = 9 R2.

1 Multiply: 9 × 5 = 45.

2 Add the remainder: 45 + 2 = 47. ✓

Answer: Correct

KEY IDEA

Quotient × divisor + remainder = dividend.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 17 ÷ 3

2. 38 ÷ 6

3. 50 ÷ 7

More practice: Workbook, Lesson 3.5.

LESSON SUMMARY

The remainder is what is left over. It must be less than the divisor. Check: quotient × divisor + remainder = dividend.

UNIT 3 · LESSON 3.6 · 4.NBT.6

Lesson 3.6: Dividing Larger Numbers

I can divide up to four-digit numbers by one-digit numbers using place value, step by step.

WORDS TO KNOW

long division	Dividing one place at a time from left to right: divide, multiply, subtract, bring down.

LET'S EXPLORE

A charity has $1,236 to split equally among 4 families. How much does each family get?

Long division divides one place at a time, starting at the greatest place. Repeat four steps: divide, multiply, subtract, bring down.

WORKED EXAMPLE · LONG DIVISION

Find 1,236 ÷ 4.

$$4\overline{)1236} \quad \text{quotient: } 309$$

1. 1 thousand ÷ 4: not enough. Use 12 hundreds instead. 12 ÷ 4 = 3. Write 3 over the hundreds. 3 × 4 = 12, 12 − 12 = 0.
2. Bring down the 3 tens. 3 ÷ 4 = 0. Write 0 over the tens. 0 × 4 = 0, 3 − 0 = 3.
3. Bring down the 6 ones: 36. 36 ÷ 4 = 9. Write 9 over the ones. 9 × 4 = 36, 36 − 36 = 0.
4. Check: 309 × 4 = 1,236. ✓

Answer: $309

WATCH OUT

When a place cannot be divided (like 3 ÷ 4), write a 0 in the quotient. Skipping it would give 39, which is far too small.

KEY IDEA

Divide, multiply, subtract, bring down. Repeat until every digit has been brought down.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 846 ÷ 2
2. 1,575 ÷ 5
3. 2,418 ÷ 6

More practice: Workbook, Lesson 3.6.

LESSON SUMMARY

Long division: divide, multiply, subtract, bring down. Write 0 when a place can't be divided.

UNIT 3 · LESSON 3.7 · 4.OA.3

Lesson 3.7: What Do I Do With the Remainder?

I can decide what a remainder means in a word problem.

WORDS TO KNOW

interpret	To explain what something means.

LET'S EXPLORE

27 people are going to the zoo. Each car holds 4 people. How many cars do they need?

27 ÷ 4 = 6 R3. But the answer is not "6 R3 cars"! The remainder means something different in different stories.

What to do with the remainder	Example (27 ÷ 4 = 6 R3)	Answer
Round up: everyone needs a spot	Cars for 27 people, 4 per car	7 cars
Drop it: only full groups count	Bags of 4 apples from 27 apples	6 full bags
The remainder is the answer	Apples left after making bags of 4	3 apples

WORKED EXAMPLE · ROUND UP

A teacher has 75 students. Each table seats 8. How many tables?

1 75 ÷ 8 = 9 R3.

2 3 students still need seats, so add a table.

Answer: 10 tables

KEY IDEA

Always ask: what does the remainder mean in this story?

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 50 photos go in an album, 6 per page. How many pages are needed?
2. 50 crayons go in boxes of 6. How many full boxes?
3. 50 crayons in boxes of 6. How many crayons are left over?

More practice: Workbook, Lesson 3.7.

LESSON SUMMARY

Round up, drop it, or use it as the answer: the story tells you what a remainder means.

UNIT 3 · LESSON 3.8 · 4.OA.3, 4.NBT.6

Lesson 3.8: Division Problem Solving

I can solve word problems with division and check my answers with multiplication.

LET'S EXPLORE

A family drives 1,248 miles in 4 days, the same distance each day. How far do they drive each day?

Look for clues for division: shared equally, each, per, how many groups.

WORKED EXAMPLE · EQUAL SHARING

1,248 miles in 4 equal days.

1 1,248 ÷ 4 = 312.

2 Check: 312 × 4 = 1,248.

Answer: 312 miles each day

WORKED EXAMPLE · TWO STEPS

A class earns $96 at a bake sale and $120 at a car wash. They split the money equally among 6 field-trip groups. How much per group?

1 Add: 96 + 120 = 216.

2 Divide: 216 ÷ 6 = 36.

Answer: $36

KEY IDEA

Multiply to check every division answer.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A rope 96 feet long is cut into 8 equal pieces. How long is each piece?
2. There are 365 days in a year. About how many weeks is that? (7 days in a week)
3. Mom buys 3 packs of 12 water bottles and splits them among 4 coolers. How many per cooler?

More practice: Workbook, Lesson 3.8.

LESSON SUMMARY

Look for equal sharing or equal groups. Divide, interpret any remainder, and check by multiplying.

UNIT 3 REVIEW

Unit 3 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
3.1	Division shares or groups equally. Use the related multiplication fact to divide.
3.2	Use the basic fact and the unit: 2,400 ÷ 6 = 24 hundreds ÷ 6 = 4 hundreds = 400.
3.3	Split the dividend into friendly parts, divide each part, and add the quotients.
3.4	Subtract easy multiples of the divisor, keep track of the groups, and add them up. Check by multiplying.
3.5	The remainder is what is left over. It must be less than the divisor. Check: quotient × divisor + remainder = dividend.
3.6	Long division: divide, multiply, subtract, bring down. Write 0 when a place can't be divided.
3.7	Round up, drop it, or use it as the answer: the story tells you what a remainder means.
3.8	Look for equal sharing or equal groups. Divide, interpret any remainder, and check by multiplying.

UNIT 4

Factors, Multiples, and Patterns

Factor pairs, prime and composite numbers, multiples, divisibility, and patterns.

Factors and multiples show how numbers are built from other numbers. Patterns show how numbers grow and change by rules. These ideas prepare you for fractions in the next units.

In this unit you will...

- Find all factor pairs of a number to 100.
- Tell whether a number is prime or composite.
- Find multiples and common multiples.
- Use divisibility rules.
- Extend number and shape patterns and find rules.

MATH IN REAL LIFE

Factors help plan equal teams and rows; multiples help plan schedules that repeat, like buses, medicine doses, and practices.

UNIT VOCABULARY

factor	A number that divides another with no remainder.
multiple	The product of a number and a whole number.
prime number	A number with exactly two factors.
composite number	A number with more than two factors.
pattern	A predictable arrangement of numbers or shapes.

Lessons in this unit

Lesson	Title
4.1	Factors and Factor Pairs
4.2	Prime and Composite Numbers

Lesson	Title
4.3	Multiples
4.4	Divisibility Rules
4.5	Number Patterns
4.6	Shape Patterns
4.7	Input-Output Tables
4.8	Factors, Multiples, and Patterns in Problems

UNIT 4 · LESSON 4.1 · 4.OA.4

Lesson 4.1: Factors and Factor Pairs

I can find all the factor pairs of a whole number from 1 to 100.

WORDS TO KNOW

factor pair	Two numbers that multiply to make a product, like 3 and 8 for 24.
factor	A number that divides another number with no remainder.

LET'S EXPLORE

Ellie wants to arrange 12 chairs in equal rows with no chairs left over. How many different ways can she do it?

Every rectangle you can make with 12 squares shows a factor pair of 12.

Rows	In each row	Factor pair
1	12	1 × 12
2	6	2 × 6
3	4	3 × 4



3 rows of 4 chairs

The factors of 12 are 1, 2, 3, 4, 6, and 12. (4 rows of 3 is the same pair as 3 rows of 4, just turned.)

WORKED EXAMPLE · FIND ALL FACTOR PAIRS

Find all the factor pairs of 36.

1. Start with 1: 1 × 36.
2. Try 2: 2 × 18. Try 3: 3 × 12. Try 4: 4 × 9. Try 5: no. Try 6: 6 × 6.
3. When the factors meet or start repeating, you are done.

Answer: 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6

KEY IDEA

Test 1, 2, 3, … in order. Stop when the pairs meet in the middle. List the factors from least to greatest.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. List the factor pairs of 18.
2. List all the factors of 20.
3. Is 7 a factor of 42?

More practice: Workbook, Lesson 4.1.

LESSON SUMMARY

Factor pairs multiply to make the number. Test numbers in order and stop when the pairs meet.

UNIT 4 · LESSON 4.2 · 4.OA.4

Lesson 4.2: Prime and Composite Numbers

I can tell whether a number is prime or composite.

WORDS TO KNOW

prime number	A whole number greater than 1 with exactly two factors: 1 and itself.
composite number	A whole number greater than 1 with more than two factors.

LET'S EXPLORE

Try to make a rectangle with 7 tiles that is more than one row. Can you?

Some numbers can only make one rectangle: a single row. Those numbers are prime. Numbers that can make more than one rectangle are composite.

Number	Factors	Prime or composite?
7	1, 7	prime
9	1, 3, 9	composite
13	1, 13	prime
15	1, 3, 5, 15	composite

WORKED EXAMPLE · PRIME OR COMPOSITE?

Is 21 prime or composite?

1 $21 = 3 \times 7$.

2 21 has more than two factors: 1, 3, 7, 21.

Answer: composite

WATCH OUT

1 is neither prime nor composite. It has only one factor. 2 is the only even prime number.

KEY IDEA

Exactly two factors → prime. More than two → composite.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Is 17 prime or composite?
2. Is 27 prime or composite?
3. List the prime numbers between 1 and 20.

More practice: Workbook, Lesson 4.2.

LESSON SUMMARY

Prime numbers have exactly two factors. Composite numbers have more. 1 is neither.

UNIT 4 · LESSON 4.3 · 4.OA.4

Lesson 4.3: Multiples

I can find multiples of a number and tell whether a number is a multiple of another number.

WORDS TO KNOW

multiple	The product of a number and a whole number. Multiples of 4 are 4, 8, 12, 16, ...

LET'S EXPLORE

A bus comes every 15 minutes, starting at 8:00. Will there be a bus at 9:00?

Skip counting gives multiples. The multiples of 15 are 15, 30, 45, 60, ... A bus comes at 8:15, 8:30, 8:45, and 9:00.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Multiples of 6 on a hundred chart.

WORKED EXAMPLE · IS IT A MULTIPLE?

Is 72 a multiple of 8?

1 Ask: $8 \times ? = 72$.

2 $8 \times 9 = 72$.

Answer: Yes

KEY IDEA

A number is a multiple of each of its factors. 72 is a multiple of 8, and 8 is a factor of 72.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. List the first 6 multiples of 7.
2. Is 54 a multiple of 6?
3. Is 46 a multiple of 4?

More practice: Workbook, Lesson 4.3.

LESSON SUMMARY

Multiples come from skip counting. If a × b = c, then c is a multiple of a and of b.

UNIT 4 · LESSON 4.4 · 4.OA.4

Lesson 4.4: Divisibility Rules

I can use divisibility rules for 2, 3, 5, 9, and 10 to find factors quickly.

WORDS TO KNOW

divisible	Can be divided with no remainder.

LET'S EXPLORE

Is 348 divisible by 3? You could divide... or you could use a shortcut.

Divisibility rules let you test for factors without dividing.

A number is divisible by...	if...	Example
2	its ones digit is even (0, 2, 4, 6, 8)	348 ✓
3	the sum of its digits is divisible by 3	3 + 4 + 8 = 15 ✓
5	its ones digit is 0 or 5	345 ✓
9	the sum of its digits is divisible by 9	3 + 4 + 2 = 9, so 342 ✓
10	its ones digit is 0	340 ✓

WORKED EXAMPLE · USE THE RULES

Which of 2, 3, 5, 9, 10 divide 135?

1 Ones digit 5: divisible by 5, not by 2 or 10.

2 Digit sum 1 + 3 + 5 = 9: divisible by 3 and by 9.

Answer: 3, 5, and 9

KEY IDEA

Even ones digit → 2. Ends in 0 or 5 → 5. Ends in 0 → 10. Digit sum divisible by 3 or 9 → 3 or 9.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Is 726 divisible by 3?

2. Is 415 divisible by 2?

3. Is 918 divisible by 9?

More practice: Workbook, Lesson 4.4.

LESSON SUMMARY

Divisibility rules test for 2, 3, 5, 9, and 10 without dividing.

UNIT 4 · LESSON 4.5 · 4.OA.5

Lesson 4.5: Number Patterns

I can make and extend a number pattern that follows a rule.

WORDS TO KNOW

pattern	Numbers or shapes that repeat or change in a predictable way.
rule	Tells how to get the next term of a pattern.
term	Each number or shape in a pattern.

LET'S EXPLORE

A savings jar starts with $5. Each week $3 is added. How much will be in the jar after 6 weeks?

A rule tells how to get each term from the one before. Rule: start at 5, add 3.

Week	0	1	2	3	4	5	6
Dollars	5	8	11	14	17	20	23

WORKED EXAMPLE · FIND THE RULE

What is the rule? 4, 9, 14, 19, ...

1 9 − 4 = 5, 14 − 9 = 5, 19 − 14 = 5.

Answer: Start at 4, add 5. Next terms: 24, 29

WORKED EXAMPLE · NOTICE FEATURES

Rule: start at 4, add 5. Which terms are odd?

1 4 (even), 9 (odd), 14 (even), 19 (odd).

2 Even + odd = odd, odd + odd = even.

Answer: Every other term is odd.

KEY IDEA

Find the change between terms to find the rule. Then look for features the rule doesn't say out loud, like odd and even.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rule: start at 7, add 4. Write the first 5 terms.

2. Find the rule: 50, 44, 38, 32

3. Next two terms: 3, 6, 12, 24, ___, ___

More practice: Workbook, Lesson 4.5.

LESSON SUMMARY

A rule tells how to get each term. Find the change between terms, and look for hidden features like odd and even.

UNIT 4 · LESSON 4.6 · 4.OA.5

Lesson 4.6: Shape Patterns

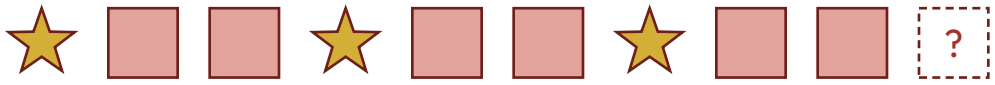
I can extend and describe a repeating or growing shape pattern.

WORDS TO KNOW

repeating pattern	A pattern with a part (the core) that repeats.
growing pattern	A pattern that gets bigger by a rule.

LET'S EXPLORE

A quilt border goes star, square, square, star, square, square, … What shape is 20th?



The core of this repeating pattern is star, square, square, 3 shapes long. Every 3rd shape is a square that ends the core.

WORKED EXAMPLE · USE DIVISION

What is the 20th shape?

1 The core has 3 shapes.

2 20 ÷ 3 = 6 R2: 6 full cores, then 2 more shapes.

3 The 2nd shape in the core is a square.

Answer: square

WORKED EXAMPLE · GROWING PATTERN

Figure 1 has 1 square, Figure 2 has 3, Figure 3 has 5. How many in Figure 10?

1 The rule is add 2.

2 Figure 10 = 1 + 9 × 2 = 19.

Answer: 19 squares

KEY IDEA

For repeating patterns, find the core and use division. For growing patterns, find the rule.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Core: circle, triangle. What is the 15th shape?
2. Core: A, B, C, D. What is the 22nd letter?
3. Growing: 4, 8, 12, ... squares. How many in Figure 7?

More practice: Workbook, Lesson 4.6.

LESSON SUMMARY

Repeating patterns: find the core and divide. Growing patterns: find the rule.

UNIT 4 · LESSON 4.7 · 4.OA.5

Lesson 4.7: Input-Output Tables

I can find and use a rule for an input-output table.

WORDS TO KNOW

input-output table — A table that shows a rule changing each input number into an output number.

LET'S EXPLORE

A machine turns 3 into 12, 5 into 20, and 8 into 32. What does it turn 10 into?

Input	3	5	8	10
Output	12	20	32	?

Each output is 4 times the input. The rule is multiply by 4, so 10 becomes 40.

WORKED EXAMPLE · FIND THE RULE

Inputs 2, 4, 7 → outputs 9, 11, 14.

1 Each output is 7 more than its input.

2 Check: 2 + 7 = 9, 4 + 7 = 11, 7 + 7 = 14.

Answer: Add 7

KEY IDEA

Test your rule on every pair in the table, not just one.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rule: × 6. Input 7 → ?
2. Inputs 10, 20, 30 → outputs 5, 15, 25. Rule?
3. Rule: ÷ 3. Output 9 → input?

More practice: Workbook, Lesson 4.7.

LESSON SUMMARY

An input-output rule changes each input into its output. Test it on every pair.

UNIT 4 · LESSON 4.8 · 4.OA.4, 4.OA.5

Lesson 4.8: Factors, Multiples, and Patterns in Problems

I can use factors, multiples, and patterns to solve problems.

LET'S EXPLORE

Hot dogs come in packs of 10 and buns in packs of 8. What is the fewest of each you can buy to have the same number of hot dogs and buns?

Common multiples answer “at the same time” or “the same number” questions. Factors answer “equal rows” and “equal groups” questions.

WORKED EXAMPLE · COMMON MULTIPLE

Packs of 10 hot dogs and 8 buns.

1 Multiples of 10: 10, 20, 30, 40.

2 Multiples of 8: 8, 16, 24, 32, 40.

3 The first match is 40.

Answer: 4 packs of hot dogs and 5 packs of buns

WORKED EXAMPLE · FACTORS

48 students need equal teams of more than 5 and fewer than 10 students. What team sizes work?

1 Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

2 Between 5 and 10: 6 and 8.

Answer: teams of 6 or 8

KEY IDEA

Same number or same time → common multiples. Equal rows or equal teams → factors.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Juice comes in packs of 6 and cups in packs of 9. Fewest of each to match?
2. Can 52 chairs be put in equal rows of 4? Of 5?
3. Is 91 prime?

More practice: Workbook, Lesson 4.8.

LESSON SUMMARY

Use common multiples for “the same number” or “at the same time.” Use factors for equal rows or teams.

UNIT 4 REVIEW

Unit 4 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
4.1	Factor pairs multiply to make the number. Test numbers in order and stop when the pairs meet.
4.2	Prime numbers have exactly two factors. Composite numbers have more. 1 is neither.
4.3	Multiples come from skip counting. If a × b = c, then c is a multiple of a and of b.
4.4	Divisibility rules test for 2, 3, 5, 9, and 10 without dividing.
4.5	A rule tells how to get each term. Find the change between terms, and look for hidden features like odd and even.
4.6	Repeating patterns: find the core and divide. Growing patterns: find the rule.
4.7	An input-output rule changes each input into its output. Test it on every pair.
4.8	Use common multiples for “the same number” or “at the same time.” Use factors for equal rows or teams.

UNIT 5

Fraction Equivalence and Comparison

Fractions on number lines, equivalent fractions, simplest form, and comparing and ordering fractions.

Fractions are numbers that name parts of a whole. In this unit you will find different names for the same fraction, simplify fractions, and compare fractions with different numerators and denominators.

In this unit you will...

- Explain fractions with models and number lines, including fractions greater than 1.
- Find fractions of a set.
- Make equivalent fractions by multiplying or dividing.
- Compare fractions with benchmarks, common denominators, and common numerators.
- Explain why comparisons need the same whole.

MATH IN REAL LIFE

Fractions are in recipes, rulers, sports statistics, sale prices, and music. Comparing fractions tells you which recipe uses more sugar or which runner went farther.

UNIT VOCABULARY

fraction	A number naming part of a whole.
numerator	Top number: parts counted.
denominator	Bottom number: equal parts in the whole.
equivalent fractions	Fractions naming the same amount.
simplest form	No common factor but 1.
benchmark fraction	Friendly fraction like 0, 1/2, 1.

Lessons in this unit

Lesson	Title
5.1	Fractions and Unit Fractions
5.2	Fractions on a Number Line
5.3	Fractions Equal to and Greater Than 1
5.4	Fractions of a Set
5.5	Equivalent Fractions with Models
5.6	Equivalent Fractions by Multiplying
5.7	Simplest Form
5.8	Equivalent Fractions on a Number Line
5.9	Comparing: Same Denominator or Same Numerator
5.10	Comparing with Benchmarks
5.11	Comparing with Common Denominators
5.12	Comparing with Common Numerators
5.13	Ordering Fractions
5.14	Comparing Fractions of the Same Whole
5.15	Fraction Comparison Word Problems
5.16	Explaining Fraction Reasoning

UNIT 5 · LESSON 5.1 · 4.NF.1 (review of 3.NF.1)

Lesson 5.1: Fractions and Unit Fractions

I can name a fraction as a number of equal parts of a whole, like $^{3}/_{4}$ is 3 of 4 equal parts.

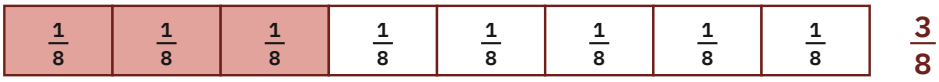
WORDS TO KNOW

fraction	A number that names part of a whole or part of a set.
numerator	The top number: how many parts we are counting.
denominator	The bottom number: how many equal parts make one whole.
unit fraction	A fraction with a numerator of 1, like $^{1}/_{5}$.

LET'S EXPLORE

A pan of cornbread is cut into 8 equal pieces. You eat 3. What fraction of the pan did you eat?

A fraction names equal parts of a whole. The denominator tells how many equal parts the whole is cut into. The numerator tells how many parts we are counting.

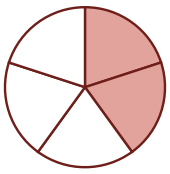


3 of 8 equal parts: $^{3}/_{8}$

Each piece is $^{1}/_{8}$ of the pan. $^{1}/_{8}$ is a unit fraction. Three pieces are 3 copies of $^{1}/_{8}$, so you ate $^{3}/_{8}$.

WORKED EXAMPLE · NAME A FRACTION

What fraction of the circle is shaded?



1 The circle has 5 equal parts, so the denominator is 5.

2 2 parts are shaded, so the numerator is 2.

Answer: $^{2}/_{5}$

WATCH OUT

The parts must be equal. A shape cut into 4 unequal pieces does not show fourths.

KEY IDEA

A fraction $\frac{a}{b}$ means a copies of the unit fraction $\frac{1}{b}$.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. What fraction of a pizza is 1 slice if the pizza has 6 equal slices?
2. How many $\frac{1}{10}$s make $\frac{7}{10}$?
3. A bar is cut into 4 equal parts with 3 shaded. Name the fraction.

More practice: Workbook, Lesson 5.1.

LESSON SUMMARY

A fraction counts equal parts. The denominator names the size of the parts; the numerator counts them.

UNIT 5 · LESSON 5.2 · 4.NF.1

Lesson 5.2: Fractions on a Number Line

I can place fractions on a number line.

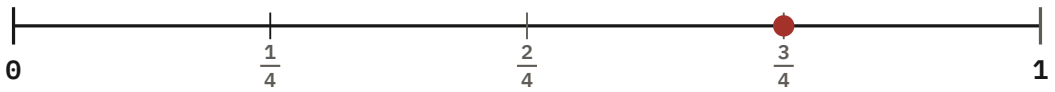
WORDS TO KNOW

number line (fractions)	A line where the distance from 0 to 1 is split into equal parts.

LET'S EXPLORE

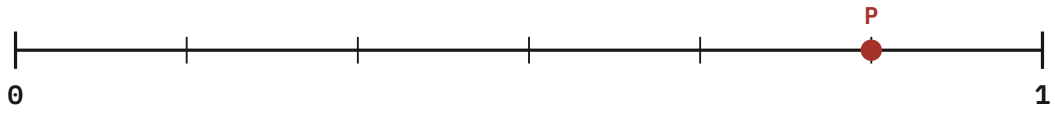
Where would $\frac{3}{4}$ be on a number line from 0 to 1?

On a number line, the whole is the distance from 0 to 1. Split it into equal parts, and count parts from 0. To show $\frac{3}{4}$, split 0 to 1 into 4 equal parts and count 3.



WORKED EXAMPLE · NAME THE POINT

What fraction is at point P?



1 0 to 1 is split into 6 equal parts.

2 P is 5 parts from 0.

Answer: $^{5}/_{6}$

KEY IDEA

Count the spaces between marks, not the marks, to find the denominator.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A line from 0 to 1 is split into 3 equal parts. What fraction is at the first mark after 0?
2. On a line split into 8 parts, how many parts from 0 is $^{5}/_{8}$?
3. What fraction is halfway between 0 and 1?

More practice: Workbook, Lesson 5.2.

LESSON SUMMARY

Split 0 to 1 into equal parts (the denominator) and count parts from 0 (the numerator).

UNIT 5 · LESSON 5.3 · 4.NF.1

Lesson 5.3: Fractions Equal to and Greater Than 1

I can show fractions equal to 1 and greater than 1, like $\frac{4}{4}$ and $\frac{5}{4}$.

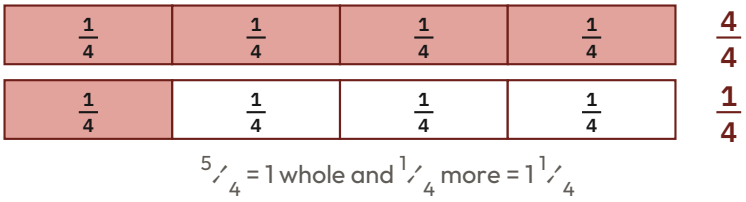
WORDS TO KNOW

improper fraction	A fraction with a numerator equal to or greater than its denominator, like $\frac{7}{4}$.
mixed number	A whole number and a fraction together, like $1\frac{3}{4}$.

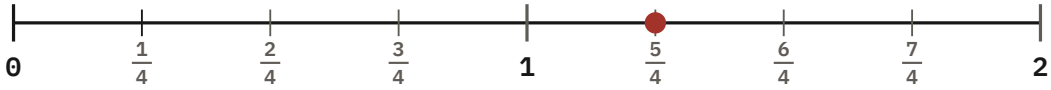
LET'S EXPLORE

Mom cut 2 pies into fourths. The family ate 5 pieces. Is that more or less than one whole pie?

When the numerator equals the denominator, the fraction equals 1 whole: $\frac{4}{4} = 1$. When the numerator is bigger, the fraction is greater than 1.



$\frac{5}{4} = 1$ whole and $\frac{1}{4}$ more = $1\frac{1}{4}$



WORKED EXAMPLE · WRITE AS A MIXED NUMBER

Write $\frac{7}{3}$ as a mixed number.

1 3 thirds make 1 whole. 7 thirds = 3 + 3 + 1.

2 That is 2 wholes and 1 third.

Answer: $2\frac{1}{3}$

KEY IDEA

Same numerator and denominator = 1 whole. To change an improper fraction to a mixed number, see how many wholes fit.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{6}/_{6}$ = ___
2. Write $^{9}/_{4}$ as a mixed number.
3. Write $1\,^{2}/_{5}$ as a fraction.

More practice: Workbook, Lesson 5.3.

LESSON SUMMARY

$^{4}/_{4}$ = 1. Improper fractions can be written as mixed numbers by finding how many wholes fit.

UNIT 5 · LESSON 5.4 · 4.NF.4 (preview), 3.NF.1

Lesson 5.4: Fractions of a Set

I can find a fraction of a group of objects.

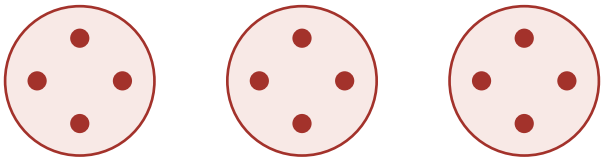
WORDS TO KNOW

set	A group of objects.

LET'S EXPLORE

There are 12 eggs in a carton. $^{1}/_{3}$ of them are brown. How many brown eggs are there?

A fraction can name part of a set. To find $^{1}/_{3}$ of 12, split the 12 into 3 equal groups. One group is $^{1}/_{3}$.



12 eggs in 3 equal groups: $^{1}/_{3}$ of 12 = 4.

WORKED EXAMPLE · MORE THAN ONE PART

Find $^{2}/_{3}$ of 12.

1 $^{1}/_{3}$ of 12 = 12 ÷ 3 = 4.

2 $^{2}/_{3}$ is 2 of those groups: 2 × 4 = 8.

Answer: 8

KEY IDEA

To find a fraction of a set: divide by the denominator to find one part, then multiply by the numerator.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{1}/_{4}$ of 20
2. $^{3}/_{5}$ of 15
3. Lillie has 18 books. $^{2}/_{6}$ are mysteries. How many mysteries?

More practice: Workbook, Lesson 5.4.

LESSON SUMMARY

Divide by the denominator to find one part; multiply by the numerator.

UNIT 5 · LESSON 5.5 · 4.NF.1

Lesson 5.5: Equivalent Fractions with Models

I can use models to show that two fractions are equivalent.

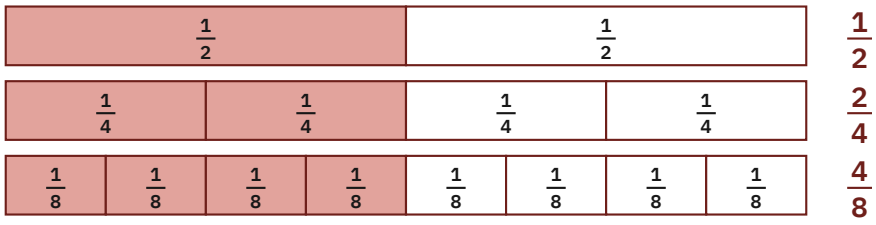
WORDS TO KNOW

equivalent fractions Fractions that name the same amount, like $^{1}/_{2}$ and $^{2}/_{4}$.

LET'S EXPLORE

Ellie ate $^{1}/_{2}$ of a sandwich. Lillie ate $^{2}/_{4}$ of the same size sandwich. Who ate more?

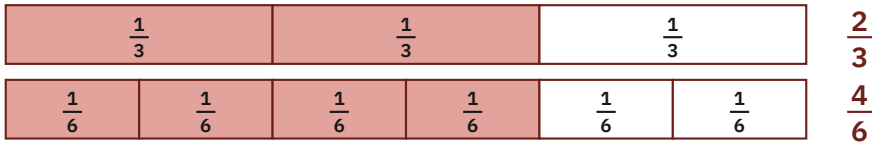
Look at the bars. $^{1}/_{2}$ and $^{2}/_{4}$ cover the same length. They are equivalent fractions: different names for the same amount.



When you cut each part into 2 smaller parts, you get twice as many parts, and each part is half as big. The amount does not change.

WORKED EXAMPLE · FIND AN EQUIVALENT FRACTION

Use a model to find a fraction equal to $^{2}/_{3}$ with sixths.



1 Draw thirds. Cut each third in half.

2 2 shaded thirds become 4 shaded sixths.

Answer: $^{2}/_{3} = {}^{4}/_{6}$

KEY IDEA

Equivalent fractions cover the same amount of the same whole.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{1}/_{3} = {}^{?}/_{6}$
2. $^{3}/_{4} = {}^{?}/_{8}$
3. Are $^{2}/_{5}$ and $^{4}/_{10}$ equivalent?

More practice: Workbook, Lesson 5.5.

LESSON SUMMARY

Equivalent fractions name the same amount. Cutting each part into more equal parts makes an equivalent fraction.

UNIT 5 · LESSON 5.6 · 4.NF.1

Lesson 5.6: Equivalent Fractions by Multiplying

I can multiply the numerator and denominator by the same number to make an equivalent fraction.

LET'S EXPLORE

Is there a way to find a fraction equal to $^{3}/_{5}$ with a denominator of 20 without drawing?

In the models, cutting every part into 4 pieces multiplies the number of parts by 4 and the number of shaded parts by 4.

	Numerator	Denominator
$^{3}/_{5}$	3	5
× 4	× 4	× 4
$^{12}/_{20}$	12	20

WORKED EXAMPLE · MULTIPLY

Find a fraction equal to $^{2}/_{3}$ with denominator 12.

1 $3 \times 4 = 12$, so multiply by 4.

2 Multiply the numerator too: $2 \times 4 = 8$.

Answer: $^{2}/_{3} = {}^{8}/_{12}$

WATCH OUT

Multiply the numerator and the denominator by the same number. Multiplying both by 4 is like multiplying by $^{4}/_{4}$, which equals 1.

KEY IDEA

$$\frac{a}{b} = \frac{n \times a}{n \times b}$$

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{1}/_{4} = {}^{?}/_{12}$
2. $^{5}/_{6} = {}^{?}/_{18}$

3. Write two fractions equivalent to $\frac{2}{5}$.

More practice: Workbook, Lesson 5.6.

LESSON SUMMARY

Multiply the numerator and denominator by the same number to make an equivalent fraction.

UNIT 5 · LESSON 5.7 · 4.NF.1

Lesson 5.7: Simplest Form

I can divide the numerator and denominator by a common factor to write a fraction in simplest form.

WORDS TO KNOW

common factor	A number that is a factor of two or more numbers.
simplest form	A fraction whose numerator and denominator have no common factor except 1.

LET'S EXPLORE

A recipe calls for $\frac{6}{8}$ of a cup of milk. Is there a simpler name for that amount?

You can also go the other way: divide the numerator and denominator by a common factor. 6 and 8 are both divisible by 2.

WORKED EXAMPLE · SIMPLIFY

Write $\frac{6}{8}$ in simplest form.

1 $6 \div 2 = 3$ and $8 \div 2 = 4$.

2 3 and 4 have no common factor but 1.

Answer: $\frac{3}{4}$

WORKED EXAMPLE · USE THE GREATEST COMMON FACTOR

Write $\frac{12}{18}$ in simplest form.

1 Factors of 12: 1, 2, 3, 4, 6, 12. Factors of 18: 1, 2, 3, 6, 9, 18.

2 The greatest common factor is 6.

3 $12 \div 6 = 2$, $18 \div 6 = 3$.

Answer: $\frac{2}{3}$

KEY IDEA

Divide the numerator and denominator by a common factor. Keep going until 1 is the only common factor.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Simplify $^{4}/_{10}$
2. Simplify $^{9}/_{12}$
3. Simplify $^{8}/_{16}$

More practice: Workbook, Lesson 5.7.

LESSON SUMMARY

Divide the numerator and denominator by a common factor until only 1 is common.

UNIT 5 · LESSON 5.8 · 4.NF.1

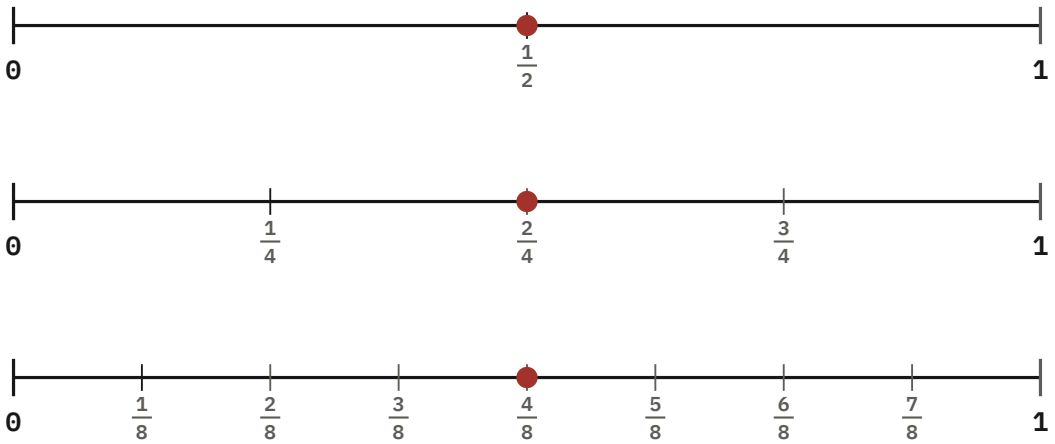
Lesson 5.8: Equivalent Fractions on a Number Line

I can show equivalent fractions at the same point on a number line.

LET'S EXPLORE

On a ruler, the $^{1}/_{2}$-inch mark is the same mark as $^{2}/_{4}$ and $^{4}/_{8}$. Why?

Equivalent fractions land on the same point of a number line, because they are the same distance from 0.



WORKED EXAMPLE · FIND THE SAME POINT

Which eighths fraction is at the same point as $\frac{3}{4}$?

1 Each fourth is 2 eighths.

2 3 fourths = 6 eighths.

Answer: $\frac{6}{8}$

KEY IDEA

Equivalent fractions are the same number, so they are at the same point on a number line.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Which sixths fraction is at the same point as $\frac{1}{3}$?
2. Which tenths fraction is at the same point as $\frac{1}{2}$?
3. Is $\frac{2}{3}$ at the same point as $\frac{4}{6}$?

More practice: Workbook, Lesson 5.8.

LESSON SUMMARY

Equivalent fractions are at the same point on a number line.

UNIT 5 · LESSON 5.9 · 4.NF.2

Lesson 5.9: Comparing: Same Denominator or Same Numerator

I can compare fractions that have the same denominator or the same numerator.

LET'S EXPLORE

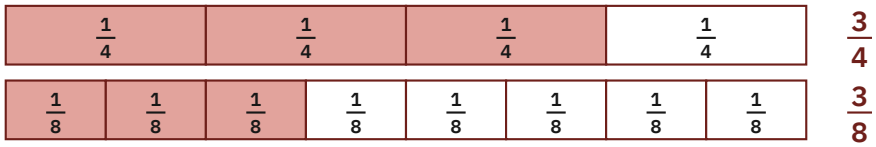
Which is more, $\frac{3}{8}$ of a pizza or $\frac{5}{8}$ of the same pizza? Which is more, $\frac{3}{4}$ or $\frac{3}{8}$?

Same denominator

When the denominators are the same, the pieces are the same size. More pieces means more: $\frac{5}{8} > \frac{3}{8}$.

Same numerator

When the numerators are the same, you have the same number of pieces. Compare the size of the pieces: fourths are bigger than eighths, so $\frac{3}{4} > \frac{3}{8}$.



WATCH OUT

A bigger denominator means smaller pieces. $\frac{1}{8}$ is less than $\frac{1}{4}$, even though 8 > 4.

KEY IDEA

Same denominator: compare numerators. Same numerator: the smaller denominator is the bigger fraction.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{2}{6}$ ◯ $\frac{5}{6}$
2. $\frac{4}{5}$ ◯ $\frac{4}{10}$
3. $\frac{1}{3}$ ◯ $\frac{1}{12}$

More practice: Workbook, Lesson 5.9.

LESSON SUMMARY

Same denominator: more pieces is greater. Same numerator: bigger pieces (smaller denominator) is greater.

UNIT 5 · LESSON 5.10 · 4.NF.2

Lesson 5.10: Comparing with Benchmarks

I can compare fractions by thinking about 0, $\frac{1}{2}$, and 1.

WORDS TO KNOW

benchmark fraction A friendly fraction like 0, $\frac{1}{2}$, or 1 used to compare.

LET'S EXPLORE

Which is greater, $\frac{3}{8}$ or $\frac{4}{6}$? They have different numerators and denominators!

Compare each fraction to $\frac{1}{2}$. $\frac{3}{8}$ is less than $\frac{4}{8}$, which is half. $\frac{4}{6}$ is more than $\frac{3}{6}$, which is half. So $\frac{3}{8} < \frac{4}{6}$.

Less than half	Exactly half	More than half
numerator less than half of denominator	numerator is half of denominator	numerator more than half of denominator
$\frac{2}{5}$, $\frac{3}{8}$, $\frac{1}{4}$	$\frac{3}{6}$, $\frac{5}{10}$	$\frac{3}{5}$, $\frac{5}{8}$, $\frac{3}{4}$

WORKED EXAMPLE · CLOSE TO 1

Compare $\frac{7}{8}$ and $\frac{9}{10}$.

1 Both are one piece away from 1.

2 $\frac{7}{8}$ is missing $\frac{1}{8}$; $\frac{9}{10}$ is missing $\frac{1}{10}$.

3 $\frac{1}{10}$ is the smaller gap, so $\frac{9}{10}$ is closer to 1.

Answer: $\frac{7}{8} < \frac{9}{10}$

KEY IDEA

Ask: Is each fraction less than, equal to, or more than $\frac{1}{2}$? Near 1, compare the missing pieces.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{2}{6} \bigcirc \frac{5}{8}$
2. $\frac{4}{7} \bigcirc \frac{1}{3}$
3. $\frac{5}{6} \bigcirc \frac{7}{8}$

More practice: Workbook, Lesson 5.10.

LESSON SUMMARY

Compare each fraction to $\frac{1}{2}$. Near 1, compare the missing pieces.

UNIT 5 · LESSON 5.11 · 4.NF.2

Lesson 5.11: Comparing with Common Denominators

I can compare fractions by renaming them with a common denominator.

WORDS TO KNOW

common denominator	A denominator that two fractions share.

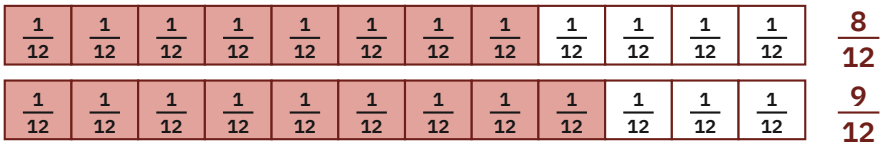
LET'S EXPLORE

Compare $\frac{2}{3}$ and $\frac{3}{4}$. Neither benchmarks nor matching numbers help much. What can we do?

Rename both fractions so they have the same denominator. Then the pieces are the same size and you can compare numerators.

WORKED EXAMPLE · COMMON DENOMINATOR

Compare $\frac{2}{3}$ and $\frac{3}{4}$.



1 Multiples of 3: 3, 6, 9, 12. Multiples of 4: 4, 8, 12. Use 12.

2 $\frac{2}{3} = \frac{8}{12}$ (× 4).

3 $\frac{3}{4} = \frac{9}{12}$ (× 3).

4 $8 < 9$.

Answer: $\frac{2}{3} < \frac{3}{4}$

WORKED EXAMPLE · ONE DENOMINATOR IS A MULTIPLE OF THE OTHER

Compare $\frac{3}{5}$ and $\frac{7}{10}$.

1 10 is a multiple of 5. $\frac{3}{5} = \frac{6}{10}$.

2 $6 < 7$.

Answer: $\frac{3}{5} < \frac{7}{10}$

KEY IDEA

Find a common multiple of the denominators. Rename both fractions, then compare numerators.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{1}{2}$ ◯ $\frac{3}{5}$
2. $\frac{5}{6}$ ◯ $\frac{3}{4}$
3. $\frac{2}{4}$ ◯ $\frac{5}{10}$

More practice: Workbook, Lesson 5.11.

LESSON SUMMARY

Rename both fractions with a common denominator, then compare numerators.

UNIT 5 · LESSON 5.12 · 4.NF.2

Lesson 5.12: Comparing with Common Numerators

I can compare fractions by renaming them with a common numerator.

WORDS TO KNOW

common numerator A numerator that two fractions share.

LET'S EXPLORE

Compare $\frac{2}{5}$ and $\frac{4}{9}$. Is there a quick way?

Sometimes it is easier to make the numerators match. $\frac{2}{5} = \frac{4}{10}$. Now compare $\frac{4}{10}$ and $\frac{4}{9}$: same number of pieces, and ninths are bigger than tenths.

WORKED EXAMPLE · COMMON NUMERATOR

Compare $\frac{2}{5}$ and $\frac{4}{9}$.

1 $\frac{2}{5} = \frac{4}{10}$.

2 4 tenths vs 4 ninths: ninths are bigger pieces.

Answer: $\frac{2}{5} < \frac{4}{9}$

KEY IDEA

If the numerators are easy to match, rename to a common numerator. Then the smaller denominator is the greater fraction.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{1}{3} \bigcirc \frac{2}{7}$
2. $\frac{3}{4} \bigcirc \frac{6}{9}$
3. $\frac{2}{3} \bigcirc \frac{4}{5}$

More practice: Workbook, Lesson 5.12.

LESSON SUMMARY

Match the numerators. With the same number of pieces, bigger pieces (smaller denominator) win.

UNIT 5 · LESSON 5.13 · 4.NF.2

Lesson 5.13: Ordering Fractions

I can order three or more fractions from least to greatest.

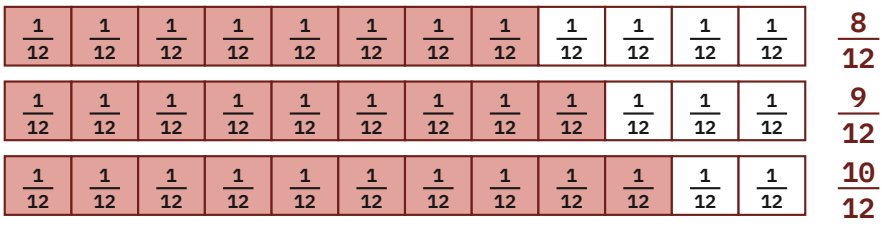
LET'S EXPLORE

Three friends ran $\frac{3}{4}$ mile, $\frac{2}{3}$ mile, and $\frac{5}{6}$ mile. Who ran farthest? Who ran the least?

Ordering fractions is comparing again and again. A common denominator lets you compare all of them at once.

WORKED EXAMPLE · ORDER

Order $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{6}$ from least to greatest.



1 Common denominator 12.

2 $\frac{3}{4} = \frac{9}{12}$, $\frac{2}{3} = \frac{8}{12}$, $\frac{5}{6} = \frac{10}{12}$.

3 $8 < 9 < 10$.

Answer: $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$

WORKED EXAMPLE · USE BENCHMARKS

Order $\frac{1}{5}$, $\frac{7}{8}$, $\frac{1}{2}$.

1 $\frac{1}{5}$ is near 0, $\frac{1}{2}$ is half, $\frac{7}{8}$ is near 1.

Answer: $\frac{1}{5}$, $\frac{1}{2}$, $\frac{7}{8}$

KEY IDEA

Use the fastest strategy: benchmarks if the fractions are far apart, a common denominator if they are close.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Order least to greatest: $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{6}$

2. Order greatest to least: $\frac{3}{5}$, $\frac{7}{10}$, $\frac{1}{2}$

3. Which is greatest: $\frac{5}{8}$, $\frac{3}{4}$, $\frac{2}{3}$?

More practice: Workbook, Lesson 5.13.

LESSON SUMMARY

Order fractions by renaming to a common denominator or by using benchmarks.

UNIT 5 · LESSON 5.14 · 4.NF.2

Lesson 5.14: Comparing Fractions of the Same Whole

I can explain why fractions can only be compared when they refer to the same whole.

WORDS TO KNOW

whole	The one complete thing or group a fraction is part of.

LET'S EXPLORE

Ellie ate $\frac{1}{2}$ of a small pizza. Lillie ate $\frac{1}{3}$ of a large pizza. Ellie says she ate more because $\frac{1}{2} > \frac{1}{3}$. Is she right?

Not necessarily! $\frac{1}{2} > \frac{1}{3}$ is only true when both fractions are of the same whole. Half of a small pizza could be less than a third of a huge pizza.

KEY IDEA

Comparisons are only valid when the two fractions refer to the same whole.

WORKED EXAMPLE · DIFFERENT WHOLES

Is $\frac{1}{2}$ of 10 more or less than $\frac{1}{3}$ of 30?

1 $\frac{1}{2}$ of 10 = 5.

2 $\frac{1}{3}$ of 30 = 10.

Answer: Less: 5 < 10

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{1}{4}$ of 40 or $\frac{1}{2}$ of 12. Which is more?
2. Can you say $\frac{2}{3} > \frac{1}{2}$ for any two pizzas?
3. $\frac{3}{4}$ of 8 vs $\frac{3}{4}$ of 20

More practice: Workbook, Lesson 5.14.

LESSON SUMMARY

You can only compare fractions directly when they are parts of the same whole.

UNIT 5 · LESSON 5.15 · 4.NF.2

Lesson 5.15: Fraction Comparison Word Problems

I can solve word problems by comparing fractions and explain my answer.

LET'S EXPLORE

Ellie practiced piano $\frac{3}{4}$ hour. Lillie practiced $\frac{5}{6}$ hour. Who practiced longer?

Decide which strategy to use, compare, then answer in a sentence.

WORKED EXAMPLE · COMPARE TIMES

$\frac{3}{4}$ hour vs $\frac{5}{6}$ hour.

1 Common denominator 12: $\frac{9}{12}$ vs $\frac{10}{12}$.

Answer: Lillie practiced longer.

WORKED EXAMPLE · MISSING PIECE

A recipe needs $\frac{2}{3}$ cup of sugar. You have $\frac{3}{4}$ cup. Do you have enough?

1 $\frac{2}{3} = \frac{8}{12}$, $\frac{3}{4} = \frac{9}{12}$.

2 $9 > 8$.

Answer: Yes, you have enough.

KEY IDEA

Answer word problems with a complete sentence and make sure the wholes are the same.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A blue ribbon is $\frac{5}{8}$ yard. A red ribbon is $\frac{1}{2}$ yard. Which is longer?
2. Ellie ate $\frac{2}{6}$ of her sandwich. Lillie ate $\frac{1}{3}$ of hers (same size). Who ate more?
3. A trail is $\frac{7}{10}$ mile. Another is $\frac{3}{5}$ mile. Which is shorter?

More practice: Workbook, Lesson 5.15.

LESSON SUMMARY

Pick a strategy, compare, and answer in a full sentence about the same whole.

UNIT 5 · LESSON 5.16 · 4.NF.1, 4.NF.2

Lesson 5.16: Explaining Fraction Reasoning

I can explain and justify my thinking about equivalent fractions and comparisons.

WORDS TO KNOW

justify	To give reasons that show why something is true.

LET'S EXPLORE

Sam says $\frac{3}{6} > \frac{1}{2}$ because 3 > 1. How would you convince Sam he is wrong?

Mathematicians explain their thinking with models, numbers, and words. A strong explanation uses at least two of these.

WORKED EXAMPLE · JUSTIFY

Show that $\frac{3}{6} = \frac{1}{2}$.

1 Model: 3 of 6 equal parts covers the same length as 1 of 2 equal parts.

2 Numbers: $\frac{1}{2} = \frac{1 \times 3}{2 \times 3} = \frac{3}{6}$.

3 Words: The pieces are 3 times smaller, but there are 3 times as many.

Answer: They are equal.

KEY IDEA

Use a model, numbers, and words to justify your answer.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Explain why $\frac{2}{8} = \frac{1}{4}$.
2. Explain why $\frac{5}{6} < \frac{7}{8}$.
3. Is $\frac{4}{10} > \frac{1}{2}$? Explain.

More practice: Workbook, Lesson 5.16.

LESSON SUMMARY

Justify fraction ideas with a model, numbers, and words.

UNIT 5 REVIEW

Unit 5 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
5.1	A fraction counts equal parts. The denominator names the size of the parts; the numerator counts them.
5.2	Split 0 to 1 into equal parts (the denominator) and count parts from 0 (the numerator).
5.3	$^4/_4 = 1$. Improper fractions can be written as mixed numbers by finding how many wholes fit.
5.4	Divide by the denominator to find one part; multiply by the numerator.
5.5	Equivalent fractions name the same amount. Cutting each part into more equal parts makes an equivalent fraction.
5.6	Multiply the numerator and denominator by the same number to make an equivalent fraction.
5.7	Divide the numerator and denominator by a common factor until only 1 is common.
5.8	Equivalent fractions are at the same point on a number line.
5.9	Same denominator: more pieces is greater. Same numerator: bigger pieces (smaller denominator) is greater.
5.10	Compare each fraction to $^1/_2$. Near 1, compare the missing pieces.
5.11	Rename both fractions with a common denominator, then compare numerators.
5.12	Match the numerators. With the same number of pieces, bigger pieces (smaller denominator) win.
5.13	Order fractions by renaming to a common denominator or by using benchmarks.
5.14	You can only compare fractions directly when they are parts of the same whole.
5.15	Pick a strategy, compare, and answer in a full sentence about the same whole.
5.16	Justify fraction ideas with a model, numbers, and words.

UNIT 6

Fraction Operations

Adding and subtracting fractions and mixed numbers, and multiplying fractions by whole numbers.

Now that you can compare fractions, you will compute with them. You will add and subtract fractions and mixed numbers with like denominators, regroup wholes, multiply fractions by whole numbers, and solve real problems with measurements and recipes.

In this unit you will…

- Decompose fractions in more than one way.
- Add and subtract fractions with like denominators.
- Add and subtract mixed numbers, regrouping when needed.
- Multiply a fraction by a whole number.
- Solve multi-step word problems with fractions and measurements.

MATH IN REAL LIFE

Cooks add fractions of cups, carpenters subtract fractions of inches, and runners add fractions of miles. Doubling a recipe is multiplying fractions by whole numbers.

UNIT VOCABULARY

decompose	Break a number into parts.
mixed number	A whole number and a fraction.
improper fraction	A fraction ≥ 1.
like denominators	Same denominators.

Lessons in this unit

Lesson	Title
6.1	Decomposing Fractions
6.2	Adding Fractions with Like Denominators

Lesson	Title
6.3	Subtracting Fractions with Like Denominators
6.4	Fraction Word Problems
6.5	Mixed Numbers and Improper Fractions
6.6	Adding Mixed Numbers
6.7	Subtracting Mixed Numbers
6.8	Subtracting Mixed Numbers with Regrouping
6.9	Fractions as Multiples of Unit Fractions
6.10	Multiplying a Fraction by a Whole Number
6.11	Models for Fraction Multiplication
6.12	Fraction Multiplication Word Problems
6.13	Fractions and Measurement
6.14	Multi-Step Fraction Problems
6.15	Estimating with Fractions
6.16	Fraction Operations Problem Solving

UNIT 6 · LESSON 6.1 · 4.NF.3b

Lesson 6.1: Decomposing Fractions

I can break a fraction into a sum of smaller fractions with the same denominator, like $\frac{3}{8} = \frac{1}{8} + \frac{2}{8}$.

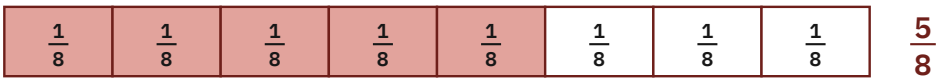
WORDS TO KNOW

decompose	To break a number into parts.
unit fraction	A fraction with numerator 1.

LET'S EXPLORE

A pizza is cut into 8 slices. Ellie and Lillie together eat $\frac{5}{8}$ of it. How might they have shared those slices?

$\frac{5}{8}$ is 5 eighths. You can split 5 eighths into parts in many ways, just like you can split 5 into 2 + 3 or 1 + 4.



Ways to decompose five-eighths
$\frac{5}{8} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$
$\frac{5}{8} = \frac{2}{8} + \frac{3}{8}$
$\frac{5}{8} = \frac{1}{8} + \frac{4}{8}$

WORKED EXAMPLE · DECOMPOSE A MIXED NUMBER

Decompose $1\frac{1}{4}$ into fourths.

1 1 whole = $\frac{4}{4}$.

2 $1\frac{1}{4} = \frac{4}{4} + \frac{1}{4} = \frac{5}{4}$.

Answer: $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

KEY IDEA

When you decompose a fraction, the denominator stays the same. The numerators add up to the original numerator.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Decompose $\frac{4}{6}$ two ways.
2. $\frac{7}{10} = \frac{3}{10} + \frac{?}{10}$
3. Write $\frac{3}{5}$ as a sum of unit fractions.

More practice: Workbook, Lesson 6.1.

LESSON SUMMARY

Decompose a fraction by splitting its numerator; the denominator stays the same.

UNIT 6 · LESSON 6.2 · 4.NF.3a, 4.NF.3d

Lesson 6.2: Adding Fractions with Like Denominators

I can add fractions that have the same denominator.

WORDS TO KNOW

like denominators	Denominators that are the same.

LET'S EXPLORE

Ellie walked $\frac{2}{5}$ mile to the park and $\frac{1}{5}$ mile to the store. How far did she walk?

Fifths plus fifths makes fifths. Add the numerators: 2 fifths + 1 fifth = 3 fifths. The denominator tells the size of the pieces, so it does not change.



WORKED EXAMPLE · ADD

$\frac{2}{5} + \frac{1}{5}$

1 Add the numerators: 2 + 1 = 3.

2 Keep the denominator: fifths.

Answer: $\frac{3}{5}$

WORKED EXAMPLE · SUM GREATER THAN 1

$\frac{5}{6} + \frac{4}{6}$

1 5 + 4 = 9 sixths.

2 $^{9}/_{6} = {}^{6}/_{6} + {}^{3}/_{6} = 1\,{}^{3}/_{6}$.

Answer: $^{9}/_{6} = 1\,{}^{3}/_{6}$

WATCH OUT

Do not add the denominators! $^{1}/_{4} + {}^{1}/_{4}$ is $^{2}/_{4}$, not $^{2}/_{8}$. Two quarters of a dollar are 2 fourths of a dollar, not 2 eighths.

KEY IDEA

Add the numerators. Keep the denominator.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{3}/_{8} + {}^{4}/_{8}$
2. $^{1}/_{3} + {}^{2}/_{3}$
3. $^{7}/_{10} + {}^{5}/_{10}$

More practice: Workbook, Lesson 6.2.

LESSON SUMMARY

Add the numerators and keep the denominator. Rename sums greater than 1 as mixed numbers.

UNIT 6 · LESSON 6.3 · 4.NF.3a, 4.NF.3d

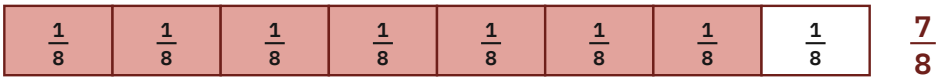
Lesson 6.3: Subtracting Fractions with Like Denominators

I can subtract fractions that have the same denominator.

LET'S EXPLORE

There was $^{7}/_{8}$ of a pie left. The family ate $^{3}/_{8}$ of the pie. How much pie is left now?

Subtracting works the same way: subtract the numerators and keep the denominator.



WORKED EXAMPLE · SUBTRACT

$^{7}/_{8} - {}^{3}/_{8}$

1 $7 - 3 = 4$ eighths.

Answer: $^{4}/_{8} = {}^{1}/_{2}$

WORKED EXAMPLE · SUBTRACT FROM 1

$1 - {}^{2}/_{5}$

1 $1 = {}^{5}/_{5}$.

2 $5 - 2 = 3$ fifths.

Answer: $^{3}/_{5}$

KEY IDEA

Subtract the numerators. Keep the denominator. To subtract from 1, rename 1 as a fraction with the same denominator.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{5}/_{6} - {}^{2}/_{6}$
2. $^{9}/_{10} - {}^{4}/_{10}$
3. $1 - {}^{3}/_{4}$

More practice: Workbook, Lesson 6.3.

LESSON SUMMARY

Subtract numerators and keep the denominator. Rename 1 as d/d when subtracting from 1.

UNIT 6 · LESSON 6.4 · 4.NF.3d

Lesson 6.4: Fraction Word Problems

I can solve word problems by adding and subtracting fractions with like denominators.

LET'S EXPLORE

A recipe uses $\frac{3}{4}$ cup of flour for the cookies and $\frac{2}{4}$ cup for the frosting. How much flour is that? Is it more than 1 cup?

Use a bar model or number line, decide whether to add or subtract, then answer in a sentence.

WORKED EXAMPLE · ADD

Flour for cookies and frosting.

1 $\frac{3}{4} + \frac{2}{4} = \frac{5}{4} = 1\frac{1}{4}$.

Answer: $1\frac{1}{4}$ cups, more than 1 cup

WORKED EXAMPLE · COMPARE BY SUBTRACTING

Lillie ran $\frac{7}{10}$ mile. Ellie ran $\frac{4}{10}$ mile. How much farther did Lillie run?

1 $\frac{7}{10} - \frac{4}{10} = \frac{3}{10}$.

Answer: $\frac{3}{10}$ mile farther

KEY IDEA

Joining → add. Taking away or "how much more" → subtract. The pieces must be the same size.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{2}{6}$ of a garden is beans and $\frac{3}{6}$ is corn. What fraction is beans and corn?
2. A jug had $\frac{7}{8}$ gallon of juice. $\frac{5}{8}$ gallon was poured out. How much is left?
3. Dad painted $\frac{3}{5}$ of a fence. How much more to finish?

More practice: Workbook, Lesson 6.4.

LESSON SUMMARY

Join → add; take away or compare → subtract. Use the same whole and like denominators.

UNIT 6 · LESSON 6.5 · 4.NF.3b

Lesson 6.5: Mixed Numbers and Improper Fractions

I can rename mixed numbers as fractions and fractions as mixed numbers.

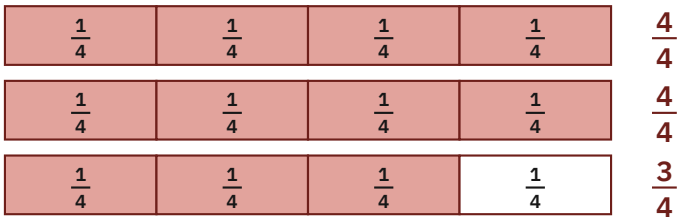
WORDS TO KNOW

mixed number	A whole number and a fraction.
improper fraction	A fraction greater than or equal to 1.

LET'S EXPLORE

A carpenter needs $2\frac{3}{4}$ feet of board. How many fourths of a foot is that?

Each whole is $\frac{4}{4}$. Two wholes are $\frac{8}{4}$. Add the $\frac{3}{4}$: $2\frac{3}{4} = \frac{11}{4}$.



WORKED EXAMPLE · MIXED NUMBER TO FRACTION

Write $3\frac{2}{5}$ as a fraction.

1 3 wholes = 3 × 5 = 15 fifths.

2 15 + 2 = 17 fifths.

Answer: $\frac{17}{5}$

WORKED EXAMPLE · FRACTION TO MIXED NUMBER

Write $\frac{23}{6}$ as a mixed number.

1 How many 6s are in 23? 23 ÷ 6 = 3 R5.

2 3 wholes and 5 sixths.

Answer: $3\frac{5}{6}$

KEY IDEA

Mixed → fraction: multiply the whole by the denominator, add the numerator. Fraction → mixed: divide the numerator by the denominator.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $1\frac{3}{8} = ?$
2. $\frac{14}{3} = ?$
3. $4\frac{1}{2} = ?$

More practice: Workbook, Lesson 6.5.

LESSON SUMMARY

Mixed to fraction: whole × denominator + numerator. Fraction to mixed: divide.

UNIT 6 · LESSON 6.6 · 4.NF.3c

Lesson 6.6: Adding Mixed Numbers

I can add mixed numbers with like denominators.

LET'S EXPLORE

Ellie practiced piano $1\frac{1}{4}$ hours on Saturday and $2\frac{2}{4}$ hours on Sunday. How long did she practice?

Add the whole numbers. Add the fractions. If the fractions add to 1 or more, regroup.

WORKED EXAMPLE · ADD

$1\frac{1}{4} + 2\frac{2}{4}$

1 Wholes: $1 + 2 = 3$.

2 Fractions: $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$.

Answer: $3\frac{3}{4}$ hours

WORKED EXAMPLE · REGROUP

$2\frac{3}{5} + 1\frac{4}{5}$

1 Wholes: $2 + 1 = 3$.

2 Fractions: $\frac{3}{5} + \frac{4}{5} = \frac{7}{5} = 1\frac{2}{5}$.

3 $3 + 1\frac{2}{5} = 4\frac{2}{5}$.

Answer: $4\frac{2}{5}$

KEY IDEA

Add wholes, add fractions, then regroup fractions that make a whole.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $3\frac{1}{6} + 1\frac{4}{6}$
2. $2\frac{5}{8} + 1\frac{6}{8}$
3. $4\frac{1}{2} + 3\frac{1}{2}$

More practice: Workbook, Lesson 6.6.

LESSON SUMMARY

Add wholes and fractions separately, then regroup any extra whole.

UNIT 6 · LESSON 6.7 · 4.NF.3c

Lesson 6.7: Subtracting Mixed Numbers

I can subtract mixed numbers with like denominators when no regrouping is needed.

LET'S EXPLORE

A rope is $5\frac{7}{8}$ feet long. You cut off $2\frac{3}{8}$ feet. How much is left?

Subtract the fractions. Subtract the wholes.

WORKED EXAMPLE · SUBTRACT

$5\frac{7}{8} - 2\frac{3}{8}$

1 Fractions: $\frac{7}{8} - \frac{3}{8} = \frac{4}{8}$.

2 Wholes: $5 - 2 = 3$.

Answer: $3\frac{4}{8} = 3\frac{1}{2}$ feet

WORKED EXAMPLE · USE FRACTIONS

$\frac{15}{4} - \frac{6}{4}$

1 $\frac{15}{4} - \frac{6}{4} = \frac{9}{4}$.

2 $\frac{9}{4} = 2\frac{1}{4}$.

Answer: $2\frac{1}{4}$

KEY IDEA

Subtract fractions, then wholes. You can also change both to fractions first.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $4\frac{5}{6} - 1\frac{2}{6}$
2. $7\frac{9}{10} - 3\frac{4}{10}$
3. $3\frac{2}{3} - 2$

More practice: Workbook, Lesson 6.7.

LESSON SUMMARY

Subtract fractions, then wholes. Check by adding.

UNIT 6 · LESSON 6.8 · 4.NF.3c

Lesson 6.8: Subtracting Mixed Numbers with Regrouping

I can regroup a whole to subtract mixed numbers, like $3\frac{1}{4} - 1\frac{3}{4}$.

LET'S EXPLORE

There are $3\frac{1}{4}$ pizzas. The family eats $1\frac{3}{4}$ pizzas. How much is left? Can you take $\frac{3}{4}$ from $\frac{1}{4}$?

You can't take $\frac{3}{4}$ from $\frac{1}{4}$. Regroup one whole as $\frac{4}{4}$: $3\frac{1}{4} = 2\frac{5}{4}$.

WORKED EXAMPLE · REGROUP

$3\frac{1}{4} - 1\frac{3}{4}$

1 Regroup: $3\frac{1}{4} = 2\frac{5}{4}$.

2 Fractions: $\frac{5}{4} - \frac{3}{4} = \frac{2}{4}$.

3 Wholes: $2 - 1 = 1$.

Answer: $1\frac{2}{4} = 1\frac{1}{2}$ pizzas

WORKED EXAMPLE · SUBTRACT FROM A WHOLE NUMBER

$4 - 1\frac{2}{3}$

1 Regroup: $4 = 3\frac{3}{3}$.

2 $\frac{3}{3} - \frac{2}{3} = \frac{1}{3}$; $3 - 1 = 2$.

Answer: $2\frac{1}{3}$

KEY IDEA

When the first fraction is too small, trade 1 whole for $^{d}/_{d}$ and add it to the fraction.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $4\,^{1}/_{5} - 2\,^{3}/_{5}$
2. $5\,^{2}/_{6} - 1\,^{5}/_{6}$
3. $3 - 1\,^{3}/_{8}$

More practice: Workbook, Lesson 6.8.

LESSON SUMMARY

If the fraction part is too small, regroup 1 whole into d/d, then subtract.

UNIT 6 · LESSON 6.9 · 4.NF.4a

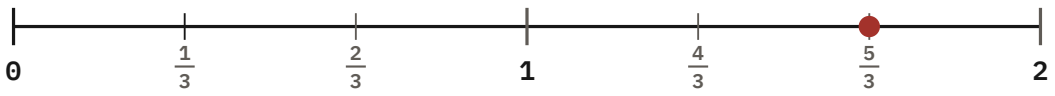
Lesson 6.9: Fractions as Multiples of Unit Fractions

I can write a fraction as a whole number times a unit fraction, like $^{5}/_{4} = 5 \times {}^{1}/_{4}$.

LET'S EXPLORE

How many $^{1}/_{3}$-cup scoops make $^{5}/_{3}$ cups?

$^{5}/_{3}$ is 5 thirds: $^{1}/_{3} + {}^{1}/_{3} + {}^{1}/_{3} + {}^{1}/_{3} + {}^{1}/_{3}$. Repeated addition is multiplication, so $^{5}/_{3} = 5 \times {}^{1}/_{3}$.



WORKED EXAMPLE · WRITE AS A MULTIPLE

Write $^{7}/_{8}$ as a multiple of a unit fraction.

1 7 eighths = 7 copies of 1 eighth.

Answer: $^{7}/_{8} = 7 \times {}^{1}/_{8}$

KEY IDEA

$^{a}/_{b} = a \times {}^{1}/_{b}$

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{4}/_{5} =$ ___ $\times\ ^{1}/_{5}$
2. $6 \times\ ^{1}/_{10} =$?
3. $^{9}/_{4} = 9 \times$?

More practice: Workbook, Lesson 6.9.

LESSON SUMMARY

Every fraction is a number of copies of a unit fraction: $^{5}/_{4} = 5 \times\ ^{1}/_{4}$.

UNIT 6 · LESSON 6.10 · 4.NF.4b

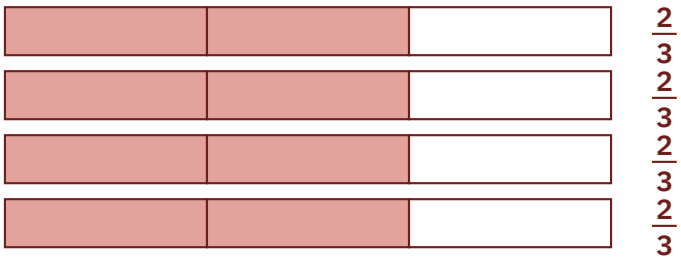
Lesson 6.10: Multiplying a Fraction by a Whole Number

I can multiply a whole number by a fraction, like $3 \times\ ^{2}/_{5}$.

LET'S EXPLORE

Each batch of muffins uses $^{2}/_{3}$ cup of oil. How much oil is needed for 4 batches?

$4 \times\ ^{2}/_{3}$ means 4 groups of $^{2}/_{3}$. That is 4 × 2 thirds = 8 thirds.



WORKED EXAMPLE · MULTIPLY

$4 \times\ ^{2}/_{3}$

1 $4 \times\ ^{2}/_{3} = 4 \times 2 \times\ ^{1}/_{3} = 8 \times\ ^{1}/_{3}$.

2 $^{8}/_{3} = 2\ ^{2}/_{3}$.

Answer: $2\ ^{2}/_{3}$ cups

WATCH OUT

Multiply the whole number by the numerator only. The size of the pieces (the denominator) does not change.

KEY IDEA

$n \times \frac{a}{b} = \frac{n \times a}{b}$

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $3 \times \frac{2}{5}$
2. $5 \times \frac{3}{4}$
3. $6 \times \frac{1}{2}$

More practice: Workbook, Lesson 6.10.

LESSON SUMMARY

Multiply the whole number by the numerator; keep the denominator.

UNIT 6 · LESSON 6.11 · 4.NF.4b

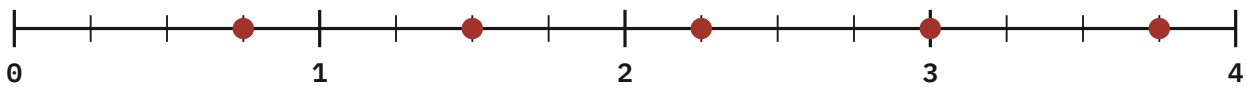
Lesson 6.11: Models for Fraction Multiplication

I can use number lines and area models to multiply fractions by whole numbers.

LET'S EXPLORE

A frog jumps $\frac{3}{4}$ meter each hop. Where is it after 5 hops?

Each hop is $\frac{3}{4}$. Five hops: $5 \times \frac{3}{4} = \frac{15}{4} = 3\frac{3}{4}$ meters.



WORKED EXAMPLE · TWO WAYS

$6 \times \frac{2}{5}$

1 Repeated addition: $\frac{2}{5} + \frac{2}{5} + \frac{2}{5} + \frac{2}{5} + \frac{2}{5} + \frac{2}{5} = \frac{12}{5}$.

2 Multiply: $6 \times 2 = 12$, so $\frac{12}{5} = 2\frac{2}{5}$.

Answer: $2\frac{2}{5}$

KEY IDEA

A number line shows equal jumps; the product is where you land.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $4 \times \frac{3}{8}$
2. $3 \times \frac{5}{6}$
3. A snail moves $\frac{2}{10}$ meter per minute. How far in 7 minutes?

More practice: Workbook, Lesson 6.11.

LESSON SUMMARY

Equal jumps on a number line show multiplication: 5 hops of $\frac{3}{4}$ lands at $\frac{15}{4}$.

UNIT 6 · LESSON 6.12 · 4.NF.4c

Lesson 6.12: Fraction Multiplication Word Problems

I can solve word problems that multiply a fraction by a whole number.

LET'S EXPLORE

Each student needs $\frac{3}{8}$ yard of ribbon for a project. There are 5 students. How much ribbon is needed? Between which two whole numbers is the answer?

WORKED EXAMPLE · RIBBON

$5 \times \frac{3}{8}$

1 $5 \times 3 = 15$, so $\frac{15}{8}$.

2 $\frac{15}{8} = 1\frac{7}{8}$: between 1 and 2 yards.

Answer: $1\frac{7}{8}$ yards

WORKED EXAMPLE · RECIPE

A recipe uses $\frac{2}{3}$ cup of sugar. Mom triples the recipe. How much sugar?

1 $3 \times \frac{2}{3} = \frac{6}{3} = 2$.

Answer: 2 cups

KEY IDEA

Equal groups of a fraction → multiply. Then think: between which whole numbers does the answer lie?

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A plant grows $\frac{3}{4}$ inch a week. How much in 6 weeks?
2. 8 friends each eat $\frac{1}{4}$ pizza. How much pizza?
3. A bottle holds $\frac{5}{6}$ liter. How much do 4 bottles hold?

More practice: Workbook, Lesson 6.12.

LESSON SUMMARY

Equal groups of a fraction mean multiply. Name the whole numbers the answer falls between.

UNIT 6 · LESSON 6.13 · 4.MD.2, 4.NF.3d

Lesson 6.13: Fractions and Measurement

I can measure lengths to the nearest half, fourth, and eighth inch and add or subtract the measurements.

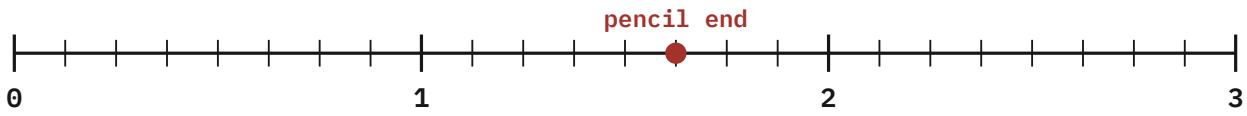
WORDS TO KNOW

inch ruler	A ruler where each inch is split into halves, fourths, and eighths.

LET'S EXPLORE

How long is your pencil? Can you measure it more exactly than "about 6 inches"?

An inch ruler splits each inch into halves, fourths, and eighths. The longest marks between inches are halves, then fourths, then eighths.



The pencil ends at 13 eighths of an inch past 0: $\frac{13}{8} = 1\frac{5}{8}$ inches.

WORKED EXAMPLE · ADD MEASUREMENTS

Two sticks are $2\frac{3}{8}$ in. and $1\frac{7}{8}$ in. Laid end to end, how long are they?

1 $2\frac{3}{8} + 1\frac{7}{8} = 3\frac{10}{8} = 4\frac{2}{8}$.

Answer: $4\frac{2}{8} = 4\frac{1}{4}$ inches

KEY IDEA

Read a ruler like a number line: count the wholes, then count the small parts.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many eighths are in $\frac{1}{2}$ inch?
2. $3\frac{1}{4}$ in. − $1\frac{3}{4}$ in.
3. Which is longer: $\frac{5}{8}$ in. or $\frac{3}{4}$ in.?

More practice: Workbook, Lesson 6.13.

LESSON SUMMARY

A ruler is a number line in inches split into halves, fourths, and eighths.

UNIT 6 · LESSON 6.14 · 4.NF.3d, 4.NF.4c

Lesson 6.14: Multi-Step Fraction Problems

I can solve multi-step word problems with fractions.

LET'S EXPLORE

Lillie needs 3 pieces of ribbon, each $\frac{3}{4}$ foot long. She has a ribbon that is 3 feet long. How much will be left?

WORKED EXAMPLE · MULTIPLY, THEN SUBTRACT

Ribbon problem.

1 Needed: $3 \times \frac{3}{4} = \frac{9}{4} = 2\frac{1}{4}$ feet.

2 Left: $3 - 2\frac{1}{4} = \frac{3}{4}$ foot.

Answer: $\frac{3}{4}$ foot

WORKED EXAMPLE · ADD, THEN COMPARE

Ellie ran $\frac{5}{8}$ mile and walked $\frac{6}{8}$ mile. Lillie ran $1\frac{2}{8}$ miles. Who went farther?

1 Ellie: $\frac{5}{8} + \frac{6}{8} = \frac{11}{8} = 1\frac{3}{8}$.

2 $1\frac{3}{8} > 1\frac{2}{8}$.

Answer: Ellie went farther.

KEY IDEA

Find the hidden first step. Write each step as an equation.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A pitcher holds 2 quarts. Mom pours 4 cups of $\frac{1}{4}$ quart each. How much is left in the pitcher?
2. Dad hiked $2\frac{3}{10}$ miles in the morning and $1\frac{9}{10}$ in the afternoon. How far in all?
3. 6 kids each eat $\frac{2}{8}$ of a pie. Is 1 pie enough?

More practice: Workbook, Lesson 6.14.

LESSON SUMMARY

Find the hidden first step, write an equation for each step, and check if the answer makes sense.

UNIT 6 · LESSON 6.15 · 4.NF.3d, 4.NF.4c

Lesson 6.15: Estimating with Fractions

I can estimate sums, differences, and products with fractions using benchmarks.

LET'S EXPLORE

About how much is $2\frac{7}{8} + 3\frac{1}{10}$? Do you need an exact answer to know it's about 6?

Round each fraction or mixed number to the nearest whole or half. $2\frac{7}{8}$ is about 3. $3\frac{1}{10}$ is about 3. So the sum is about 6.

WORKED EXAMPLE · ESTIMATE A PRODUCT

About how much is $4 \times \frac{7}{8}$?

1 $\frac{7}{8}$ is close to 1.

2 $4 \times 1 = 4$, a little less.

Answer: a little less than 4

KEY IDEA

Round fractions to 0, $\frac{1}{2}$, or 1 (and mixed numbers to the nearest whole or half) to estimate.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Estimate $5\frac{1}{6} - 1\frac{5}{6}$
2. Estimate $6 \times \frac{4}{9}$
3. Is $1\frac{3}{8} + 2\frac{1}{8}$ more or less than 4?

More practice: Workbook, Lesson 6.15.

LESSON SUMMARY

Round to 0, $\frac{1}{2}$, or 1 (or to the nearest whole) to estimate and to check answers.

UNIT 6 · LESSON 6.16 · 4.NF.3, 4.NF.4

Lesson 6.16: Fraction Operations Problem Solving

I can choose the right operation and solve fraction problems from start to finish.

LET'S EXPLORE

Plan a fruit salad: it needs fractions of cups of different fruits for a group. How much fruit do you need?

This lesson pulls together adding, subtracting, and multiplying fractions.

WORKED EXAMPLE · FRUIT SALAD

Each serving: $\frac{3}{4}$ cup grapes and $\frac{2}{4}$ cup berries. Make 4 servings. How much fruit?

1 One serving: $\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$.

2 Four servings: $4 \times \frac{5}{4} = \frac{20}{4} = 5$.

Answer: 5 cups

KEY IDEA

Read, plan, solve, check.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $4 \times \frac{3}{4} + 4 \times \frac{2}{4}$
2. $3\frac{1}{3} - 1\frac{2}{3}$
3. $\frac{5}{6} + \frac{5}{6} + \frac{5}{6}$

More practice: Workbook, Lesson 6.16.

LESSON SUMMARY

Add and subtract like fractions, regroup mixed numbers, and multiply fractions by whole numbers to solve problems.

UNIT 6 REVIEW

Unit 6 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
6.1	Decompose a fraction by splitting its numerator; the denominator stays the same.
6.2	Add the numerators and keep the denominator. Rename sums greater than 1 as mixed numbers.
6.3	Subtract numerators and keep the denominator. Rename 1 as d/d when subtracting from 1.
6.4	Join → add; take away or compare → subtract. Use the same whole and like denominators.
6.5	Mixed to fraction: whole × denominator + numerator. Fraction to mixed: divide.
6.6	Add wholes and fractions separately, then regroup any extra whole.
6.7	Subtract fractions, then wholes. Check by adding.
6.8	If the fraction part is too small, regroup 1 whole into d/d, then subtract.
6.9	Every fraction is a number of copies of a unit fraction: $^{5}/_{4} = 5 \times {}^{1}/_{4}$.
6.10	Multiply the whole number by the numerator; keep the denominator.
6.11	Equal jumps on a number line show multiplication: 5 hops of $^{3}/_{4}$ lands at $^{15}/_{4}$.
6.12	Equal groups of a fraction mean multiply. Name the whole numbers the answer falls between.
6.13	A ruler is a number line in inches split into halves, fourths, and eighths.
6.14	Find the hidden first step, write an equation for each step, and check if the answer makes sense.
6.15	Round to 0, $^{1}/_{2}$, or 1 (or to the nearest whole) to estimate and to check answers.
6.16	Add and subtract like fractions, regroup mixed numbers, and multiply fractions by whole numbers to solve problems.

UNIT 7

Decimals

Tenths and hundredths, equivalent decimals, comparing decimals, and money.

Decimals are another way to write fractions with denominators of 10 and 100. You will connect decimals to fractions, place value, number lines, and money.

In this unit you will...

- Write tenths and hundredths as fractions and decimals.
- Show that tenths and hundredths can be equivalent.
- Add tenths and hundredths.
- Compare and order decimals.
- Add and subtract money amounts and make change.

MATH IN REAL LIFE

Decimals are in prices, gas pumps, sports times, and measurements. Every time you pay for something, you are using decimals.

UNIT VOCABULARY

decimal	A number written with a decimal point.
tenth	1 of 10 equal parts.
hundredth	1 of 100 equal parts.
decimal point	Separates wholes from parts.

Lessons in this unit

Lesson	Title
7.1	Tenths
7.2	Decimals Greater Than 1
7.3	Tenths on a Number Line

Lesson	Title
7.4	Decimal Place Value
7.5	Hundredths
7.6	Equivalent Tenths and Hundredths
7.7	Adding Tenths and Hundredths
7.8	Decimals in Different Forms
7.9	Comparing Decimals with Models
7.10	Comparing Decimals with Place Value
7.11	Ordering Decimals
7.12	Decimals and Fractions on a Number Line
7.13	Money and Decimals
7.14	Adding Money
7.15	Subtracting Money and Making Change
7.16	Decimal and Money Problem Solving

UNIT 7 · LESSON 7.1 · 4.NF.6

Lesson 7.1: Tenths

I can write tenths as fractions and as decimals, like $^{3}/_{10} = 0.3$.

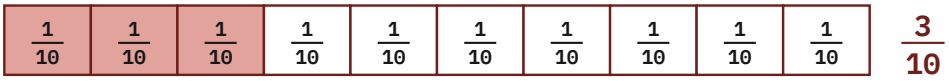
WORDS TO KNOW

decimal	A number with a decimal point, used to show parts of a whole in tenths, hundredths, and so on.
tenth	One of 10 equal parts of a whole: $^{1}/_{10}$ or 0.1.
decimal point	The dot that separates the whole number from the parts of a whole.

LET'S EXPLORE

A candy bar is split into 10 equal pieces. You eat 3 pieces. How can you write the amount you ate without a fraction bar?

When a whole is split into 10 equal parts, each part is one tenth. Three tenths can be written as the fraction $^{3}/_{10}$ or the decimal 0.3.



Ones	Tenths
0 .	3

0.3: 0 ones and 3 tenths

The decimal point separates whole numbers from parts of a whole. The first place after the decimal point is the tenths place. We read 0.3 as "three tenths."

WORKED EXAMPLE · WRITE AS A DECIMAL

Write $^{7}/_{10}$ as a decimal.

1 7 tenths.

2 Write 7 in the tenths place.

Answer: 0.7

KEY IDEA

$^{n}/_{10}$ = 0.n. The tenths place is the first place to the right of the decimal point.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Write $^{9}/_{10}$ as a decimal.
2. Write 0.4 as a fraction.
3. How do you read 0.6?

More practice: Workbook, Lesson 7.1.

LESSON SUMMARY

Tenths are 10 equal parts of a whole: $^{3}/_{10}$ = 0.3.

UNIT 7 · LESSON 7.2 · 4.NF.6

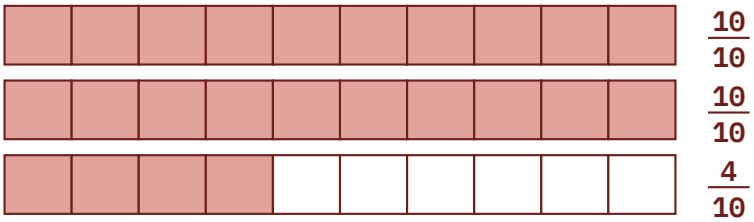
Lesson 7.2: Decimals Greater Than 1

I can write mixed numbers with tenths as decimals, like $2\,^{4}/_{10} = 2.4$.

LET'S EXPLORE

A recipe needs 2 and 4 tenths cups of flour. How can you write that as a decimal?

The whole number goes to the left of the decimal point. The tenths go to the right. $2\,^{4}/_{10}$ = 2.4, read "two and four tenths." The word and stands for the decimal point.



Tens	Ones		Tenths
	2	.	4

WORKED EXAMPLE · WRITE AS A DECIMAL

Write $13\,^{6}/_{10}$ as a decimal.

1 13 wholes, 6 tenths.

Answer: 13.6

KEY IDEA

Say “and” only for the decimal point: 13.6 is “thirteen and six tenths.”

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $5\frac{9}{10}$ as a decimal
2. 4.3 as a mixed number
3. Read 20.7 aloud. Write the words.

More practice: Workbook, Lesson 7.2.

LESSON SUMMARY

Wholes go left of the decimal point, tenths go right. Say “and” for the decimal point.

UNIT 7 · LESSON 7.3 · 4.NF.6

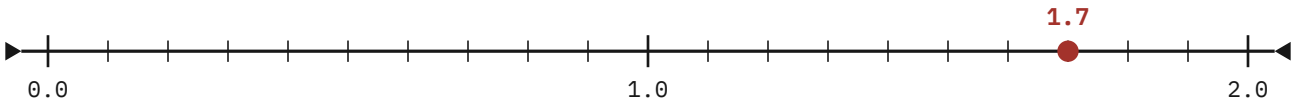
Lesson 7.3: Tenths on a Number Line

I can place and name decimals in tenths on a number line.

LET'S EXPLORE

A walking trail has markers every tenth of a mile. Lillie stops at the 7th marker after the 1-mile sign. Where is she?

A number line from 0 to 1 split into 10 equal parts shows tenths. 1.7 is 7 tenths past 1.



WORKED EXAMPLE · NAME THE POINT

What decimal is at point A?



1 Each jump is 0.1.

2 A is 4 jumps past 3.

Answer: 3.4

KEY IDEA

Count jumps of 0.1 from the whole number.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. What decimal is halfway between 0 and 1?
2. What decimal is 3 tenths more than 2?
3. Which whole number is 6.8 closer to?

More practice: Workbook, Lesson 7.3.

LESSON SUMMARY

Split each whole into 10 equal jumps; each jump is 0.1.

UNIT 7 · LESSON 7.4 · 4.NF.6

Lesson 7.4: Decimal Place Value

I can tell the place and value of digits in a decimal.

WORDS TO KNOW

tenths place	The first place to the right of the decimal point.
hundredths place	The second place to the right of the decimal point.

LET'S EXPLORE

In 45.7, the 4, the 5, and the 7 all have different values. What is each worth?

Place value continues to the right of the decimal point. Each place is still 10 times the place to its right: 1 one = 10 tenths.

Tens	Ones		Tenths
4	5	.	7

WORKED EXAMPLE · VALUE OF A DIGIT

What is the value of each digit in 45.7?

1 4 is in the tens place: 40.

2 5 is in the ones place: 5.

3 7 is in the tenths place: 0.7 ($^{7}/_{10}$).

Answer: 40 + 5 + 0.7

KEY IDEA

Ones, then the decimal point, then tenths, then hundredths. 10 tenths = 1 one.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Value of the 3 in 6.3
2. Value of the 6 in 6.3
3. Write 20 + 8 + 0.1 as a decimal.

More practice: Workbook, Lesson 7.4.

LESSON SUMMARY

Each place is 10 times the place to its right, even after the decimal point.

UNIT 7 · LESSON 7.5 · 4.NF.6

Lesson 7.5: Hundredths

I can write hundredths as fractions and decimals, like $^{37}/_{100} = 0.37$.

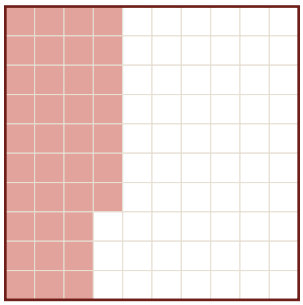
WORDS TO KNOW

hundredth — One of 100 equal parts of a whole: $^{1}/_{100}$ or 0.01.

LET'S EXPLORE

A dollar is 100 cents. If you have 37 cents, what part of a dollar do you have?

Split a whole into 100 equal parts. Each part is one hundredth. 37 hundredths = $^{37}/_{100}$ = 0.37.



37 of 100 squares shaded: 0.37

Ones		Tenths	Hundredths
0	.	3	7

WORKED EXAMPLE · HUNDREDTHS WITH A ZERO

Write $^{5}/_{100}$ as a decimal.

1 5 hundredths: 0 tenths and 5 hundredths.

2 Write a 0 in the tenths place.

Answer: 0.05

WATCH OUT

0.5 and 0.05 are different! 0.5 is 5 tenths (50 cents). 0.05 is 5 hundredths (5 cents).

KEY IDEA

Hundredths use two places after the decimal point.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{48}/_{100}$ as a decimal
2. 0.09 as a fraction
3. $^{3}/_{100}$ as a decimal

More practice: Workbook, Lesson 7.5.

LESSON SUMMARY

Hundredths use two decimal places: $^{37}/_{100}$ = 0.37; $^{5}/_{100}$ = 0.05.

UNIT 7 · LESSON 7.6 · 4.NF.5, 4.NF.6

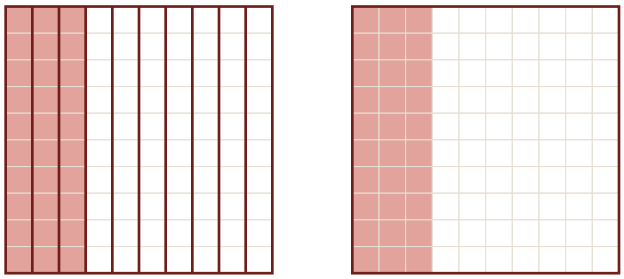
Lesson 7.6: Equivalent Tenths and Hundredths

I can show that tenths and hundredths can be equivalent, like $\frac{3}{10} = \frac{30}{100}$ and 0.3 = 0.30.

LET'S EXPLORE

Is 3 dimes the same amount of money as 30 pennies?

Yes! 3 dimes = 30 cents = 30 pennies. In the same way, $\frac{3}{10} = \frac{30}{100}$: each tenth is 10 hundredths.



0.3 (3 columns) = 0.30 (30 squares)

WORKED EXAMPLE · RENAME TENTHS AS HUNDREDTHS

$\frac{6}{10} = \frac{?}{100}$

1 Multiply top and bottom by 10.

Answer: $\frac{60}{100}$; 0.6 = 0.60

KEY IDEA

A zero at the end of a decimal does not change its value: 0.3 = 0.30.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{9}{10} = \frac{?}{100}$
2. $\frac{40}{100} = \frac{?}{10}$
3. 0.5 = 0.___

More practice: Workbook, Lesson 7.6.

LESSON SUMMARY

Each tenth equals 10 hundredths: 0.3 = 0.30.

UNIT 7 · LESSON 7.7 · 4.NF.5

Lesson 7.7: Adding Tenths and Hundredths

I can add a fraction in tenths to a fraction in hundredths, like $\frac{3}{10} + \frac{4}{100}$.

LET'S EXPLORE

You have 3 dimes and 4 pennies. What part of a dollar is that?

To add tenths and hundredths, first make the denominators the same. Change tenths to hundredths.

WORKED EXAMPLE · ADD

$\frac{3}{10} + \frac{4}{100}$

1 $\frac{3}{10} = \frac{30}{100}$.

2 $\frac{30}{100} + \frac{4}{100} = \frac{34}{100}$.

Answer: $\frac{34}{100} = 0.34$

WORKED EXAMPLE · ADD WITH REGROUPING

$\frac{7}{10} + \frac{45}{100}$

1 $\frac{7}{10} = \frac{70}{100}$.

2 $\frac{70}{100} + \frac{45}{100} = \frac{115}{100} = 1\frac{15}{100}$.

Answer: 1.15

KEY IDEA

Rename tenths as hundredths, then add the numerators.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{2}{10} + \frac{5}{100}$
2. $\frac{6}{10} + \frac{30}{100}$
3. $\frac{8}{10} + \frac{35}{100}$

More practice: Workbook, Lesson 7.7.

LESSON SUMMARY

Rename tenths as hundredths, then add.

UNIT 7 · LESSON 7.8 · 4.NF.6

Lesson 7.8: Decimals in Different Forms

I can write decimals in standard, word, fraction, and expanded form.

LET'S EXPLORE

The fastest 100-meter dash time at a school meet was 13.45 seconds. How do you say that number?

Form	13.45
Standard	13.45
Word	thirteen and forty-five hundredths
Fraction / mixed number	$13\frac{45}{100}$
Expanded	10 + 3 + 0.4 + 0.05

WORKED EXAMPLE · WRITE IN WORDS

Write 6.08 in words.

1. Whole number: six.
2. "and" for the point.
3. 08 hundredths = eight hundredths.

Answer: six and eight hundredths

KEY IDEA

Read the decimal part as a whole number, then say the last place: 0.45 is "forty-five hundredths."

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Write 2.7 in words.
2. Write "nine and twelve hundredths" as a decimal.
3. Write 5.36 in expanded form.

More practice: Workbook, Lesson 7.8.

LESSON SUMMARY

Decimals can be written in standard, word, fraction, and expanded form.

UNIT 7 · LESSON 7.9 · 4.NF.7

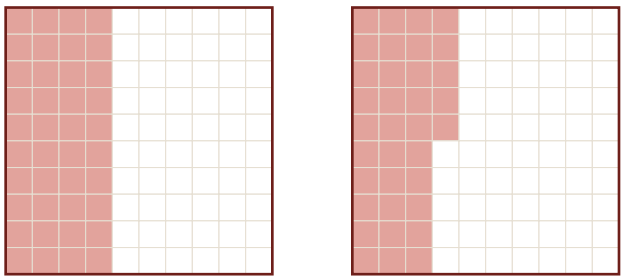
Lesson 7.9: Comparing Decimals with Models

I can compare two decimals using hundred grids and number lines.

LET'S EXPLORE

Is 0.4 or 0.35 greater? 35 is bigger than 4... but is 0.35 bigger than 0.4?

Models show the truth: 0.4 shades 40 squares, and 0.35 shades only 35 squares. So 0.4 > 0.35.



0.4 (left) and 0.35 (right)

WATCH OUT

Longer decimals are not always greater. Compare the same places.

KEY IDEA

Decimals must refer to the same whole to be compared. Shade both on the same-size grid.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 0.6 ◯ 0.59
2. 0.25 ◯ 0.3
3. 0.7 ◯ 0.70

More practice: Workbook, Lesson 7.9.

LESSON SUMMARY

Shade both decimals on the same grid; more shaded is greater.

UNIT 7 · LESSON 7.10 · 4.NF.7

Lesson 7.10: Comparing Decimals with Place Value

I can compare decimals by lining up place values.

LET'S EXPLORE

Two runners finished in 12.48 seconds and 12.5 seconds. Who was faster (less time)?

Line up the decimal points. Write a zero so both have hundredths. Compare from the left like whole numbers.

	Tens	Ones	.	Tenths	Hundredths
12.48	1	2	.	4	8
12.50	1	2	.	5	0

WORKED EXAMPLE · COMPARE

12.48 vs 12.5

1 Tens and ones are the same.

2 Tenths: 4 < 5.

Answer: 12.48 < 12.5; the 12.48 runner was faster

KEY IDEA

Line up the decimal points, fill in zeros, compare from the left.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3.09 ◯ 3.1

2. 5.6 ◯ 5.06

3. 10.4 ◯ 9.99

More practice: Workbook, Lesson 7.10.

LESSON SUMMARY

Line up decimal points, add zeros, and compare from the left.

UNIT 7 · LESSON 7.11 · 4.NF.7

Lesson 7.11: Ordering Decimals

I can order three or more decimals.

LET'S EXPLORE

Three bean plants grew 2.35 in., 2.5 in., and 2.07 in. List them from shortest to tallest growth.

WORKED EXAMPLE · ORDER

2.35, 2.5, 2.07 from least to greatest

1 Write with hundredths: 2.35, 2.50, 2.07.

2 Compare tenths: 0 < 3 < 5.

Answer: 2.07, 2.35, 2.5

KEY IDEA

Write all decimals with the same number of places, then order like whole numbers.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Order least to greatest: 0.6, 0.06, 0.66

2. Order greatest to least: 1.2, 1.12, 1.21

3. Which is least: 4.4, 4.04, 4.40?

More practice: Workbook, Lesson 7.11.

LESSON SUMMARY

Write the same number of decimal places, then order like whole numbers.

UNIT 7 · LESSON 7.12 · 4.NF.6, 4.NF.7

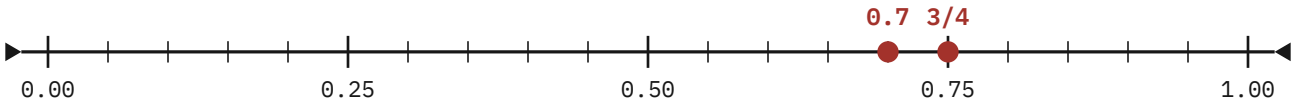
Lesson 7.12: Decimals and Fractions on a Number Line

I can place decimals and fractions on the same number line and compare them.

LET'S EXPLORE

Which is greater, $\frac{3}{4}$ or 0.7?

Write both in hundredths. $\frac{3}{4} = \frac{75}{100} = 0.75$. $0.75 > 0.70$, so $\frac{3}{4} > 0.7$.



WORKED EXAMPLE · COMMON BENCHMARK DECIMALS

Write $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{5}$ as decimals.

1 $\frac{1}{2} = \frac{50}{100} = 0.5$.

2 $\frac{1}{4} = \frac{25}{100} = 0.25$.

3 $\frac{1}{5} = \frac{2}{10} = 0.2$.

Answer: 0.5, 0.25, 0.2

KEY IDEA

Change fractions to tenths or hundredths, then compare them as decimals.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{1}{2}$ ◯ 0.45
2. $\frac{2}{5}$ ◯ 0.4
3. 0.8 ◯ $\frac{3}{4}$

More practice: Workbook, Lesson 7.12.

LESSON SUMMARY

Change fractions to tenths or hundredths to compare them with decimals.

UNIT 7 · LESSON 7.13 · 4.NF.6, 4.MD.2

Lesson 7.13: Money and Decimals

I can write money amounts as decimals and connect dollars, dimes, and pennies to ones, tenths, and hundredths.

WORDS TO KNOW

cent	One hundredth of a dollar: $0.01.

LET'S EXPLORE

You have 2 one-dollar bills, 3 dimes, and 4 pennies. How do you write that amount?



Dollars are ones, dimes are tenths of a dollar, and pennies are hundredths of a dollar. 2 dollars, 3 dimes, 4 pennies = $2.34.

Coin or bill	Value	Decimal
dollar	100¢	$1.00
quarter	25¢	$0.25
dime	10¢	$0.10
nickel	5¢	$0.05
penny	1¢	$0.01

WORKED EXAMPLE · COUNT MONEY

3 quarters and 2 dimes

1. Quarters: 75¢.
2. Dimes: 20¢.
3. Total: 95¢.

Answer: $0.95

KEY IDEA

Write money with a dollar sign and two decimal places: $3.05, not $3.5 for three dollars and five cents.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4 dollars, 1 dime, 7 pennies
2. 2 quarters, 1 nickel
3. What fraction of a dollar is 1 quarter?

More practice: Workbook, Lesson 7.13.

LESSON SUMMARY

Dollars = ones, dimes = tenths, pennies = hundredths. Write money with two decimal places.

UNIT 7 · LESSON 7.14 · 4.MD.2

Lesson 7.14: Adding Money

I can add money amounts by lining up the decimal points.

LET'S EXPLORE

A notebook costs $2.75 and a pack of pens costs $3.49. What is the total?

Add money like whole numbers: line up the decimal points (dollars under dollars, cents under cents), add from the right, regroup, and bring the decimal point straight down.

WORKED EXAMPLE · ADD MONEY

$2.75 + $3.49

$$\begin{array}{r} {\scriptstyle 1\ 1\ \ } \\ 275 \\ +349 \\ \hline 624 \end{array}$$

1 Think in cents: 275 + 349 = 624 cents.

2 Write the decimal point: $6.24.

Answer: $6.24

WATCH OUT

Line up the decimal points, not the right edges, when one amount has no cents: $5 + $1.25 → $5.00 + $1.25.

KEY IDEA

Line up the decimal points, add like whole numbers, then place the decimal point in the answer.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $4.20 + $3.65
2. $1.89 + $2.47
3. $10 + $3.75

More practice: Workbook, Lesson 7.14.

LESSON SUMMARY

Line up decimal points and add like whole numbers.

UNIT 7 · LESSON 7.15 · 4.MD.2

Lesson 7.15: Subtracting Money and Making Change

I can subtract money amounts and make change.

WORDS TO KNOW

change	The money you get back when you pay more than the price.

LET'S EXPLORE

A game costs $13.65. You pay with a $20 bill. How much change do you get?

WORKED EXAMPLE · SUBTRACT

$20.00 − $13.65

$$\begin{array}{r} 2{,}000 \\ -1{,}365 \\ \hline 635 \end{array}$$

1 Write $20 as $20.00.

2 2000 − 1365 = 635 cents.

Answer: $6.35

WORKED EXAMPLE · COUNT UP

Make change for $13.65 from $20.

1 $13.65 + $0.35 = $14.00.

2 $14.00 + $6.00 = $20.00.

3 $0.35 + $6.00 = $6.35.

Answer: $6.35

KEY IDEA

Subtract to find change, or count up from the price to the amount paid.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $9.50 − $4.25

2. $10.00 − $6.79

3. Change from $5 for a $2.38 snack

More practice: Workbook, Lesson 7.15.

LESSON SUMMARY

Subtract or count up to find change.

UNIT 7 · LESSON 7.16 · 4.MD.2, 4.NF.7

Lesson 7.16: Decimal and Money Problem Solving

I can solve multi-step problems with decimals and money.

LET'S EXPLORE

You have $20 for the school book fair. Books cost $6.95, $4.50, and $7.25. Can you buy all three?

WORKED EXAMPLE · ADD, THEN COMPARE

Book fair

1 $6.95 + $4.50 + $7.25 = $18.70.

2 $18.70 < $20.00.

Answer: Yes, with $1.30 left

KEY IDEA

Estimate first ($7 + $5 + $7 = $19), then find the exact answer.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3 notebooks at $1.25 each. Total?

2. $10 − ($3.49 + $2.75)

3. Which costs less: 4 pens at $0.60 or a 4-pack for $2.25?

More practice: Workbook, Lesson 7.16.

LESSON SUMMARY

Estimate, then add and subtract money carefully, lining up decimal points.

UNIT 7 REVIEW

Unit 7 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
7.1	Tenths are 10 equal parts of a whole: $^{3}/_{10}$ = 0.3.
7.2	Wholes go left of the decimal point, tenths go right. Say “and” for the decimal point.
7.3	Split each whole into 10 equal jumps; each jump is 0.1.
7.4	Each place is 10 times the place to its right, even after the decimal point.
7.5	Hundredths use two decimal places: $^{37}/_{100}$ = 0.37; $^{5}/_{100}$ = 0.05.
7.6	Each tenth equals 10 hundredths: 0.3 = 0.30.
7.7	Rename tenths as hundredths, then add.
7.8	Decimals can be written in standard, word, fraction, and expanded form.
7.9	Shade both decimals on the same grid; more shaded is greater.
7.10	Line up decimal points, add zeros, and compare from the left.
7.11	Write the same number of decimal places, then order like whole numbers.
7.12	Change fractions to tenths or hundredths to compare them with decimals.
7.13	Dollars = ones, dimes = tenths, pennies = hundredths. Write money with two decimal places.
7.14	Line up decimal points and add like whole numbers.
7.15	Subtract or count up to find change.
7.16	Estimate, then add and subtract money carefully, lining up decimal points.

UNIT 8

Geometry, Angles, and Line Plots

Lines and rays, classifying shapes, symmetry, measuring and drawing angles, and line plots with fractions.

Geometry is the math of shapes and space. You will name lines and angles, sort shapes by their attributes, find lines of symmetry, measure angles with a protractor, and show measurement data on line plots.

In this unit you will…

- Draw and identify points, lines, segments, rays, and angles.
- Identify parallel and perpendicular lines.
- Classify triangles and quadrilaterals; find lines of symmetry.
- Measure, draw, add, and find unknown angles.
- Make and use line plots with fractional measurements.

MATH IN REAL LIFE

Builders check right angles, designers use symmetry, skateboarders talk in degrees, and scientists use line plots to show measurements.

UNIT VOCABULARY

angle	Two rays with a shared endpoint.
degree	Unit for measuring angles.
parallel	Never meet.
perpendicular	Meet at right angles.
line of symmetry	Folds a figure into matching halves.
line plot	Number line with X marks for data.

Lessons in this unit

Lesson	Title
8.1	Points, Lines, Segments, and Rays
8.2	Parallel and Perpendicular Lines
8.3	Classifying Triangles and Quadrilaterals
8.4	Lines of Symmetry
8.5	Angles: Right, Acute, and Obtuse
8.6	Angles and Turns
8.7	Estimating Angle Measures
8.8	Classifying Shapes by Angles
8.9	Measuring Angles with a Protractor
8.10	Drawing Angles
8.11	Adding Angle Measures
8.12	Finding Unknown Angles
8.13	Line Plots with Fractions
8.14	Solving Problems with Line Plots
8.15	Geometry Review: Attributes of Shapes
8.16	Geometry and Data Problem Solving

UNIT 8 · LESSON 8.1 · 4.G.1

Lesson 8.1: Points, Lines, Segments, and Rays

I can draw and identify points, lines, line segments, and rays.

WORDS TO KNOW

point	An exact location, shown with a dot.
line	A straight path that goes on forever in both directions.
line segment	Part of a line with two endpoints.
ray	Part of a line with one endpoint that goes on forever in one direction.

LET'S EXPLORE

A flashlight beam starts at the flashlight and shines out. A piece of string has two ends. A railroad track seems to go on forever. How are these alike and different?

Geometry has its own vocabulary for straight paths. The difference is in the endpoints.



Name	Endpoints	How to write it
point A	—	A
line segment AB	2	AB with a bar on top
ray CD	1 (C)	CD with an arrow on top (start at the endpoint)
line EF	0	EF with arrows both ways on top

WORKED EXAMPLE · NAME IT

A path starts at point G, goes through H, and keeps going forever. What is it?

1 It has one endpoint (G).

2 It goes on forever in one direction.

Answer: ray GH

KEY IDEA

Line: no endpoints. Ray: one endpoint. Segment: two endpoints.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many endpoints does a ray have?
2. A pencil's edge from tip to eraser is like a ___.
3. Name the figure: two arrows, through points P and Q.

More practice: Workbook, Lesson 8.1.

LESSON SUMMARY

Line: no endpoints. Ray: one endpoint. Segment: two endpoints.

UNIT 8 · LESSON 8.2 · 4.G.1

Lesson 8.2: Parallel and Perpendicular Lines

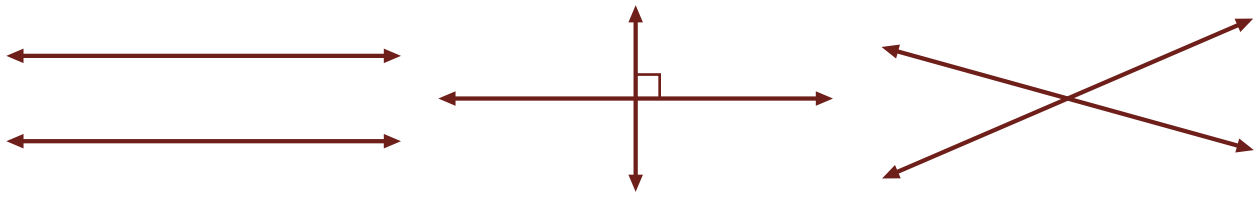
I can identify and draw parallel, perpendicular, and intersecting lines.

WORDS TO KNOW

parallel lines	Lines that never cross and are always the same distance apart.
perpendicular lines	Lines that cross to make square corners (right angles).
intersecting lines	Lines that cross at a point.

LET'S EXPLORE

Look at a notebook page and a window frame. Which lines never meet? Which lines meet in square corners?



parallel perpendicular intersecting

Parallel lines are like railroad tracks: they never meet. Perpendicular lines meet at a square corner, marked with a small square. All perpendicular lines are intersecting, but not all intersecting lines are perpendicular.

WORKED EXAMPLE · FIND THEM IN A SHAPE

In a rectangle, which sides are parallel? Which are perpendicular?



1 Opposite sides never meet: parallel.

2 Sides that meet at corners make right angles: perpendicular.

Answer: Opposite sides are parallel; touching sides are perpendicular.

KEY IDEA

Parallel: never meet. Perpendicular: meet at right angles.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Are the two rails of a ladder parallel or perpendicular?

2. The corner of a book shows ___ lines.

3. Can parallel lines intersect?

More practice: Workbook, Lesson 8.2.

LESSON SUMMARY

Parallel lines never meet. Perpendicular lines meet at right angles.

UNIT 8 · LESSON 8.3 · 4.G.2

Lesson 8.3: Classifying Triangles and Quadrilaterals

I can sort triangles and quadrilaterals by their sides and angles.

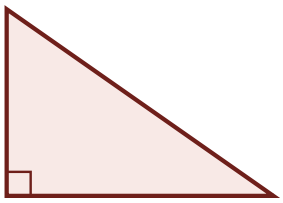
WORDS TO KNOW

right triangle	A triangle with one right angle.
parallelogram	A quadrilateral with two pairs of parallel sides.
trapezoid	A quadrilateral with at least one pair of parallel sides.
rhombus	A parallelogram with four equal sides.

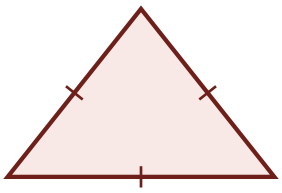
LET'S EXPLORE

A square is a rectangle. Is a rectangle always a square?

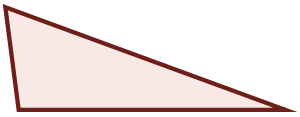
Triangles



right triangle (one right angle)

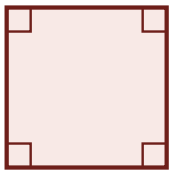


equilateral (all sides equal; all angles acute)



obtuse triangle (one obtuse angle)

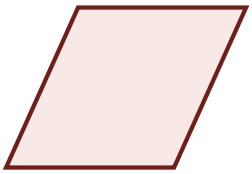
Quadrilaterals



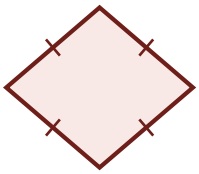
square



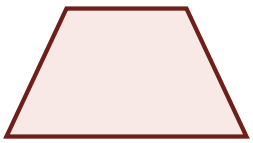
rectangle



parallelogram



rhombus



trapezoid

Shapes can belong to more than one group. A square has 4 right angles (so it is a rectangle), 4 equal sides (so it is a rhombus), and 2 pairs of parallel sides (so it is a parallelogram).

KEY IDEA

Classify by sides (equal? parallel? perpendicular?) and by angles (right, acute, obtuse).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Name a quadrilateral with 4 right angles and 4 equal sides.
2. Does a trapezoid always have parallel sides?
3. Is every rectangle a square?

More practice: Workbook, Lesson 8.3.

LESSON SUMMARY

Sort shapes by their sides and angles. Some shapes have more than one name.

UNIT 8 · LESSON 8.4 · 4.G.3

Lesson 8.4: Lines of Symmetry

I can find and draw lines of symmetry.

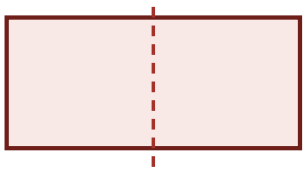
WORDS TO KNOW

line of symmetry	A line that folds a figure into two matching halves.
line-symmetric	Having at least one line of symmetry.

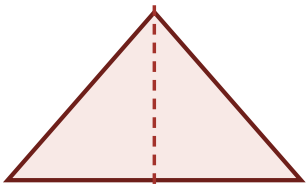
LET'S EXPLORE

Fold a paper heart in half. Do the two halves match exactly?

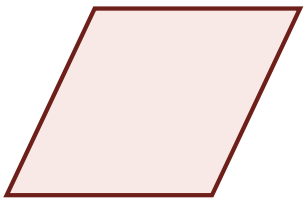
If you can fold a figure so the two halves match exactly, the fold line is a line of symmetry.



rectangle: 2 lines (one shown)



equilateral triangle: 3 lines (one shown)



parallelogram: no lines of symmetry

WORKED EXAMPLE · COUNT LINES OF SYMMETRY

How many lines of symmetry does a square have?

1 Fold top to bottom: match.

2 Fold left to right: match.

3 Fold along both diagonals: match.

Answer: 4

WATCH OUT

A diagonal fold of a rectangle (that isn't a square) does not make matching halves.

KEY IDEA

Imagine folding. If the halves match exactly, it's a line of symmetry.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many lines of symmetry does the letter A have?
2. Does the letter Z have a line of symmetry?
3. How many lines of symmetry does a circle have?

More practice: Workbook, Lesson 8.4.

LESSON SUMMARY

A line of symmetry folds a figure into matching halves.

UNIT 8 · LESSON 8.5 · 4.G.1, 4.G.2

Lesson 8.5: Angles: Right, Acute, and Obtuse

I can identify right, acute, obtuse, and straight angles.

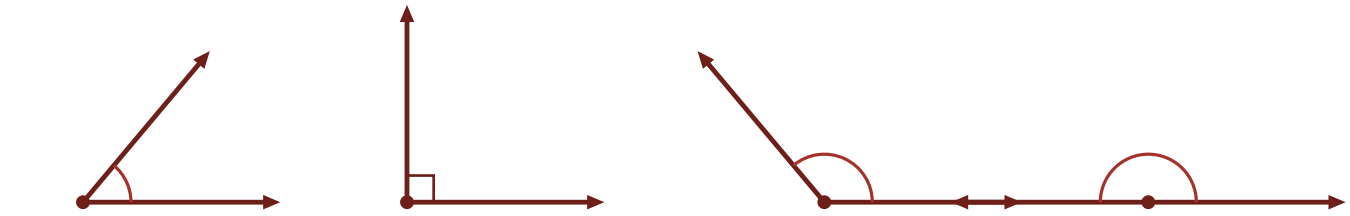
WORDS TO KNOW

angle	Two rays that share an endpoint.
vertex	The shared endpoint of an angle.
right angle	A square corner, 90°.
acute angle	Smaller than a right angle.
obtuse angle	Larger than a right angle but smaller than a straight angle.
straight angle	A straight line, 180°.

LET'S EXPLORE

Open a book a little, then wider, then flat on the table. How does the corner between the covers change?

An angle is made of two rays with a shared endpoint called the vertex. We compare angles to a right angle (a square corner).



acute right obtuse straight

WORKED EXAMPLE · CLASSIFY

A clock shows 3:00. What kind of angle do the hands make?

1 The hands point to 12 and 3.

2 That is a square corner.

Answer: right angle

KEY IDEA

Use the corner of a paper as a right-angle tester: smaller is acute, bigger is obtuse.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. The hands of a clock at 1:00 make what kind of angle?
2. At 6:00?
3. At 4:00?

More practice: Workbook, Lesson 8.5.

LESSON SUMMARY

Acute < right (90°) < obtuse < straight (180°).

UNIT 8 · LESSON 8.6 · 4.MD.5

Lesson 8.6: Angles and Turns

I can explain angles as turns and fractions of a full circle (360°).

WORDS TO KNOW

degree (°) A unit for measuring angles. A full turn is 360°; one degree is $\frac{1}{360}$ of a turn.

LET'S EXPLORE

A skateboarder does a "180" and a "360." What do those numbers mean?

An angle measures a turn. A full turn all the way around is 360 degrees (360°). A half turn is 180°. A quarter turn is 90°: a right angle.

Turn	Fraction of a circle	Degrees
quarter turn	$\frac{1}{4}$	90°
half turn	$\frac{1}{2}$	180°
three-quarter turn	$\frac{3}{4}$	270°
full turn	1	360°

WORKED EXAMPLE · FRACTION TO DEGREES

How many degrees is $\frac{1}{3}$ of a full turn?

1 360 ÷ 3 = 120.

Answer: 120°

KEY IDEA

A full circle is 360°. Find a fraction of 360 to find the angle.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many degrees in a half turn?
2. $\frac{1}{6}$ of a turn is how many degrees?
3. A clock's minute hand moves 15 minutes. How many degrees?

More practice: Workbook, Lesson 8.6.

LESSON SUMMARY

A full turn is 360°. Angles are fractions of a full turn.

UNIT 8 · LESSON 8.7 · 4.MD.5

Lesson 8.7: Estimating Angle Measures

I can estimate the measure of an angle using benchmark angles of 45°, 90°, 135°, and 180°.

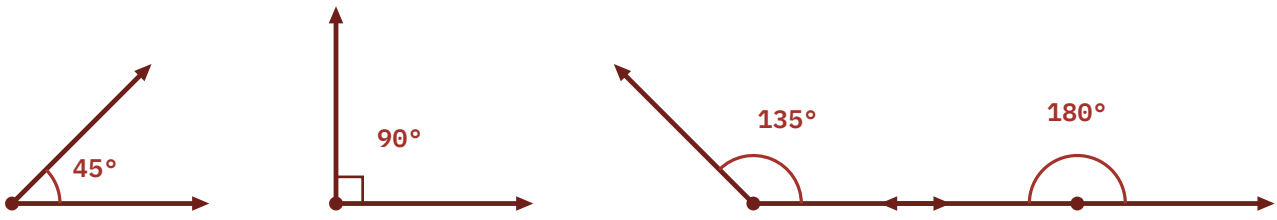
WORDS TO KNOW

benchmark angle	A familiar angle like 45°, 90°, or 180° used to estimate.

LET'S EXPLORE

Without a tool, about how many degrees is the angle a door makes when it is half open?

Use benchmarks: a right angle is 90°. Half of a right angle is 45°. A right angle plus 45° is 135°. A straight line is 180°.



WORKED EXAMPLE · ESTIMATE

An angle is a bit bigger than a right angle but much smaller than a straight angle.

1 Between 90° and 180°.

2 Close to the 135° benchmark? It looks a little less.

Answer: about 120°

KEY IDEA

Compare to 45°, 90°, 135°, and 180° to estimate.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Is a 40° angle closer to 45° or 90°?
2. Estimate an angle a little less than a straight angle.
3. Which benchmark is closest to 100°?

More practice: Workbook, Lesson 8.7.

LESSON SUMMARY

Estimate angles using 45°, 90°, 135°, and 180° benchmarks.

UNIT 8 · LESSON 8.8 · 4.G.2

Lesson 8.8: Classifying Shapes by Angles

I can sort shapes by their angles and tell how many right, acute, or obtuse angles they have.

WORDS TO KNOW

acute triangle	A triangle with three acute angles.
obtuse triangle	A triangle with one obtuse angle.

LET'S EXPLORE

Can a triangle have two obtuse angles?

Look at each angle in a shape and classify it. Triangles are named by their largest angle.

Triangle	Angles
acute triangle	3 acute angles
right triangle	1 right angle, 2 acute
obtuse triangle	1 obtuse angle, 2 acute

WORKED EXAMPLE · TRY TO DRAW IT

Can a triangle have two obtuse angles?

1 If two sides spread wider than a square corner at both ends, they can never meet to close the triangle.

Answer: No

KEY IDEA

Every triangle has at least two acute angles.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many right angles does a rectangle have?
2. A triangle has angles of 60°, 60°, 60°. What kind of triangle?
3. Can a triangle have 2 right angles?

More practice: Workbook, Lesson 8.8.

LESSON SUMMARY

Name triangles by their largest angle. Every triangle has at least two acute angles.

UNIT 8 · LESSON 8.9 · 4.MD.6

Lesson 8.9: Measuring Angles with a Protractor

I can use a protractor to measure angles in whole degrees.

WORDS TO KNOW

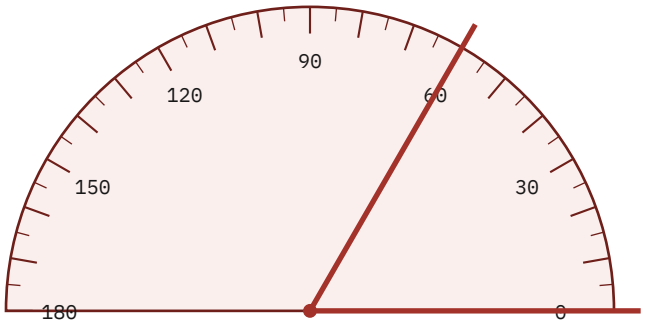
protractor A tool for measuring angles in degrees.

LET'S EXPLORE

How can we find the exact size of an angle?

A protractor is a half circle marked from 0° to 180°.

- Put the center point of the protractor on the vertex.
- Line up the 0° line with one ray.
- Read where the other ray crosses the scale. Use the scale that starts at 0 on your first ray.



This angle measures 60°.

WATCH OUT

Most protractors have two scales. If your angle is obviously acute but you read 120°, you used the wrong scale.

KEY IDEA

Center on the vertex, zero on one ray, read the other ray. Check with an estimate.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. An angle looks a little smaller than a right angle. The protractor shows 80° and 100°. Which is it?
2. Where does the protractor's center go?
3. A straight angle measures ___.

More practice: Workbook, Lesson 8.9.

LESSON SUMMARY

Center on the vertex, line up zero with one ray, and read the other ray.

UNIT 8 · LESSON 8.10 · 4.MD.6

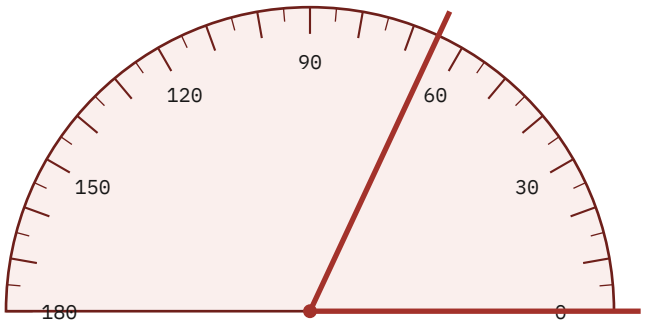
Lesson 8.10: Drawing Angles

I can draw an angle of a given measure using a protractor.

LET'S EXPLORE

How would you draw an angle of exactly 65°?

- Draw a ray. Its endpoint is the vertex.
- Place the protractor's center on the vertex and line up 0° with the ray.
- Make a dot at 65° on the scale.
- Draw a ray from the vertex through the dot. Label the angle.



Drawing a 65° angle.

KEY IDEA

Draw a ray, place the protractor, mark the degree, connect.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Draw a 30° angle.

2. Draw a 120° angle.

3. Draw a 90° angle. What is another name for it?

More practice: Workbook, Lesson 8.10.

LESSON SUMMARY

Draw a ray, place the protractor, mark the degree, and draw the second ray.

UNIT 8 · LESSON 8.11 · 4.MD.7

Lesson 8.11: Adding Angle Measures

I can find the measure of an angle made of smaller angles by adding.

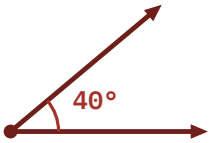
WORDS TO KNOW

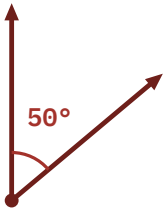
additive	An angle made of non-overlapping parts has a measure equal to the sum of the parts.

LET'S EXPLORE

Two slices of pie meet at the center. One slice makes a 40° angle and the next makes 50°. What angle do they make together?

When an angle is split into parts that don't overlap, the whole angle equals the sum of the parts. 40° + 50° = 90°.





WORKED EXAMPLE · ADD PARTS

An angle is split into parts of 35°, 25°, and 60°. What is the whole angle?

1 35 + 25 + 60 = 120.

Answer: 120°

KEY IDEA

Angle measures are additive: whole = sum of the parts.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 30° + 45°

2. Three 30° angles side by side make ___.

3. Two right angles together make ___.

More practice: Workbook, Lesson 8.11.

LESSON SUMMARY

When angles don't overlap, add their measures to get the whole angle.

UNIT 8 · LESSON 8.12 · 4.MD.7

Lesson 8.12: Finding Unknown Angles

I can find an unknown angle by adding or subtracting.

LET'S EXPLORE

A right angle is split into two parts. One part is 35°. How big is the other part?

WORKED EXAMPLE · SUBTRACT FROM 90°

Right angle: 35° + ? = 90°

1 90 − 35 = 55.

Answer: 55°

WORKED EXAMPLE · SUBTRACT FROM 180°

A straight angle is split into 110° and ?.

1 180 − 110 = 70.

Answer: 70°

KEY IDEA

Write an equation with a letter for the unknown: 35 + x = 90.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Right angle split into 60° and x. Find x.

2. Straight angle split into 45° and x.

3. Full turn split into 100°, 120°, and x.

More practice: Workbook, Lesson 8.12.

LESSON SUMMARY

Use 90°, 180°, or 360° and subtract the known parts to find the unknown angle.

UNIT 8 · LESSON 8.13 · 4.MD.4

Lesson 8.13: Line Plots with Fractions

I can make a line plot to show measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$).

WORDS TO KNOW

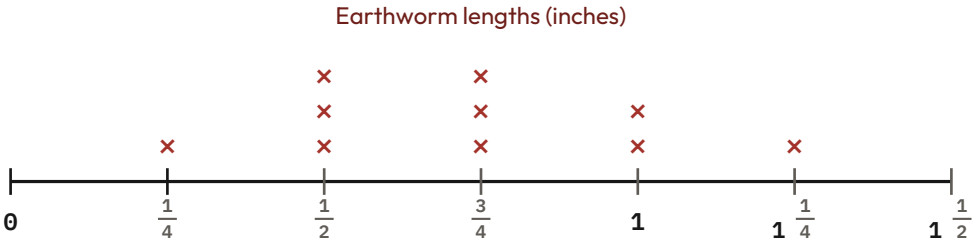
line plot	A number line with X marks that shows how many times each value appears.
data	Information collected, such as measurements.

LET'S EXPLORE

Ellie measured 10 earthworms after it rained. How can she show all the lengths in one picture?

A line plot shows data on a number line. Each X stands for one measurement.

Worm lengths (inches)
$\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}, 1, \frac{5}{4}, \frac{1}{2}, 1$



- Draw a number line that covers the smallest to the largest value.
- Mark it in the right fraction (here, fourths).
- Put an X above the value for each measurement.
- Give it a title and units.

KEY IDEA

Each X is one data value. Tall stacks show the most common values.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many worms were $3/4$ inch?
2. What was the longest worm?
3. How many worms were longer than $1/2$ inch?

More practice: Workbook, Lesson 8.13.

LESSON SUMMARY

A line plot shows each measurement as an X above a number line marked in fractions.

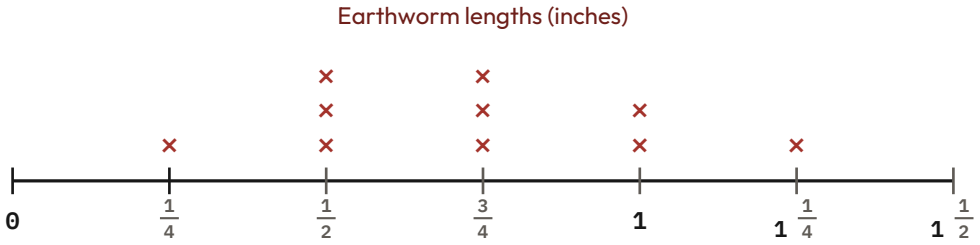
UNIT 8 · LESSON 8.14 · 4.MD.4

Lesson 8.14: Solving Problems with Line Plots

I can add and subtract fractions to answer questions about a line plot.

LET'S EXPLORE

What is the difference between the longest and shortest earthworm? What is the total length of all the $3/4$-inch worms?



WORKED EXAMPLE · DIFFERENCE

Longest − shortest

1 Longest: $1\ 1/4 = 5/4$. Shortest: $1/4$.

2 $5/4 - 1/4 = 4/4 = 1$.

Answer: 1 inch

WORKED EXAMPLE · TOTAL OF A GROUP

Total length of the $3/4$-inch worms

1 There are 3 of them: $3 \times 3/4 = 9/4 = 2\ 1/4$.

Answer: $2\frac{1}{4}$ inches

KEY IDEA

Use the line plot to find the values, then add, subtract, or multiply fractions.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Total length of the $\frac{1}{2}$-inch worms?
2. How many worms in all?
3. How much longer is the longest worm than a $\frac{1}{2}$-inch worm?

More practice: Workbook, Lesson 8.14.

LESSON SUMMARY

Read values from the line plot, then compute with fractions.

UNIT 8 · LESSON 8.15 · 4.G.1, 4.G.2, 4.G.3

Lesson 8.15: Geometry Review: Attributes of Shapes

I can describe shapes using lines, angles, and symmetry.

WORDS TO KNOW

attribute	A feature of a shape, such as number of sides or types of angles.

LET'S EXPLORE

Play “Guess My Shape”: I have 4 sides, 2 pairs of parallel sides, no right angles, and 4 equal sides. What am I?

Describe shapes by their attributes: number of sides, parallel and perpendicular sides, types of angles, equal sides, and lines of symmetry.

WORKED EXAMPLE · GUESS MY SHAPE

4 sides, 2 pairs of parallel sides, no right angles, 4 equal sides.

1 Parallelogram with equal sides.

2 No right angles, so not a square.

Answer: rhombus

KEY IDEA

Use exact vocabulary: parallel, perpendicular, acute, right, obtuse, symmetric.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. I have 3 sides and one right angle. What am I?
2. I have 4 right angles and 2 lines of symmetry, and my sides are not all equal.
3. I have exactly one pair of parallel sides.

More practice: Workbook, Lesson 8.15.

LESSON SUMMARY

Describe shapes by sides, parallel and perpendicular lines, angles, and symmetry.

UNIT 8 · LESSON 8.16 · 4.MD.4, 4.MD.7, 4.G.2

Lesson 8.16: Geometry and Data Problem Solving

I can solve problems with angles, shapes, and line plots.

LET'S EXPLORE

Design a kite: it must have one line of symmetry and two pairs of equal sides. What angles might it have?

Today you'll use everything from this unit to solve design and data problems.

WORKED EXAMPLE · KITE ANGLES

A kite has angles of 110°, 70°, 110°, and x°. The four angles of a quadrilateral add to 360°.

1 110 + 70 + 110 = 290.

2 360 − 290 = 70.

Answer: x = 70°

KEY IDEA

Use angle facts (90°, 180°, 360°), shape attributes, and line plot data together.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Three angles of a quadrilateral are 90°, 90°, and 120°. Find the fourth.

2. How many lines of symmetry does a kite have?

3. Can a trapezoid have a right angle?

More practice: Workbook, Lesson 8.16.

LESSON SUMMARY

Use angle totals, shape attributes, and line plots to solve problems.

UNIT 8 REVIEW

Unit 8 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
8.1	Line: no endpoints. Ray: one endpoint. Segment: two endpoints.
8.2	Parallel lines never meet. Perpendicular lines meet at right angles.
8.3	Sort shapes by their sides and angles. Some shapes have more than one name.
8.4	A line of symmetry folds a figure into matching halves.
8.5	Acute < right (90°) < obtuse < straight (180°).
8.6	A full turn is 360°. Angles are fractions of a full turn.
8.7	Estimate angles using 45°, 90°, 135°, and 180° benchmarks.
8.8	Name triangles by their largest angle. Every triangle has at least two acute angles.
8.9	Center on the vertex, line up zero with one ray, and read the other ray.
8.10	Draw a ray, place the protractor, mark the degree, and draw the second ray.
8.11	When angles don't overlap, add their measures to get the whole angle.
8.12	Use 90°, 180°, or 360° and subtract the known parts to find the unknown angle.
8.13	A line plot shows each measurement as an X above a number line marked in fractions.
8.14	Read values from the line plot, then compute with fractions.
8.15	Describe shapes by sides, parallel and perpendicular lines, angles, and symmetry.
8.16	Use angle totals, shape attributes, and line plots to solve problems.

UNIT 9

Measurement, Area, and Perimeter

Perimeter and area of rectangles, combined shapes, unit conversions, and elapsed time.

Measurement connects math to everyday life. You will find perimeter and area, solve for unknown sides, convert units of length, weight, capacity, and time, and solve real problems with measurements.

In this unit you will...

- Find perimeter and area with formulas.
- Find unknown side lengths.
- Find the area of combined rectangles.
- Convert customary, metric, and time units.
- Solve elapsed time and multi-step measurement problems.

MATH IN REAL LIFE

Carpenters, cooks, gardeners, nurses, and pilots all measure and convert units every day.

UNIT VOCABULARY

perimeter	Distance around a figure.
area	Square units that cover a surface.
convert	Change to a different unit.
elapsed time	Time that passes between start and end.

Lessons in this unit

Lesson	Title
9.1	Perimeter
9.2	Area
9.3	Area and Perimeter Formulas

Lesson	Title
9.4	Same Perimeter, Different Area
9.5	Finding an Unknown Side
9.6	Area of Combined Rectangles
9.7	Area and Perimeter Word Problems
9.8	Area or Perimeter?
9.9	Customary Length
9.10	Customary Weight and Capacity
9.11	Metric Units
9.12	Time Conversions
9.13	Elapsed Time
9.14	Measurement Problems with Fractions and Decimals
9.15	Multi-Step Measurement Problems
9.16	Measurement Problem Solving

UNIT 9 · LESSON 9.1 · 4.MD.3

Lesson 9.1: Perimeter

I can find the perimeter of a shape by adding the lengths of its sides.

WORDS TO KNOW

perimeter	The distance around a figure.

LET'S EXPLORE

Dad is putting a fence around a garden that is 12 feet long and 8 feet wide. How much fencing does he need?

Perimeter is the distance around a shape. Add the lengths of all the sides.



A garden 12 ft by 8 ft.

WORKED EXAMPLE · ADD THE SIDES

Find the perimeter of the garden.

1 A rectangle has two lengths and two widths.

2 12 + 8 + 12 + 8 = 40.

Answer: 40 feet of fencing

WORKED EXAMPLE · ANY POLYGON

A triangle has sides 7 cm, 9 cm, and 5 cm. Find the perimeter.

1 7 + 9 + 5 = 21.

Answer: 21 cm

KEY IDEA

Perimeter: add all the side lengths. The answer is in units (cm, ft), not square units.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rectangle 10 m by 4 m: perimeter?
2. Square with 6-inch sides: perimeter?
3. Pentagon with all sides 3 ft: perimeter?

More practice: Workbook, Lesson 9.1.

LESSON SUMMARY

Perimeter is the distance around: add all the side lengths.

UNIT 9 · LESSON 9.2 · 4.MD.3

Lesson 9.2: Area

I can find the area of a rectangle by counting square units and by multiplying.

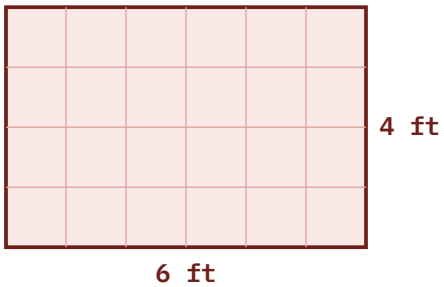
WORDS TO KNOW

area	The number of square units that cover a flat surface.
square unit	A square 1 unit long on each side, like a square inch or square meter.

LET'S EXPLORE

How many 1-foot square tiles cover a floor that is 6 feet long and 4 feet wide?

Area measures how much surface a shape covers. We count square units.



4 rows of 6 squares = 24 square feet.

The tiles form an array: 4 rows of 6. Multiplication is faster than counting: 6 × 4 = 24 square feet.

WORKED EXAMPLE · MULTIPLY

Find the area of a rectangle 9 m long and 7 m wide.

1 Area = length × width.

2 $9 \times 7 = 63$.

Answer: 63 square meters

WATCH OUT

Area is in square units (sq ft, sq m). Perimeter is in plain units (ft, m).

KEY IDEA

Area of a rectangle = length × width.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rectangle 8 in. by 5 in.: area?
2. Square with 9 cm sides: area?
3. A room 12 ft by 10 ft: area?

More practice: Workbook, Lesson 9.2.

LESSON SUMMARY

Area counts square units: length × width.

UNIT 9 · LESSON 9.3 · 4.MD.3

Lesson 9.3: Area and Perimeter Formulas

I can use the formulas $P = 2 \times l + 2 \times w$ and $A = l \times w$.

WORDS TO KNOW

formula	A rule written with symbols, like A = l × w.

LET'S EXPLORE

Is there a shortcut for adding 2 lengths and 2 widths?

Mathematicians write rules as formulas. For a rectangle with length l and width w:

Measure	Formula	Example: 9 × 5
Perimeter	P = 2 × l + 2 × w	2 × 9 + 2 × 5 = 28
Area	A = l × w	9 × 5 = 45

WORKED EXAMPLE · USE BOTH FORMULAS

A garden is 16 ft by 9 ft. Find P and A.

1 P = 2 × 16 + 2 × 9 = 32 + 18 = 50.

2 A = 16 × 9 = 144.

Answer: P = 50 ft, A = 144 sq ft

KEY IDEA

P = 2 × l + 2 × w. A = l × w.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 13 cm by 6 cm

2. 20 m by 15 m

3. Square, 11 in. sides

More practice: Workbook, Lesson 9.3.

LESSON SUMMARY

P = 2 × l + 2 × w and A = l × w.

UNIT 9 · LESSON 9.4 · 4.MD.3

Lesson 9.4: Same Perimeter, Different Area

I can show that rectangles with the same perimeter can have different areas.

LET'S EXPLORE

You have 24 feet of fence for a rabbit pen. Which rectangle gives the rabbit the most room?

Length	Width	Perimeter	Area
1	11	24 ft	11 sq ft
2	10	24 ft	20 sq ft
3	9	24 ft	27 sq ft
4	8	24 ft	32 sq ft
5	7	24 ft	35 sq ft
6	6	24 ft	36 sq ft

All these pens use 24 feet of fence, but the areas are very different. The 6 × 6 square gives the most room: 36 square feet.

KEY IDEA

For a fixed perimeter, the closer the rectangle is to a square, the bigger its area.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Two rectangles have perimeter 16. One is 7 × 1, one is 4 × 4. Which has more area?
2. A rectangle has area 12. Name two possible perimeters.
3. Same area 36: which has the smaller perimeter, 9 × 4 or 6 × 6?

More practice: Workbook, Lesson 9.4.

LESSON SUMMARY

Rectangles with the same perimeter can have different areas; squares give the most area.

UNIT 9 · LESSON 9.5 · 4.MD.3

Lesson 9.5: Finding an Unknown Side

I can find a missing side length when I know the area or perimeter.

LET'S EXPLORE

A rectangle has an area of 48 square feet and a length of 8 feet. How wide is it?

WORKED EXAMPLE · FROM AREA

A = 48 sq ft, l = 8 ft. Find w.

1 $8 \times w = 48$.

2 $48 \div 8 = 6$.

Answer: 6 feet

WORKED EXAMPLE · FROM PERIMETER

P = 30 m, l = 9 m. Find w.

1 Two lengths: $2 \times 9 = 18$.

2 $30 - 18 = 12$ for the two widths.

3 $12 \div 2 = 6$.

Answer: 6 m

KEY IDEA

Write the formula with a letter for the unknown, then use the inverse operation.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A = 56 sq in., l = 7 in. Find w.
2. P = 26 cm, l = 8 cm. Find w.
3. Square with A = 64 sq m. Side?

More practice: Workbook, Lesson 9.5.

LESSON SUMMARY

Use the formula and the inverse operation to find a missing side.

UNIT 9 · LESSON 9.6 · 4.MD.3

Lesson 9.6: Area of Combined Rectangles

I can find the area of a shape made of rectangles by breaking it apart.

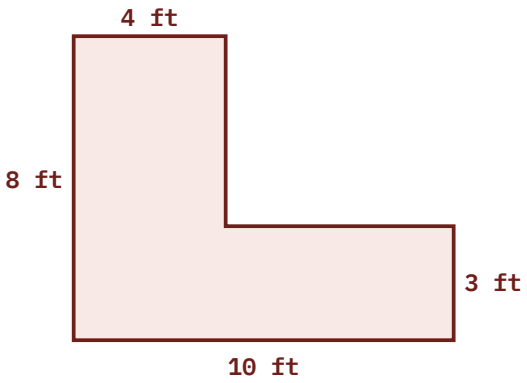
WORDS TO KNOW

composite figure — A shape made by joining simpler shapes.

LET'S EXPLORE

An L-shaped room needs new carpet. How can we find its area?

Break the shape into rectangles, find each area, and add.



An L-shaped room

WORKED EXAMPLE · BREAK IT APART

Find the area of the L-shape.

1. Bottom rectangle: 10 × 3 = 30.
2. Top rectangle: 4 × (8 − 3) = 4 × 5 = 20.
3. Add: 30 + 20 = 50.

Answer: 50 sq ft

KEY IDEA

Split into rectangles, find each area, add. Find missing side lengths by subtracting.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Two rectangles: 5 × 4 and 3 × 2. Total area?
2. A 10 × 10 square with a 2 × 2 corner cut out. Area?
3. L-shape: bottom 8 × 2, top 3 × 4. Area?

More practice: Workbook, Lesson 9.6.

LESSON SUMMARY

Break a combined shape into rectangles, find each area, and add.

UNIT 9 · LESSON 9.7 · 4.MD.3

Lesson 9.7: Area and Perimeter Word Problems

I can solve real-world problems with area and perimeter.

LET'S EXPLORE

Tile costs $3 per square foot. How much does it cost to tile a 9-foot by 6-foot bathroom?

WORKED EXAMPLE · AREA, THEN COST

Tile a 9 ft × 6 ft room at $3 per sq ft.

1 Area: 9 × 6 = 54 sq ft.

2 Cost: 54 × 3 = 162.

Answer: $162

WORKED EXAMPLE · PERIMETER, THEN COST

Border trim costs $2 per foot. Trim the same room.

1 Perimeter: 2 × 9 + 2 × 6 = 30 ft.

2 Cost: 30 × 2 = 60.

Answer: $60

KEY IDEA

Covering → area. Going around → perimeter.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Carpet for a 12 ft × 10 ft room at $4 per sq ft?
2. Fence around a 20 m × 15 m field at $5 per meter?

3. How many 1-ft tiles for a 7 ft × 8 ft floor?

More practice: Workbook, Lesson 9.7.

LESSON SUMMARY

Decide: covering (area) or going around (perimeter)? Then compute.

UNIT 9 · LESSON 9.8 · 4.MD.3

Lesson 9.8: Area or Perimeter?

I can decide whether a problem needs area or perimeter and explain why.

LET'S EXPLORE

Are you finding area or perimeter when you put a frame on a picture? When you paint the picture?

Situation	Area or perimeter?
fence around a yard	perimeter
carpet for a room	area
trim around a door	perimeter
paint for a wall	area
lace around a pillow	perimeter
grass seed for a lawn	area

KEY IDEA

Covering a surface → area (square units). Edge or border → perimeter (units).

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. A border of tape around a table top
2. Wrapping paper to cover a box's top
3. A path around a pond

More practice: Workbook, Lesson 9.8.

LESSON SUMMARY

Covering → area. Edge or border → perimeter.

UNIT 9 · LESSON 9.9 · 4.MD.1

Lesson 9.9: Customary Length

I can convert between inches, feet, yards, and miles.

WORDS TO KNOW

convert	To change a measurement to a different unit.
customary units	Units used in the U.S.: inches, feet, yards, miles, ounces, pounds, cups, gallons.

LET'S EXPLORE

Lillie is 4 feet 6 inches tall. How many inches tall is she?

Unit	Equals
1 foot (ft)	12 inches (in.)
1 yard (yd)	3 feet = 36 inches
1 mile (mi)	5,280 feet

To change a larger unit to a smaller unit, multiply. There are more of the small units.

WORKED EXAMPLE · FEET AND INCHES TO INCHES

4 ft 6 in. = ___ in.

1 4 × 12 = 48 inches.

2 48 + 6 = 54.

Answer: 54 inches

WORKED EXAMPLE · MAKE A TABLE

Yards to feet

1 Multiply by 3: 1 → 3, 2 → 6, 3 → 9, 4 → 12.

Answer: yards × 3 = feet

KEY IDEA

Big unit → small unit: multiply by how many small units are in one big unit.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 5 ft = ___ in.

2. 7 yd = ___ ft

3. 2 mi = ___ ft

More practice: Workbook, Lesson 9.9.

LESSON SUMMARY

1 ft = 12 in., 1 yd = 3 ft, 1 mi = 5,280 ft. Big to small: multiply.

UNIT 9 · LESSON 9.10 · 4.MD.1

Lesson 9.10: Customary Weight and Capacity

I can convert ounces and pounds, and cups, pints, quarts, and gallons.

WORDS TO KNOW

weight	How heavy something is.
capacity	How much a container holds.

LET'S EXPLORE

A recipe needs 3 quarts of water. How many cups is that?

Weight	Capacity
1 pound (lb) = 16 ounces (oz)	1 pint (pt) = 2 cups (c)
	1 quart (qt) = 2 pints = 4 cups
	1 gallon (gal) = 4 quarts = 16 cups

WORKED EXAMPLE · QUARTS TO CUPS

3 qt = ___ c

1 1 qt = 4 c.

2 3 × 4 = 12.

Answer: 12 cups

WORKED EXAMPLE · POUNDS TO OUNCES

2 lb 5 oz = ___ oz

1 2 × 16 = 32.

2 32 + 5 = 37.

Answer: 37 oz

KEY IDEA

Learn the facts: 16 oz = 1 lb; 2 c = 1 pt; 2 pt = 1 qt; 4 qt = 1 gal.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4 lb = ___ oz
2. 2 gal = ___ qt
3. 5 pt = ___ c

More practice: Workbook, Lesson 9.10.

LESSON SUMMARY

16 oz = 1 lb; 2 c = 1 pt; 2 pt = 1 qt; 4 qt = 1 gal.

UNIT 9 · LESSON 9.11 · 4.MD.1

Lesson 9.11: Metric Units

I can convert metric units of length, mass, and capacity.

WORDS TO KNOW

metric units	Units based on 10s: meters, grams, liters.
mass	The amount of matter in an object, measured in grams and kilograms.

LET'S EXPLORE

A race is 5 kilometers long. How many meters is that?

Length	Mass	Capacity
1 cm = 10 mm	1 kg = 1,000 g	1 L = 1,000 mL
1 m = 100 cm		
1 km = 1,000 m		

Metric units use place value: kilo- means 1,000, centi- means one hundredth, milli- means one thousandth.

WORKED EXAMPLE · KILOMETERS TO METERS

5 km = ___ m

1 1 km = 1,000 m.

2 5 × 1,000 = 5,000.

Answer: 5,000 m

KEY IDEA

Metric conversions multiply by 10, 100, or 1,000.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3 m = ___ cm

2. 4 kg = ___ g

3. 2 L = ___ mL

More practice: Workbook, Lesson 9.11.

LESSON SUMMARY

Metric units convert by 10, 100, or 1,000: 1 m = 100 cm, 1 km = 1,000 m, 1 kg = 1,000 g, 1 L = 1,000 mL.

UNIT 9 · LESSON 9.12 · 4.MD.1

Lesson 9.12: Time Conversions

I can convert hours, minutes, seconds, days, and weeks.

LET'S EXPLORE

Ellie's soccer practice is 90 minutes. Is that more or less than 2 hours?

Unit	Equals
1 minute	60 seconds
1 hour	60 minutes
1 day	24 hours
1 week	7 days
1 year	12 months = 52 weeks (about) = 365 days

WORKED EXAMPLE · HOURS AND MINUTES

2 hours = ___ minutes

1 2 × 60 = 120.

Answer: 120 minutes, so 90 minutes is less

WORKED EXAMPLE · MIXED UNITS

1 hr 25 min = ___ min

1 60 + 25 = 85.

Answer: 85 minutes

KEY IDEA

60 seconds = 1 minute; 60 minutes = 1 hour; 24 hours = 1 day; 7 days = 1 week.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 3 hr = ___ min
2. 4 min = ___ sec
3. 2 weeks = ___ days

More practice: Workbook, Lesson 9.12.

LESSON SUMMARY

60 sec = 1 min; 60 min = 1 hr; 24 hr = 1 day; 7 days = 1 week.

UNIT 9 · LESSON 9.13 · 4.MD.2

Lesson 9.13: Elapsed Time

I can find how much time has passed and find start or end times.

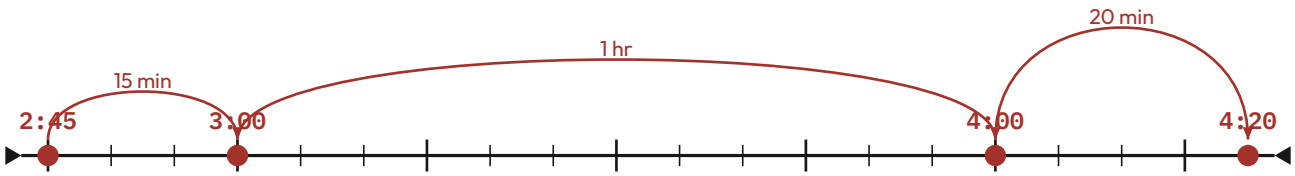
WORDS TO KNOW

elapsed time	The amount of time that passes from the start to the end of an event.

LET'S EXPLORE

A movie starts at 2:45 p.m. and ends at 4:20 p.m. How long is the movie?

Count up from the start time in friendly jumps: to the next hour, then whole hours, then minutes.



WORKED EXAMPLE · COUNT UP

2:45 to 4:20

1. 2:45 → 3:00 is 15 minutes.
2. 3:00 → 4:00 is 1 hour.
3. 4:00 → 4:20 is 20 minutes.
4. Total: 1 hr 35 min.

Answer: 1 hour 35 minutes

WORKED EXAMPLE · FIND THE END TIME

Start 9:30 a.m., lasts 2 hr 45 min.

1. 9:30 + 2 hr = 11:30.
2. 11:30 + 45 min = 12:15 p.m.

Answer: 12:15 p.m.

KEY IDEA

Jump to the next hour, then by hours, then by minutes. Watch for a.m. and p.m.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 1:20 p.m. to 3:50 p.m.
2. Start 10:15, 1 hr 50 min later?
3. Ends 5:00 p.m., lasted 2 hr 20 min. Start?

More practice: Workbook, Lesson 9.13.

LESSON SUMMARY

Count up from the start time to the next hour, then by hours, then minutes.

UNIT 9 · LESSON 9.14 · 4.MD.2

Lesson 9.14: Measurement Problems with Fractions and Decimals

I can solve measurement problems that use fractions and decimals.

LET'S EXPLORE

A recipe needs $3/4$ cup of milk per batch. How much for 3 batches? How many pints is that?

WORKED EXAMPLE · FRACTIONS

3 batches of $3/4$ cup

1 $3 \times 3/4 = 9/4 = 2\,1/4$ cups.

2 2 cups = 1 pint, so it is a little more than 1 pint.

Answer: $2\,1/4$ cups

WORKED EXAMPLE · DECIMALS

A bottle holds 1.5 L. How many mL?

1 1 L = 1,000 mL. 1.5 L = 1,000 + 500.

Answer: 1,500 mL

KEY IDEA

Convert to the same unit, then compute with fractions or decimals.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{1}{2}$ ft = ___ in.
2. $\frac{1}{4}$ hour = ___ minutes
3. 2.5 m = ___ cm

More practice: Workbook, Lesson 9.14.

LESSON SUMMARY

Convert to the same unit, then compute with fractions or decimals.

UNIT 9 · LESSON 9.15 · 4.MD.2

Lesson 9.15: Multi-Step Measurement Problems

I can solve multi-step problems with measurement and conversions.

LET'S EXPLORE

Lillie needs 3 pieces of rope, each 2 feet 4 inches long. She has a 9-foot rope. Is it enough?

WORKED EXAMPLE · CONVERT, MULTIPLY, COMPARE

3 pieces of 2 ft 4 in. from 9 ft

1 2 ft 4 in. = 28 in.

2 3 × 28 = 84 in.

3 9 ft = 108 in.

4 108 − 84 = 24 in. left.

Answer: Yes, with 2 feet left

KEY IDEA

Change to the smallest unit first. Then solve step by step.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4 boards of 1 ft 6 in. Total in feet?
2. A 2-gallon jug fills how many 1-cup glasses?
3. 3 movies of 1 hr 40 min. Total time?

More practice: Workbook, Lesson 9.15.

LESSON SUMMARY

Convert to one unit first, then solve step by step.

UNIT 9 · LESSON 9.16 · 4.MD.1, 4.MD.2, 4.MD.3

Lesson 9.16: Measurement Problem Solving

I can solve a variety of measurement, area, and perimeter problems.

LET'S EXPLORE

Design a garden: 20 feet of fence, the biggest area possible, and a 2-hour planting job starting at 9:45 a.m. What do you need?

WORKED EXAMPLE · GARDEN DESIGN

20 ft of fence, biggest area.

1 Perimeter 20: try 5 × 5 = 25 sq ft (largest).

2 Planting: 9:45 + 2 hr = 11:45 a.m.

Answer: 5 ft × 5 ft garden; done at 11:45 a.m.

KEY IDEA

Use formulas, conversions, and elapsed time together.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Area of a 15 ft × 4 ft garden
2. 6 qt = ___ gal ___ qt
3. Start 1:50, end 4:05. Elapsed?

More practice: Workbook, Lesson 9.16.

LESSON SUMMARY

Combine formulas, conversions, and elapsed time to solve real problems.

UNIT 9 REVIEW

Unit 9 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
9.1	Perimeter is the distance around: add all the side lengths.
9.2	Area counts square units: length × width.
9.3	P = 2 × l + 2 × w and A = l × w.
9.4	Rectangles with the same perimeter can have different areas; squares give the most area.
9.5	Use the formula and the inverse operation to find a missing side.
9.6	Break a combined shape into rectangles, find each area, and add.
9.7	Decide: covering (area) or going around (perimeter)? Then compute.
9.8	Covering → area. Edge or border → perimeter.
9.9	1 ft = 12 in., 1 yd = 3 ft, 1 mi = 5,280 ft. Big to small: multiply.
9.10	16 oz = 1 lb; 2 c = 1 pt; 2 pt = 1 qt; 4 qt = 1 gal.
9.11	Metric units convert by 10, 100, or 1,000: 1 m = 100 cm, 1 km = 1,000 m, 1 kg = 1,000 g, 1 L = 1,000 mL.
9.12	60 sec = 1 min; 60 min = 1 hr; 24 hr = 1 day; 7 days = 1 week.
9.13	Count up from the start time to the next hour, then by hours, then minutes.
9.14	Convert to the same unit, then compute with fractions or decimals.
9.15	Convert to one unit first, then solve step by step.
9.16	Combine formulas, conversions, and elapsed time to solve real problems.

UNIT 10

Year in Review

Pulling the whole year together: numbers, operations, fractions, decimals, geometry, and measurement.

In the last four weeks you will review every big idea from Grade 4, practice the skills that need it most, and show what you have learned on the Year-End Test.

In this unit you will...

- Review place value and all four operations.
- Review factors, multiples, and patterns.
- Review fractions and decimals.
- Review geometry, measurement, and data.
- Show your learning on the Year-End Test.

MATH IN REAL LIFE

Everything you learned this year is a tool you will use in Grade 5 and in real life.

UNIT VOCABULARY

review	To study again so you remember.
strategy	A plan for solving a problem.

Lessons in this unit

Lesson	Title
10.1	Review: Place Value and Rounding
10.2	Review: Adding and Subtracting
10.3	Review: Multiplying
10.4	Review: Dividing
10.5	Review: Factors and Multiples

Lesson	Title
10.6	Review: Patterns
10.7	Review: Multi-Step Word Problems
10.8	Review: Whole-Number Challenge
10.9	Review: Equivalent Fractions
10.10	Review: Comparing Fractions
10.11	Review: Fraction Operations
10.12	Review: Decimals
10.13	Review: Geometry and Angles
10.14	Review: Area, Perimeter, and Conversions
10.15	Review: Data and Line Plots
10.16	Celebrating a Year of Math

UNIT 10 · LESSON 10.1 · 4.NBT.1–3

Lesson 10.1: Review: Place Value and Rounding

I can read, write, compare, and round numbers to one million.

LET'S EXPLORE

What is the greatest six-digit number you can write? Round it to the nearest hundred thousand.

REMEMBER

Each place is 10 times the place to its right. To round, look one place to the right: 5 or more rounds up.

WORKED EXAMPLE · ROUND AND COMPARE

Round 638,452 to the nearest ten thousand. Is it greater than 638,524?

1 Ten thousands digit 3; look right: 8 → round up.

2 640,000.

3 638,452 < 638,524 (hundreds: 4 < 5).

Answer: 640,000; no

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Write 407,019 in words.

2. Round 85,349 to the nearest thousand.

3. Value of the 6 in 961,000

More practice: Workbook, Lesson 10.1.

LESSON SUMMARY

Place value, comparing, and rounding work the same way for every size of number.

UNIT 10 · LESSON 10.2 · 4.NBT.4

Lesson 10.2: Review: Adding and Subtracting

I can add and subtract multi-digit numbers fluently and check with estimation.

LET'S EXPLORE

A truck carried 12,486 pounds on Monday and 9,758 pounds on Tuesday. How much more on Monday? How much in all?

REMEMBER

Line up places. Add or subtract from right to left, regrouping as needed. Estimate to check.

WORKED EXAMPLE · BOTH OPERATIONS

12,486 and 9,758

1 Sum: 12,486 + 9,758 = 22,244.

2 Difference: 12,486 − 9,758 = 2,728.

3 Check: 12,000 + 10,000 = 22,000; 12,000 − 10,000 = 2,000.

Answer: 22,244 in all; 2,728 more

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 46,327 + 38,895

2. 70,000 − 26,418

3. Estimate 6,812 − 2,390.

More practice: Workbook, Lesson 10.2.

LESSON SUMMARY

Line up places, regroup carefully, and estimate to check.

UNIT 10 · LESSON 10.3 · 4.NBT.5

Lesson 10.3: Review: Multiplying

I can multiply up to four digits by one digit and two digits by two digits.

LET'S EXPLORE

A stadium has 48 sections with 50 seats each. How many seats is that? How could you find it quickly?

REMEMBER

Use area models, partial products, or the standard algorithm. Estimate to check.

WORKED EXAMPLE · TWO STRATEGIES

Find 6 × 2,347 and 34 × 27.

1 6 × 2,347: 6 × 2,000 + 6 × 300 + 6 × 40 + 6 × 7 = 12,000 + 1,800 + 240 + 42 = 14,082.

2 34 × 27: 600 + 80 + 210 + 28 = 918.

Answer: 14,082 and 918

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 7 × 3,058

2. 45 × 23

3. 80 × 60

More practice: Workbook, Lesson 10.3.

LESSON SUMMARY

Break factors into place-value parts, multiply each part, and add.

UNIT 10 · LESSON 10.4 · 4.NBT.6, 4.OA.3

Lesson 10.4: Review: Dividing

I can divide up to four digits by one digit and interpret remainders.

LET'S EXPLORE

A farmer packs 1,350 eggs into cartons of 6. How many full cartons? Any left over?

REMEMBER

Divide, multiply, subtract, bring down. Check: quotient × divisor + remainder = dividend. Decide what the remainder means.

WORKED EXAMPLE · LONG DIVISION

1,350 ÷ 6

1 13 ÷ 6 = 2 R1; 15 ÷ 6 = 2 R3; 30 ÷ 6 = 5.

2 Check: 225 × 6 = 1,350.

Answer: 225 cartons, none left

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 2,184 ÷ 7
2. 605 ÷ 4
3. 50 people, 8 per table. How many tables?

More practice: Workbook, Lesson 10.4.

LESSON SUMMARY

Divide step by step, check by multiplying, and interpret the remainder.

UNIT 10 · LESSON 10.5 · 4.OA.4

Lesson 10.5: Review: Factors and Multiples

I can find factors, multiples, and tell prime from composite.

LET'S EXPLORE

Is 91 prime? (Many people think so!)

REMEMBER

Factor pairs multiply to the number. Prime: exactly 2 factors. Multiples come from skip counting.

WORKED EXAMPLE · PRIME OR COMPOSITE

Is 91 prime?

1 Test small factors: 2 no, 3 no (9 + 1 = 10), 5 no, 7: 7 × 13 = 91.

Answer: Composite

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Factors of 36
2. Is 53 prime?
3. First common multiple of 6 and 8

More practice: Workbook, Lesson 10.5.

LESSON SUMMARY

Factors divide evenly; multiples come from skip counting; primes have exactly two factors.

UNIT 10 · LESSON 10.6 · 4.OA.5

Lesson 10.6: Review: Patterns

I can extend number and shape patterns and describe their rules and features.

LET'S EXPLORE

Rule: start at 5, add 5. Will 101 be in the pattern?

REMEMBER

Find the change between terms. Look for features like odd/even. For repeating patterns, use division.

WORKED EXAMPLE · FEATURES

Start at 5, add 5: 5, 10, 15, 20...

1 All terms are multiples of 5.

2 They end in 5 or 0, alternating.

Answer: 101 is not a term

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Rule: start at 3, multiply by 2. First 5 terms.

2. Next: 99, 88, 77, ___

3. Core: A B B C. 18th letter?

More practice: Workbook, Lesson 10.6.

LESSON SUMMARY

Find the rule, then look for hidden features.

UNIT 10 · LESSON 10.7 · 4.OA.3

Lesson 10.7: Review: Multi-Step Word Problems

I can solve multi-step word problems with all four operations.

LET'S EXPLORE

A class of 28 students goes on a trip. Tickets are $9 each. Buses hold 12 students. How many buses, and how much do tickets cost?

REMEMBER

Find the hidden steps. Write an equation for each, with a letter for the unknown. Check reasonableness.

WORKED EXAMPLE · TWO QUESTIONS

28 students, $9 tickets, buses of 12

1 Tickets: 28 × 9 = 252.

2 Buses: 28 ÷ 12 = 2 R4 → 3 buses.

Answer: $252 and 3 buses

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 4 packs of 18 pencils shared by 6 students. Each gets?

2. $50 − 3 × $12

3. A 6-week summer camp, 5 days a week, 3 hours a day. Hours?

More practice: Workbook, Lesson 10.7.

LESSON SUMMARY

Find the hidden steps, write equations, and check that your answer makes sense.

UNIT 10 · LESSON 10.8 · 4.NBT, 4.OA

Lesson 10.8: Review: Whole-Number Challenge

I can solve challenging whole-number problems and explain my strategies.

LET'S EXPLORE

Use the digits 2, 4, 6, 8 once each to make the greatest product of a 2-digit × 2-digit problem.

REMEMBER

Try, check, and improve. Organized guessing is a real strategy.

WORKED EXAMPLE · ORGANIZED GUESSING

Greatest product with 2, 4, 6, 8

1 Put the big digits in the tens places: 8_ × 6_.

2 Try 82 × 64 = 5,248 and 84 × 62 = 5,208.

3 82 × 64 is greater.

Answer: 82 × 64 = 5,248

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Least product with the same digits?

2. Greatest 4-digit number divisible by 5 using 1, 3, 5, 7 once each

3. Two numbers multiply to 96 and add to 20. What are they?

More practice: Workbook, Lesson 10.8.

LESSON SUMMARY

Organized trying, checking, and improving solves challenging problems.

UNIT 10 · LESSON 10.9 · 4.NF.1

Lesson 10.9: Review: Equivalent Fractions

I can find equivalent fractions and simplify.

LET'S EXPLORE

Is $\frac{6}{9}$ the same as $\frac{2}{3}$? How can you prove it?

REMEMBER

Multiply or divide the numerator and denominator by the same number: $\frac{a}{b} = \frac{n \times a}{n \times b}$.

WORKED EXAMPLE · PROVE IT

$\frac{6}{9}$ and $\frac{2}{3}$

1 $\frac{2}{3} \times \frac{3}{3} = \frac{6}{9}$.

2 A model shows the same amount.

Answer: They are equivalent.

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{3}{5} = \frac{?}{20}$
2. Simplify $\frac{16}{20}$
3. Write two fractions equal to $\frac{1}{3}$.

More practice: Workbook, Lesson 10.9.

LESSON SUMMARY

Equivalent fractions name the same amount; multiply or divide top and bottom by the same number.

UNIT 10 · LESSON 10.10 · 4.NF.2

Lesson 10.10: Review: Comparing Fractions

I can compare and order fractions with different denominators.

LET'S EXPLORE

Which is greater: $\frac{5}{8}$ or $\frac{3}{5}$?

REMEMBER

Use benchmarks, common denominators, or common numerators. Compare only when the wholes are the same.

WORKED EXAMPLE · COMMON DENOMINATOR

$\frac{5}{8}$ vs $\frac{3}{5}$

1 $\frac{25}{40}$ vs $\frac{24}{40}$.

Answer: $\frac{5}{8} > \frac{3}{5}$

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $\frac{4}{9} \bigcirc \frac{1}{2}$
2. $\frac{7}{10} \bigcirc \frac{2}{3}$
3. Order: $\frac{1}{2}, \frac{3}{8}, \frac{3}{4}$

More practice: Workbook, Lesson 10.10.

LESSON SUMMARY

Pick the fastest comparison strategy and make sure the wholes match.

UNIT 10 · LESSON 10.11 · 4.NF.3, 4.NF.4

Lesson 10.11: Review: Fraction Operations

I can add and subtract fractions and mixed numbers and multiply fractions by whole numbers.

LET'S EXPLORE

A recipe needs $1\frac{3}{4}$ cups of flour. You triple it. How much flour?

REMEMBER

Like denominators: add or subtract numerators. Mixed numbers: regroup when needed. $n \times \frac{a}{b} = \frac{n \times a}{b}$.

WORKED EXAMPLE · TRIPLE A MIXED NUMBER

$3 \times 1\frac{3}{4}$

1 $1\frac{3}{4} = \frac{7}{4}$.

2 $3 \times \frac{7}{4} = \frac{21}{4} = 5\frac{1}{4}$.

Answer: $5\frac{1}{4}$ cups

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $2\frac{5}{8} + 1\frac{6}{8}$
2. $5\frac{1}{3} - 2\frac{2}{3}$
3. $4 \times \frac{5}{6}$

More practice: Workbook, Lesson 10.11.

LESSON SUMMARY

Add and subtract like fractions, regroup mixed numbers, and multiply the numerator by the whole number.

UNIT 10 · LESSON 10.12 · 4.NF.5–7

Lesson 10.12: Review: Decimals

I can write, compare, and add with tenths and hundredths, including money.

LET'S EXPLORE

Which is more money: $0.75 or $0.8?

REMEMBER

Tenths and hundredths: 0.8 = 0.80. Line up decimal points to compare and to add money.

WORKED EXAMPLE · COMPARE

$0.75 vs $0.8

1 $0.8 = $0.80.

2 80 > 75.

Answer: $0.80 is more

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. $^{7}/_{100}$ as a decimal
2. $^{4}/_{10} + {}^{35}/_{100}$
3. Order: 0.5, 0.45, 0.54

More practice: Workbook, Lesson 10.12.

LESSON SUMMARY

Decimals are fractions in tenths and hundredths; line up the decimal points.

UNIT 10 · LESSON 10.13 · 4.G.1–3, 4.MD.5–7

Lesson 10.13: Review: Geometry and Angles

I can classify shapes and measure, draw, and find angles.

LET'S EXPLORE

A pizza is cut into 6 equal slices. What angle does each slice make at the center?

REMEMBER

A full turn is 360°. Angles are additive. Classify shapes by sides, angles, and symmetry.

WORKED EXAMPLE · ANGLES IN A CIRCLE

6 equal slices

1 360 ÷ 6 = 60.

Answer: 60° each

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Right angle split into 25° and x
2. Lines of symmetry in a rectangle (not a square)
3. A triangle with one 90° angle is a ___ triangle.

More practice: Workbook, Lesson 10.13.

LESSON SUMMARY

Use angle facts (90°, 180°, 360°) and shape attributes.

UNIT 10 · LESSON 10.14 · 4.MD.1–3

Lesson 10.14: Review: Area, Perimeter, and Conversions

I can solve area, perimeter, and measurement conversion problems.

LET'S EXPLORE

A rectangular garden has an area of 72 square feet and a width of 6 feet. How much fence goes around it?

REMEMBER

A = l × w; P = 2 × l + 2 × w. Big unit to small unit: multiply.

WORKED EXAMPLE · TWO STEPS

A = 72, w = 6. Find P.

1 l = 72 ÷ 6 = 12.

2 P = 2 × 12 + 2 × 6 = 36.

Answer: 36 feet of fence

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. 5 yd = ___ in.

2. Area of 14 m × 9 m

3. 2 hr 45 min = ___ min

More practice: Workbook, Lesson 10.14.

LESSON SUMMARY

Use formulas and conversion facts; convert first, then compute.

UNIT 10 · LESSON 10.15 · 4.MD.4

Lesson 10.15: Review: Data and Line Plots

I can make and use line plots with fractions.

LET'S EXPLORE

Five friends' jump distances in feet: $4\frac{1}{2}$, 5, $4\frac{3}{4}$, 5, $4\frac{1}{2}$. Make a line plot. What is the difference between the longest and shortest jumps?

REMEMBER

Each X is one data value. Use fraction operations to answer questions.

WORKED EXAMPLE · READ A LINE PLOT

Longest − shortest

1 $5 - 4\frac{1}{2} = \frac{1}{2}$.

Answer: $\frac{1}{2}$ foot

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. How many jumps were 5 feet?
2. Total of the $4\frac{1}{2}$-foot jumps
3. Most common distance(s)?

More practice: Workbook, Lesson 10.15.

LESSON SUMMARY

Line plots show measurement data; use fraction operations to answer questions.

UNIT 10 · LESSON 10.16 · Grade 4 review

Lesson 10.16: Celebrating a Year of Math

I can show what I learned in Grade 4 math and set goals for Grade 5.

LET'S EXPLORE

Look back at Lesson 1.1. What can you do now that you couldn't do then?

REMEMBER

You learned place value to one million, all four operations with big numbers, fractions, decimals, geometry, and measurement.

WORKED EXAMPLE · TEACH IT BACK

Choose your favorite lesson and teach it to someone at home.

1 Pick a lesson.

2 Explain the Key Idea.

3 Work one example together.

Answer: Answers vary

TRY IT

Work these in your notebook or on scrap paper. Then check with your teacher.

1. Name one thing you learned in each unit.

2. What was the hardest unit? What helped you?

3. Set one math goal for Grade 5.

More practice: Workbook, Lesson 10.16.

LESSON SUMMARY

You can do a lot of math now. Keep practicing and keep asking “why?”

UNIT 10 REVIEW

Unit 10 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember…
10.1	Place value, comparing, and rounding work the same way for every size of number.
10.2	Line up places, regroup carefully, and estimate to check.
10.3	Break factors into place-value parts, multiply each part, and add.
10.4	Divide step by step, check by multiplying, and interpret the remainder.
10.5	Factors divide evenly; multiples come from skip counting; primes have exactly two factors.
10.6	Find the rule, then look for hidden features.
10.7	Find the hidden steps, write equations, and check that your answer makes sense.
10.8	Organized trying, checking, and improving solves challenging problems.
10.9	Equivalent fractions name the same amount; multiply or divide top and bottom by the same number.
10.10	Pick the fastest comparison strategy and make sure the wholes match.
10.11	Add and subtract like fractions, regroup mixed numbers, and multiply the numerator by the whole number.
10.12	Decimals are fractions in tenths and hundredths; line up the decimal points.
10.13	Use angle facts (90°, 180°, 360°) and shape attributes.
10.14	Use formulas and conversion facts; convert first, then compute.
10.15	Line plots show measurement data; use fraction operations to answer questions.
10.16	You can do a lot of math now. Keep practicing and keep asking “why?”

Glossary

acute angle	Smaller than a right angle.
acute triangle	A triangle with three acute angles.
additive	An angle made of non-overlapping parts has a measure equal to the sum of the parts.
additive comparison	A comparison using “more than” or “fewer than.”
angle	Two rays with a shared endpoint.
area	Square units that cover a surface.
area model	A rectangle split into parts to show multiplication.
attribute	A feature of a shape, such as number of sides or types of angles.
bar model	A drawing with bars that shows how the numbers in a problem are related.
base-ten	Our number system, where each place is worth 10 times the place to its right.
benchmark	A friendly number, like a multiple of 10, 100, or 1,000, used to round or estimate.
benchmark angle	A familiar angle like 45°, 90°, or 180° used to estimate.
benchmark fraction	Friendly fraction like 0, 1/2, 1.
capacity	How much a container holds.
cent	One hundredth of a dollar: $0.01.
change	The money you get back when you pay more than the price.
common denominator	A denominator that two fractions share.
common factor	A number that is a factor of two or more numbers.
common numerator	A numerator that two fractions share.
composite figure	A shape made by joining simpler shapes.
composite number	A number with more than two factors.
convert	Change to a different unit.
customary units	Units used in the U.S.: inches, feet, yards, miles, ounces, pounds, cups, gallons.
data	Information collected, such as measurements.
decimal	A number written with a decimal point.
decimal point	Separates wholes from parts.
decompose	Break a number into parts.

degree	Unit for measuring angles.
degree (°)	A unit for measuring angles. A full turn is 360°; one degree is $\frac{1}{360}$ of a turn.
denominator	Bottom number: equal parts in the whole.
difference	The answer to a subtraction problem.
digit	One of the ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 used to write numbers.
distributive property	Breaking a factor into parts, multiplying each part, and adding: 4 × 23 = 4 × 20 + 4 × 3.
dividend	The number being divided.
divisible	Can be divided with no remainder.
divisor	The number you divide by.
elapsed time	Time that passes between start and end.
equal groups	Groups that all have the same number of items.
equal to (=)	Shows two amounts are the same.
equivalent fractions	Fractions naming the same amount.
estimate	A close answer found with rounded numbers.
estimate (product)	Round a factor, then multiply to get a close answer.
expanded form	A number written as the sum of the values of its digits.
fact family	Related multiplication and division facts, like 6 × 4 = 24, 4 × 6 = 24, 24 ÷ 4 = 6, 24 ÷ 6 = 4.
factor	A number being multiplied.
factor pair	Two numbers that multiply to make a product, like 3 and 8 for 24.
formula	A rule written with symbols, like A = l × w.
fraction	A number naming part of a whole.
greater than (>)	Shows that the first number is larger: 9 > 4.
greatest	The largest number in a group.
growing pattern	A pattern that gets bigger by a rule.
halfway	The point exactly in the middle of two benchmarks, like 450 between 400 and 500.
hundreds part	In 324, the hundreds part is 300.
hundredth	1 of 100 equal parts.
hundredths place	The second place to the right of the decimal point.

improper fraction	A fraction with a numerator equal to or greater than its denominator, like $^{7}/_{4}$.
inch ruler	A ruler where each inch is split into halves, fourths, and eighths.
input-output table	A table that shows a rule changing each input number into an output number.
interpret	To explain what something means.
intersecting lines	Lines that cross at a point.
interval	The amount between two marks next to each other on a number line.
justify	To give reasons that show why something is true.
least	The smallest number in a group.
less than (<)	Shows that the first number is smaller: 4 < 9.
like denominators	Same denominators.
line	A straight path that goes on forever in both directions.
line of symmetry	Folds a figure into matching halves.
line plot	Number line with X marks for data.
line segment	Part of a line with two endpoints.
line-symmetric	Having at least one line of symmetry.
long division	Dividing one place at a time from left to right: divide, multiply, subtract, bring down.
mass	The amount of matter in an object, measured in grams and kilograms.
metric units	Units based on 10s: meters, grams, liters.
mixed number	A whole number and a fraction together, like $1\,^{3}/_{4}$.
multi-step problem	A problem that takes more than one calculation to solve.
multiple	The product of a number and a whole number.
multiple of 10	A number you get by multiplying a whole number by 10, like 30, 70, or 120.
multiplicative comparison	A comparison using “times as many.”
number line	A line with evenly spaced marks that shows numbers in order.
number line (fractions)	A line where the distance from 0 to 1 is split into equal parts.
numerator	Top number: parts counted.
obtuse angle	Larger than a right angle but smaller than a straight angle.
obtuse triangle	A triangle with one obtuse angle.
one million	1,000,000. It is 10 times 100,000 and 1,000 times 1,000.

order	To arrange numbers from least to greatest or from greatest to least.
parallel	Never meet.
parallel lines	Lines that never cross and are always the same distance apart.
parallelogram	A quadrilateral with two pairs of parallel sides.
partial products	Smaller products added to find the whole product.
partial quotients	A way to divide by subtracting easy multiples of the divisor and adding up how many groups you took away.
pattern	A predictable arrangement of numbers or shapes.
perimeter	Distance around a figure.
period	A group of three places separated by a comma.
perpendicular	Meet at right angles.
perpendicular lines	Lines that cross to make square corners (right angles).
place value	The value a digit has because of its place in a number.
point	An exact location, shown with a dot.
prime number	A number with exactly two factors.
product	The answer to a multiplication problem.
protractor	A tool for measuring angles in degrees.
quotient	The answer to a division problem.
ray	Part of a line with one endpoint that goes on forever in one direction.
reasonable	Makes sense; close to what we expect from an estimate.
regroup	To trade 10 of one unit for 1 of the next bigger unit, or the reverse.
regroup (subtraction)	Trade 1 of a bigger unit for 10 of the next smaller unit so you have enough to subtract.
remainder	The amount left over.
rename	To write the same amount using different units, such as 1 thousand = 10 hundreds.
repeating pattern	A pattern with a part (the core) that repeats.
review	To study again so you remember.
rhombus	A parallelogram with four equal sides.
right angle	A square corner, 90°.
right triangle	A triangle with one right angle.

round	To change a number to a nearby friendly number.
rounding place	The place you are rounding to. If you round to the nearest thousand, the rounding place is the thousands place.
rule	Tells how to get the next term of a pattern.
set	A group of objects.
simplest form	No common factor but 1.
square unit	A square 1 unit long on each side, like a square inch or square meter.
standard algorithm	A step-by-step way to add or subtract by lining up places and working from right to left.
standard algorithm (multiplication)	Multiplying each place from right to left and regrouping, writing the product on one line.
standard form	The usual way to write a number with digits, such as 452,318.
straight angle	A straight line, 180°.
strategy	A plan or method for solving a problem.
subtract across zeros	Regrouping through one or more zeros to get enough to subtract.
sum	The answer to an addition problem.
tenth	1 of 10 equal parts.
tenths place	The first place to the right of the decimal point.
term	Each number or shape in a pattern.
times as many	A comparison that uses multiplication: 3 times as many as 8 is 24.
trapezoid	A quadrilateral with at least one pair of parallel sides.
unit fraction	A fraction with a numerator of 1, like $\frac{1}{5}$.
value	How much a digit is worth in a number. In 7,482 the 4 has a value of 400.
vertex	The shared endpoint of an angle.
weight	How heavy something is.
whole	The one complete thing or group a fraction is part of.
word form	A number written with words, such as thirty-two thousand, five.

Grade 4 Mathematics

Student Textbook · 2026–2027

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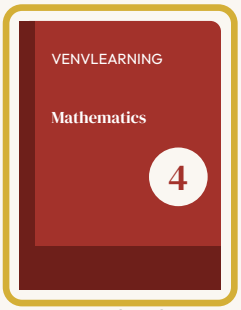
INSIDE THIS YEAR

01 Place Value, Rounding, and Whole Numbers
02 Multiplication
03 Division
04 Factors, Multiples, and Patterns
05 Fraction Equivalence and Comparison
06 Fraction Operations
07 Decimals
08 Geometry, Angles, and Line Plots
09 Measurement, Area, and Perimeter
10 Year in Review

IN THIS BOOK

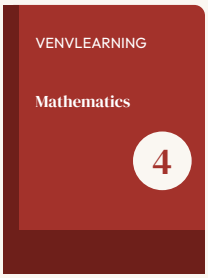
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- Models, examples, and Key Idea summaries
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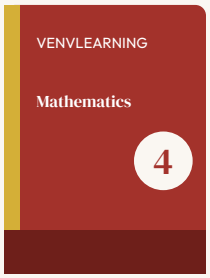


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